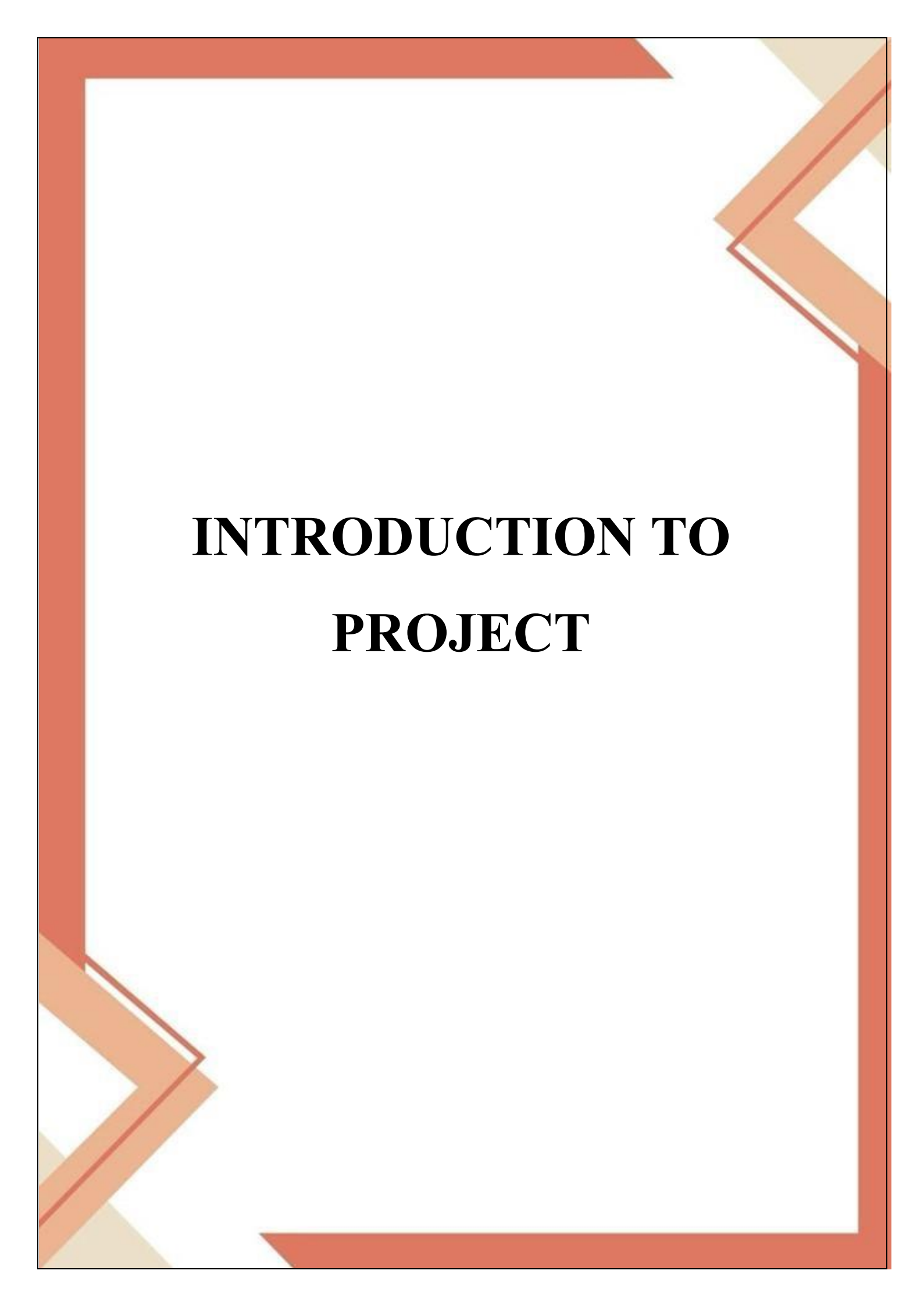




INTRODUCTION TO PROJECT

INTRODUCTION TO PROJECT

This project focuses on developing a comprehensive Social Media Performance Comparison Dashboard to analyze and compare the performance of six major digital platforms — Facebook, Instagram, YouTube, LinkedIn, Twitter, and Snapchat. The objective of this project is to transform raw social media data into meaningful business insights using SQL for data aggregation and Power BI for interactive visualization. The dashboard enables stakeholders to monitor key performance indicators such as Total Posts, Total Reach, Total Engagement, Total Views, and Engagement Rate %, allowing data-driven decision-making for digital marketing strategies.

The dataset consists of structured content-level information, including Post ID, Platform, Category, Post Type, Reach, Engagement, Views, and Post Date. Using SQL queries, the data was cleaned, aggregated, and validated to ensure accuracy in calculating performance metrics. Key measures such as Engagement Rate % were calculated using logical formulas to avoid divide-by-zero errors and ensure reliable results. The processed data was then imported into Power BI, where DAX measures were created to dynamically calculate KPIs based on applied filters and slicers.

The dashboard is designed with interactive slicers such as Date, Platform, Category, and Post Type to allow dynamic filtering and detailed comparison. Visualizations such as stacked column charts, bar charts, KPI cards, scatter plots, and matrix tables provide multi-dimensional analysis of platform performance. The comparison page enables users to evaluate which platform generates the highest reach, which category performs best on specific platforms, and which platform delivers the highest engagement efficiency.

This project demonstrates strong skills in data modeling, SQL aggregation, DAX calculations, KPI development, dashboard design, and business insight generation. It highlights the ability to convert raw data into actionable insights, helping organizations optimize their content strategy and improve audience engagement across multiple social media platforms.

Key Performance Indicators (KPIs), including:

The following KPIs were created using DAX:

- **Total Posts** → Measures content volume
- **Total Reach** → Measures audience exposure
- **Total Engagement** → Measures user interaction
- **Total Views** → Measures content consumption
- **Engagement Rate %** → Measures content effectiveness

Engagement Rate % was calculated using a safe division formula to avoid errors and ensure accuracy.

The **Social Media Performance Comparison Dashboard** is a comprehensive data analytics project developed to evaluate and compare the performance of six major social media platforms — Facebook, Instagram, YouTube, LinkedIn, Twitter, and Snapchat — using a unified data model. The primary purpose of this project is to help stakeholders understand platform-wise audience behavior, content effectiveness, and engagement efficiency through structured data analysis and interactive visualization. In today's competitive digital environment, organizations publish content across multiple platforms, but without centralized analysis, it becomes difficult to measure which platform, category, or content type delivers the best results. This project addresses that challenge by integrating cross-platform data into a single analytical dashboard.

The dataset used in this project contains post-level records including Platform name, Post ID, Content Category, Post Type (Live, Short, Image, Video, etc.), Post Date, Reach, Engagement, and Views. The project began with data cleaning and transformation using SQL to ensure data consistency, remove duplicates, handle missing values, and validate aggregation logic. SQL queries were written to calculate aggregated metrics such as total engagement by category and platform, total reach by platform, and engagement rate percentages. These SQL results were used to cross-verify Power BI calculations to maintain data accuracy.

In Power BI, a structured data model was created, and DAX measures were developed to calculate key performance indicators such as Total Posts (using `DISTINCTCOUNT`), Total Reach (SUM of Reach), Total Engagement (SUM of Engagement), Total Views (SUM of Views), and Engagement Rate % (Engagement divided by Reach with error

handling using DIVIDE function). These measures dynamically respond to slicers and filters, enabling real-time comparison across different dimensions. The dashboard includes interactive slicers such as Date (Month-Year hierarchy), Platform, Category, and Post Type to allow users to analyze performance trends, filter specific content strategies, and compare multiple platforms simultaneously.

The dashboard design focuses on clear storytelling and decision support. Visual elements such as stacked column charts, clustered bar charts, matrix tables, KPI cards, line charts for trend analysis, and percentage-based comparisons help identify which platform generates the highest reach, which category drives maximum engagement, and which content type produces the best engagement rate. The comparison page consolidates all major KPIs into a single view, allowing management to quickly assess overall performance and make strategic marketing decisions.

Overall, this project demonstrates practical expertise in SQL querying, data validation, DAX calculations, KPI development, data modeling, interactive dashboard creation, and analytical storytelling. It reflects the ability to transform raw multi-platform social media data into structured insights that support performance optimization and digital strategy improvement.

OBJECTIVE

To Centralize Multi-Platform Social Media Data

The primary objective of this project is to integrate data from six different social media platforms — Facebook, Instagram, YouTube, LinkedIn, Twitter, and Snapchat — into a single unified analytical dashboard. Since each platform generates data in different formats and structures, this project ensures that all datasets are standardized and modeled into one structured table format. This centralization eliminates scattered reporting and enables consistent cross-platform comparison within a single environment.

To Measure and Compare Key Performance Indicators (KPIs)

Another major objective is to calculate and compare important KPIs such as:

- Total Posts
- Total Reach
- Total Engagement
- Total Views
- Engagement Rate %

These KPIs help evaluate platform performance from different perspectives — content quantity, audience exposure, user interaction, and content effectiveness. By using DAX measures and SQL validation, the project ensures accurate KPI calculation for meaningful comparison.

To Analyze Platform-Wise Performance Efficiency

The project aims to determine which platform performs best based on engagement efficiency rather than just volume. Engagement Rate % ($\text{Engagement} \div \text{Reach}$) is calculated to understand how effectively each platform converts audience reach into interaction. This helps in identifying whether high reach actually leads to high engagement or not.

To Evaluate Category-Based Content Performance

This objective focuses on analyzing how different content categories (e.g., Educational, Promotional, Entertainment, Informative) perform across platforms. By comparing Total Engagement and Reach by Category and Platform, the project identifies:

- Which category performs best on each platform

- Which platform is most suitable for specific content types

This supports better content strategy planning.

To Assess Post Type Effectiveness

Different post formats such as Live, Short, Video, and Image generate varying engagement levels. The project aims to analyze engagement and reach by Post Type to determine:

- Which format drives the highest engagement
- Which format delivers better engagement rate
- Platform-specific content format preferences

This enables optimization of content publishing strategy.

To Enable Time-Based Performance Analysis

The dashboard includes month-wise and date-wise analysis to track performance trends over time. This objective ensures identification of:

- Growth or decline in engagement
- Seasonal spikes in reach
- Monthly content performance patterns

Time-based insights help in understanding campaign effectiveness and long-term trends.

To Support Data-Driven Decision Making

The project is designed to provide actionable insights to marketing teams and decision-makers. Instead of relying on assumptions, the dashboard enables performance comparison based on actual data metrics. This supports decisions such as:

- Budget allocation across platforms
- Platform prioritization
- Content strategy adjustments

To Ensure Data Accuracy and Validation

An important objective is to maintain data reliability by validating Power BI calculations with SQL aggregation queries. Error handling techniques such as DIVIDE function in DAX and NULLIF in SQL are used to prevent incorrect results. This ensures that all KPIs are calculated accurately and consistently.

To Develop Interactive and User-Friendly Dashboard Design

The dashboard is built with interactive slicers (Date, Platform, Category, Post Type) allowing users to dynamically filter and drill down into data. The objective is to create a clean, professional, and intuitive interface that provides clarity without overwhelming users.

To Demonstrate Technical and Analytical Skills

Finally, the project aims to showcase practical expertise in:

- SQL querying and aggregation
- Data cleaning and transformation
- Data modeling in Power BI
- DAX measure development
- KPI design
- Analytical storytelling

This objective reflects the ability to convert raw data into structured business insights.

To Identify High-Performing Platforms Based on Audience Behaviour

One of the objectives is to analyze user interaction patterns across platforms to determine where audiences are most active. By comparing engagement metrics such as likes, comments, and shares, the dashboard helps identify which platform encourages stronger audience participation. This allows marketers to focus on platforms where users are more responsive rather than just where reach is high.

To Measure Content Productivity and Posting Efficiency

This objective focuses on evaluating how effectively each platform converts posted content into measurable engagement. By comparing Total Posts with Total Engagement and Engagement Rate %, the dashboard identifies whether a higher number of posts actually results in better performance or if fewer, high-quality posts perform better. This helps optimize posting frequency strategies.

To Detect Performance Gaps Across Platforms

The project aims to identify underperforming platforms by comparing reach and engagement metrics side-by-side. If a platform shows high reach but low engagement rate, it indicates a content relevance issue. If it shows low reach but high engagement rate, it

suggests strong audience loyalty but limited exposure. Identifying such gaps supports improvement planning.

To Support Strategic Budget Allocation Decisions

Another objective is to provide performance-based evidence for marketing investment decisions. By analyzing which platform generates better engagement efficiency, management can allocate advertising budgets and promotional efforts more strategically instead of distributing resources equally across all platforms.

To Monitor Content Performance Trends Over Time

The dashboard is designed to track growth patterns in engagement, reach, and views month-over-month. This objective helps determine whether performance improvements are consistent, seasonal, or campaign-based. Trend analysis assists in evaluating long-term digital marketing effectiveness.

To Enable Multi-Dimensional Data Analysis

This project allows data to be analyzed across multiple dimensions simultaneously, including Platform, Category, Post Type, and Time. The objective is to provide flexibility in analysis so users can drill down into specific combinations, such as “Engagement Rate of Video posts on Instagram during a specific month.”

To Improve Content Strategy Optimization

By identifying which categories and post types generate higher engagement rates on specific platforms, the dashboard supports content planning decisions. The objective is to refine digital strategy based on data insights rather than assumptions, improving overall marketing ROI.

To Build Scalable Analytical Framework

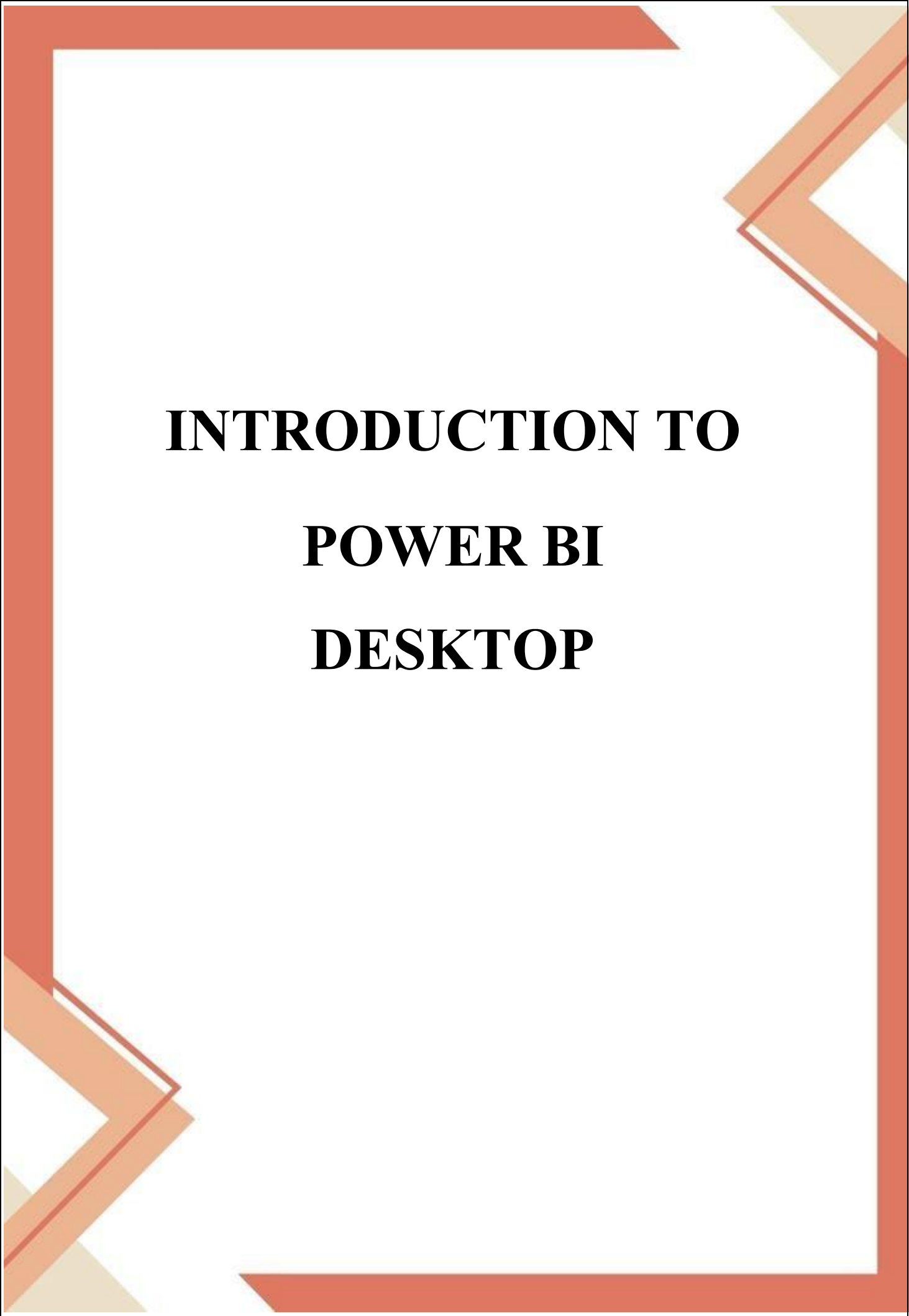
The project is structured in a way that new platforms or additional KPIs can be added easily without redesigning the entire dashboard. The objective is to create a scalable model that supports future expansion, such as adding paid campaign data or demographic insights.

To Ensure Performance Transparency and Reporting Consistency

The dashboard provides a standardized reporting format for all platforms. This objective ensures that stakeholders receive consistent metrics and definitions across reports, reducing confusion and improving clarity in performance evaluation.

To Strengthen Data Storytelling and Visualization Skills

Beyond technical calculation, the project aims to present insights through meaningful visual storytelling. The objective is to design a dashboard that not only shows numbers but also clearly communicates trends, comparisons, and performance insights in an understandable way for business users.



INTRODUCTION TO POWER BI DESKTOP

INTRODUCTION TO POWER BI DESKTOP

Power BI Desktop is a powerful business intelligence and data visualization application developed by Microsoft. It allows users to connect to various data sources, transform raw data into meaningful insights, and create interactive reports and dashboards.

Power BI Stands for Power Business Intelligence which is a technology-driven business intelligence tool.

It is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights. Your data might be an Excel spreadsheet, or a collection of cloud-based and on-premises hybrid data warehouses.

Power BI lets you easily connect to your data sources, visualize and discover what's important, and share that with anyone or everyone you want.

Power BI Desktop is widely used by businesses and data professionals to analyse data and support data-driven decision-making. The tool provides a user-friendly interface, making it accessible to both technical and non-technical users. One of the core strengths of Power BI Desktop is its ability to connect with multiple data sources.

It supports various file types like Excel, CSV, and XML, along with direct connections to databases such as SQL Server, Oracle, and MySQL. Additionally, Power BI Desktop can connect to cloud services like Azure, Google Analytics, and SharePoint. Once the data is imported, users can use Power Query Editor to clean, transform, and shape the data according to business requirements. This includes filtering rows, removing duplicates, merging tables, and adding custom calculated columns.

Data modelling is another key feature of Power BI Desktop, enabling users to create relationships between different data tables. It allows the creation of calculated columns, measures, and hierarchies using DAX (Data Analysis Expressions) — a formula language that helps perform advanced calculations. These models serve as the backbone of the report, ensuring that data is organized efficiently and relationships between different datasets are properly established.

Visualization is at the heart of Power BI Desktop. It provides a wide range of interactive visual elements like bar charts, pie charts, line graphs, maps, and tables. Users can customize the visuals to make the reports more appealing and easier to interpret. The drag-

and-drop interface makes it simple to create visualizations without requiring coding skills. Additionally, custom visuals from the Power BI marketplace can be integrated to enhance the report further.

Once reports are created, they can be published to the Power BI Service, a cloud-based platform where users can share reports with colleagues, collaborate, and schedule data refreshes. Power BI Desktop also allows exporting reports to formats like PDF or PowerPoint for offline sharing. The combination of Power BI Desktop with Power BI Service creates a comprehensive business intelligence solution for organizations.

Overall, Power BI Desktop empowers businesses to make data-driven decisions by turning raw data into visually appealing insights. It combines data integration, transformation, modelling, and visualization into one unified platform, making it one of the leading tools in the business intelligence landscape.

Benefits Of Power BI

- **User-friendly interface:**

Power BI has an intuitive interface allowing users to visualize and analyse data easily.

- **Data integration:**

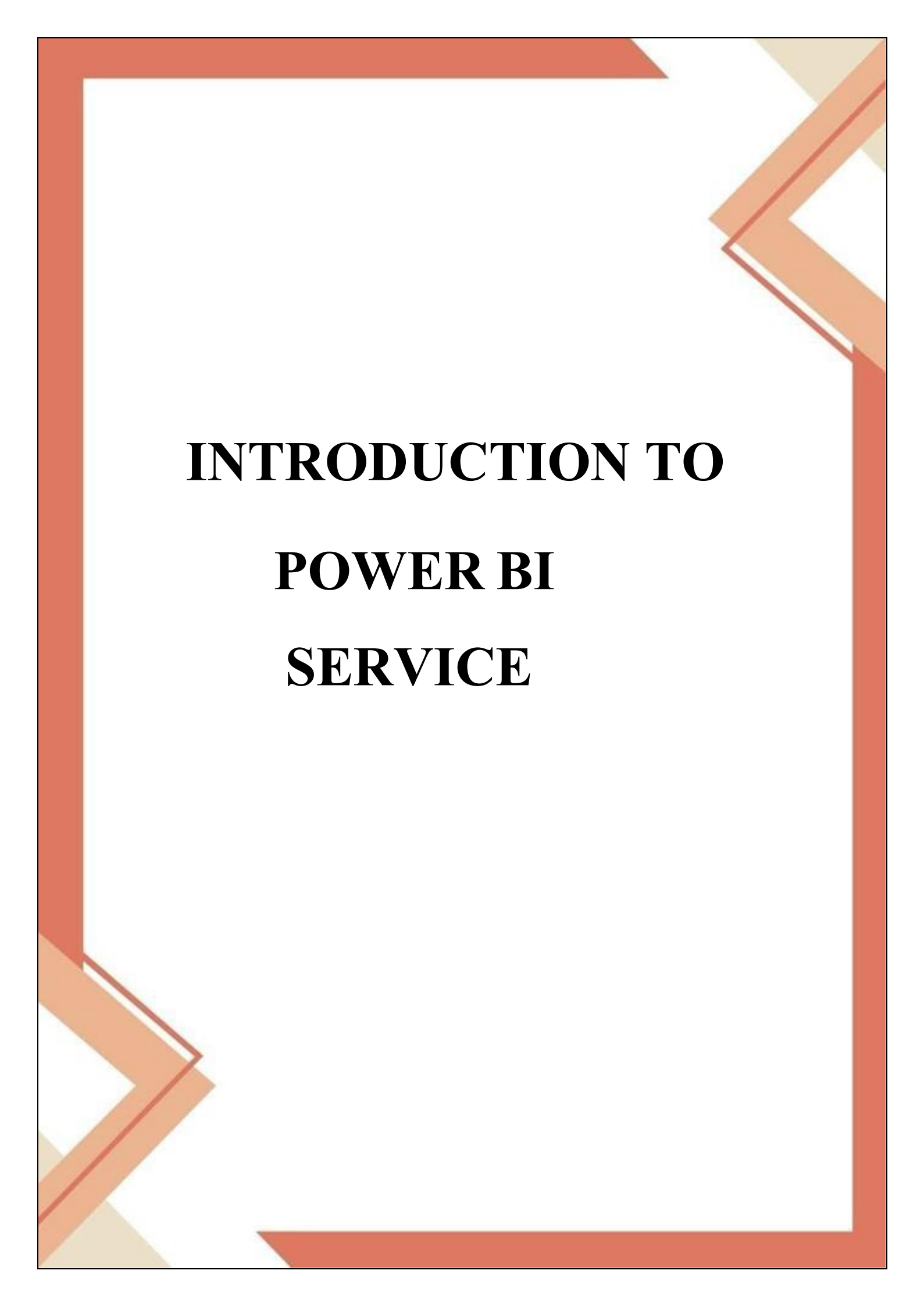
Power BI allows users to easily integrate data from various sources, including Excel, SQL Server, and cloud-based sources like Azure and Salesforce.

- **Advanced Analytics:**

Power BI integrates with advanced analytics tools like R and Python and DAX functions, allowing users to perform complex statistical analysis and create advanced visualizations.

- **Mobile Access:**

Power BI has mobile apps for iOS, Android, and Windows devices, allowing users to access their dashboards and reports.



INTRODUCTION TO POWER BI SERVICE

INTRODUCTION TO POWER BI SERVICE

Power BI Service is a cloud-based business intelligence (BI) platform by Microsoft that enables users to publish, share, and collaborate on interactive reports and dashboards. It is an essential part of the Power BI ecosystem, working alongside Power BI Desktop to provide a complete data analytics solution. While Power BI Desktop is primarily used for creating reports, Power BI Service is designed for sharing, collaboration, and online accessibility. It allows users to access reports and dashboards from anywhere via a web browser or mobile app, ensuring seamless data-driven decision-making.

One of the key features of Power BI Service is its ability to provide real-time data updates and scheduled refreshes. Unlike Power BI Desktop, where data updates must be done manually, Power BI Service can automatically refresh datasets at set intervals. This ensures that reports always display the most current data without requiring manual intervention. The platform also integrates with various cloud and on-premises data sources, such as Azure, SQL Server, Google Analytics, and Excel, allowing users to pull data directly from these sources and visualize it in real-time.

Collaboration is a major advantage of Power BI Service. Users can share reports and dashboards with team members, assign access roles, and collaborate in workspaces. Workspaces in Power BI Service act as shared environments where multiple users can contribute to report development and analysis. This makes it an ideal solution for organizations that require multiple users to work on the same datasets and reports. Additionally, the platform supports Power BI Apps, which allow businesses to package multiple dashboards and reports into an easy-to-access app for end-users.

Another critical aspect of Power BI Service is its ability to provide AI-powered insights and data alerts. The platform leverages artificial intelligence and machine learning capabilities to identify trends, detect anomalies, and suggest insights based on the data. Users can also set up data alerts, which notify them when key business metrics cross predefined thresholds. This feature helps organizations monitor business performance and take immediate action when necessary.

Power BI Service also integrates with Microsoft tools like Teams, Excel, and SharePoint, making it easier to incorporate data analysis into daily workflows. It offers robust security features, such as role-based access control (RBAC), data encryption, and integration with

Microsoft Azure Active Directory, ensuring that sensitive business data remains secure. Additionally, Power BI Service includes Power BI Premium, which provides enhanced performance, larger data capacity, and advanced AI capabilities for enterprise users.

Overall, Power BI Service transforms raw data into actionable insights by providing a centralized, cloud-based platform for data visualization, analysis, and collaboration. It bridges the gap between data analysts and business users by making interactive reports and dashboards accessible from anywhere. With features like real-time data updates, AI-driven analytics, and enterprise-level security, Power BI Service is a powerful tool for organizations looking to enhance their data-driven decision-making process.

Benefits of Power BI Service

- **Data Sharing and Collaboration:**

By enabling users to publish their Power BI Desktop reports and dashboards to the cloud, users can collaborate in real-time, add comments, and work together on the same reports and dashboards, enhancing teamwork and decision-making

- **Interactive Dashboards:**

Dashboards can combine multiple visualizations from different reports, allowing users to monitor key performance indicators (KPIs) and track metrics in one place.

- **Real-Time Data:**

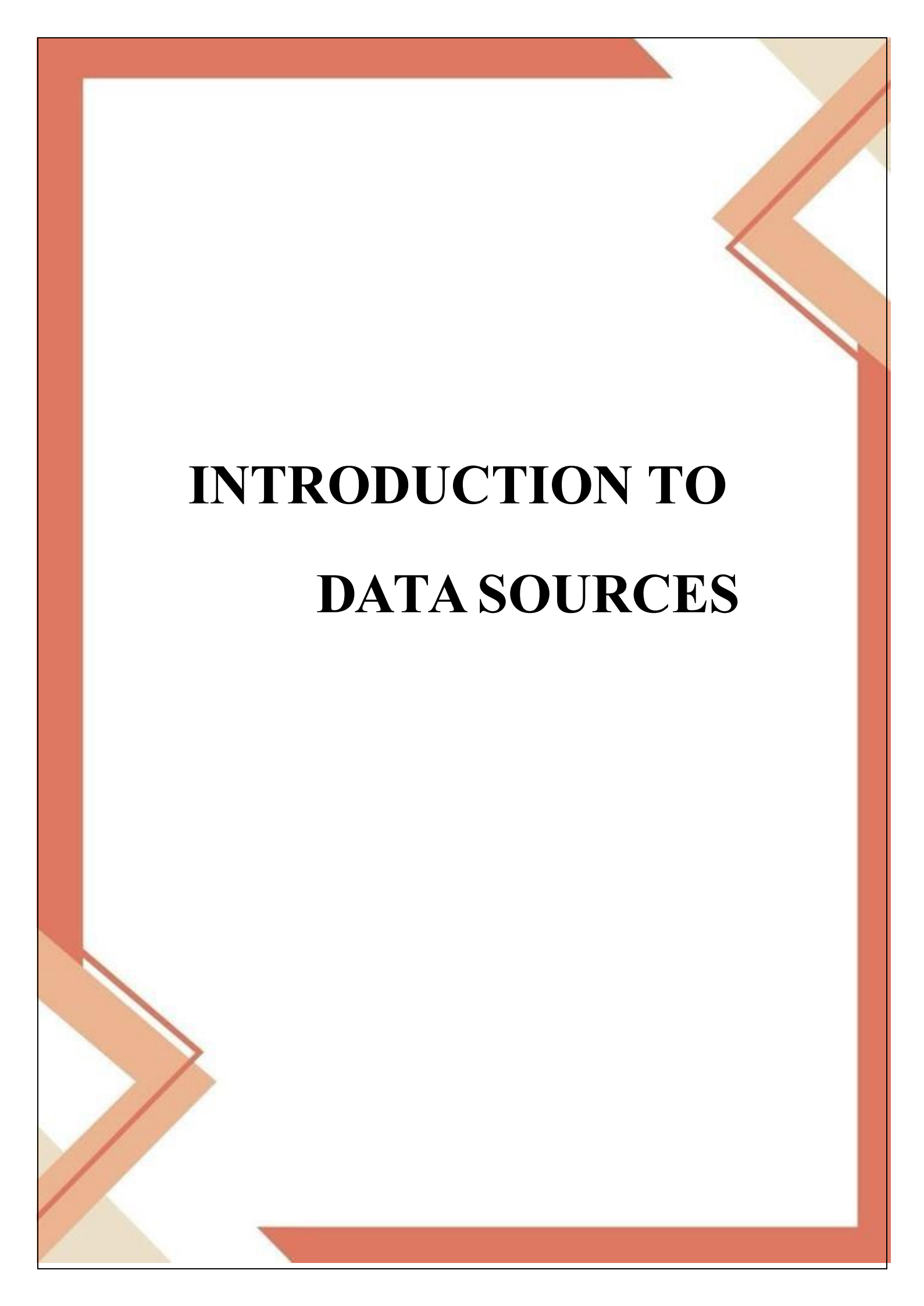
Power BI Service supports real-time data streaming, enabling dashboards to be updated in real-time as data changes. This is particularly useful for monitoring live data and making timely decisions.

- **Security and Access Control:**

Power BI Service provides robust security features, including row-level security, which allows users to control access to data based on user roles.

- **Data Refresh:**

The service supports scheduled data refreshes, ensuring that data in reports and dashboards is always up-to-date. Users can set refresh schedules according to needs.



INTRODUCTION TO DATA SOURCES

INTRODUCTION TO DATA SOURCES

In Power BI, a "data source" refers to the location or system where raw data originates, which can be a database, file (like Excel), web page, or another application, that Power BI can connect to and extract data from to create reports and visualizations; essentially, it's the source of information used to build your analysis within Power BI.

In a Power BI project, the ability to connect to various data sources is paramount for comprehensive analysis and reporting. Data sources serve as the foundation for any business intelligence (BI) solution, providing the raw data necessary to generate meaningful insights, drive decision-making, and optimize operations. Power BI, is a powerful business analytics tool by Microsoft, that enable users to import and integrate data from multiple sources seamlessly. This capability is crucial for ensuring accurate and thorough analysis, allowing users to consider all relevant data points when making informed decisions.

A data source in the context of Power BI is any system, file, or application that provides data to the BI solution. These sources can vary significantly in format, structure, and origin, including structured databases, flat files, web services, and cloud-based data platforms. The diversity of data sources reflects the complexity of modern business environments, where data is generated from various touchpoints, such as sales transactions, customer interactions, online activities, and external market data.

Power BI excels in its ability to connect to a wide variety of data sources, both on-premises and cloud-based. This flexibility ensures that users can integrate data from their existing systems and maximize the potential of their information assets.

Here are detailed explanations of the data sources available in Power BI:

- **Excel Workbook:**

Power BI can import data from Excel files, commonly used for structured data storage. Users can bring in data from Excel tables, ranges, or pivot tables, enabling easy integration of spreadsheets into Power BI for analysis and visualization. This allows for the seamless utilization of existing Excel data in dynamic reports and dashboards.

- **Power BI Semantic Models:**

It simplifies the interaction with complex datasets by providing a structured and user-friendly layer. Created using SQL Server Analysis Services (SSAS) or Power BI's data modeling features, these models define relationships between data entities like tables and columns. This makes it easier to perform advanced analytics and create rich visualizations. They support features like calculated columns, measures, and hierarchies, enabling detailed analysis without needing to understand all the underlying data complexities. By connecting to a semantic model, Power BI users can leverage powerful analytical capabilities to make well-informed business decisions.

- **Dataflows:**

Dataflows in Power BI enable users to create reusable data preparation processes that can be shared across multiple reports and dashboards. By connecting to dataflows, users can standardize data transformation steps and maintain a single source of truth. This ensures that data is consistently prepared and readily available for analysis.

- **Dataverse:**

Microsoft Dataverse is a cloud-based data platform that allows users to securely store and manage data used by business applications. By connecting to Dataverse, users can integrate data from various Microsoft applications like Dynamics 365 and Power Apps into Power BI. This integration facilitates comprehensive analysis by combining data from different business processes.

- **SQL Server:**

Power BI can connect to SQL Server databases, providing access to relational databases that store large amounts of data. SQL Server is known for its robustness, scalability, and ability to handle complex queries, making it ideal for enterprise-level data analysis. Users can efficiently query and visualize data stored in SQL Server within Power BI.

- **SQL Server Analysis Services (SSAS):**

Connect to SSAS to perform advanced analytics using multidimensional or tabular data models. SSAS enables users to create complex data models and run sophisticated queries, making it a powerful tool for data analysis and reporting. This connection allows for in-depth analysis of data using pre-built cubes or tabular models.

- **Text/CSV:**

Text and CSV files are simple, flat-file formats used for data exchange. They store data in plain text with fields separated by delimiters like commas or tabs. In Power BI, these files are easily imported and transformed using the Power Query Editor. Their simplicity and compatibility make them ideal for sharing and integrating data from various sources. Common uses include exporting data from systems, sharing data between teams, and archiving historical data. Leveraging Text/CSV files in Power BI enhances data portability and streamlines data preparation for analysis. Power BI allows users to easily transform and analyze this data within their reports.

- **Web:**

Connect to data from web sources, enabling users to integrate online data into Power BI reports and dashboards. This can include data from websites, APIs, or other web-based sources, allowing for real-time data integration and analysis. Users can create dynamic reports that incorporate up-to-date online information.

- **OData Feed:**

OData (Open Data Protocol) is a standardized protocol for building and consuming RESTful APIs. It allows for the querying and updating of data using a uniform and consistent approach. In Power BI, OData feeds serve as a data source that enables users to connect to and import data from web services that support the OData protocol. Once connected, users can select the tables or entities they want to import and use the Power Query Editor to transform and shape the data as needed. By leveraging OData feeds in Power BI, users can seamlessly integrate data from various web services, enhance data interoperability, and ensure their reports and dashboards are always current and relevant.

- **Blank Query:**

In Power BI, a Blank Query is considered a data source because it serves as an initial point of entry for creating and importing data into the Power BI environment. A Blank Query starts with no predefined data, it provides the flexibility to define custom data connections and transformations using the Power Query Editor and the M language. This allows users to connect to various data sources, shape the data according to their requirements, and then import it into Power BI.

- **Power BI Template Apps:**

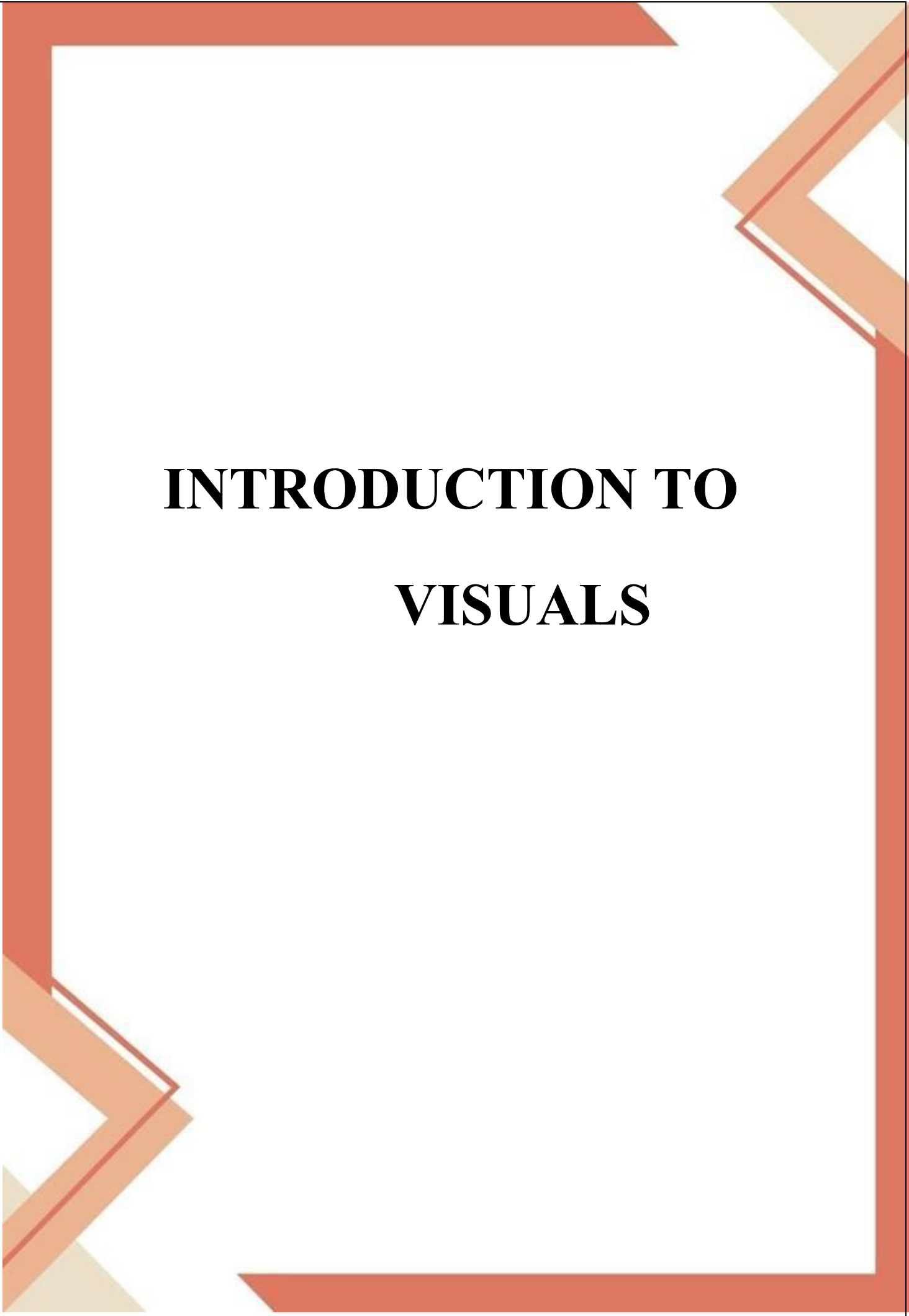
Connect to Power BI template apps, which are pre-built solutions designed to address specific business needs. These template apps often include pre-configured reports and dashboards, making it easier for users to get started with common use cases and quickly gain insights. This accelerates the deployment of Power BI solutions for various scenarios.

- **Python Script:**

Power BI allows users to import data processed or analyzed using Python scripts. Python, with libraries like pandas, numpy, and matplotlib, is well-suited for data manipulation, cleaning, and advanced analytics. By using Python scripts as data sources, users can bring in refined data directly into Power BI for further visualization and analysis.

- **R Script:**

Similar to Python, R scripts can also be used as data sources in Power BI. R is commonly used for statistical analysis and data visualization, leveraging libraries like ggplot2 and dplyr. Using R scripts, users can integrate data processed and analyzed within R into Power BI, combining the strengths of both tools for comprehensive analysis.



INTRODUCTION TO VISUALS

INTRODUCTION TO VISUALS

Visuals in Power BI are the graphical representations that bring data to life. They are essential tools for transforming raw data into meaningful insights, making complex information easier to understand and analyze. By using visuals, users can highlight key metrics, identify trends, uncover patterns, and make data-driven decisions. Power BI offers a diverse range of visualizations, from basic charts to advanced custom visuals, each designed to cater to different analytical needs and preferences.

Visuals play a crucial role in summarizing with data. They enable users to present their findings in a clear and compelling manner, engaging the audience and facilitating better communication of insights. Whether you are presenting sales performance, customer demographics, financial metrics, or operational efficiency, the right visual can make a significant difference in how the information is perceived and understood.

One of the key strengths of Power BI is its flexibility in allowing users to choose from a wide variety of visuals, each with its own unique features and capabilities. From simple bar and column charts to intricate maps and custom visuals, Power BI provides the tools needed to create visually appealing and informative reports.

The primary function of visuals in Power BI is to convey data in a manner that is both intuitive and comprehensible. By leveraging different types of visualizations, users can highlight trends, uncover patterns, and make data-driven decisions with greater precision. The dynamic nature of Power BI visuals allows for interactive data exploration, enabling users to drill down into details, apply filters, and modify the scope of analysis in real-time. Visuals in Power BI are governed by data modeling principles that define the relationships and hierarchies within the dataset. This ensures that the visual representations are not only accurate but also contextually relevant.

Furthermore, Power BI's integration with other Microsoft products and services, such as Excel, Azure, and Power Automate, enhances the functionality of visuals, enabling users to leverage advanced analytics, automate workflows, and seamlessly integrate data from multiple sources. This integration ensures that users can create comprehensive and actionable reports that drive better business outcomes.

Visuals in Power BI are indispensable tools that facilitate the transformation of data into actionable insights. They enable users to communicate findings effectively, engage their audience, and make informed decisions. By harnessing the power of advanced data

modeling, interactive features, and custom visuals, users can create compelling and informative reports that drive better business outcomes.

Visuals in Power BI

Here are the visuals available in Power BI, each explained in detail:

- **Stacked Bar Chart:**

A stacked bar chart displays data in horizontal bars stacked on top of each other. It is useful for comparing the composition of multiple categories within a single bar. Each segment of the bar represents a different category, allowing users to see the distribution of values across categories. This visual is ideal for comparing the proportion of different categories within a total.

- **Stacked Column Chart:**

A stacked column chart is similar to a stacked bar chart but displays data in vertical columns. It is helpful for showing the composition of multiple categories over a period. Each segment of the column represents a different category, enabling users to see how values are distributed within each column. This visual is ideal for comparing the proportion of categories over time.

- **Clustered Bar Chart:**

A clustered bar chart displays data in horizontal bars grouped by category. Each group of bars represents a different category, making it easy to compare individual data points across categories. This visual is suitable for analyzing the performance of different categories side by side, allowing for a clear comparison of values.

- **Clustered Column Chart:**

A clustered column chart displays data in vertical columns grouped by category. Each group of columns represents a different category, allowing for easy comparison of individual data points across categories. This visual is perfect for analyzing the performance of different categories over time, providing a clear comparison of values.

- **100% Stacked Bar Chart:**

A 100% stacked bar chart displays the relative percentage of multiple data series in horizontal bars, with each bar summing to 100%. This visual is used to compare the proportion contribution of each category to the whole across different categories. It is particularly useful for showing the composition of categories and understanding their relative significance within the total.

- **100% Stacked Column Chart:**

A 100% stacked column chart is similar to a 100% stacked bar chart but displays data in vertical columns. Each column represents the relative percentage of multiple data series, summing to 100%. This visual is ideal for comparing the proportional contribution of each category to the whole across different time periods or groups. It helps users understand the composition and relative importance of categories within the total.

- **Line Chart:**

A line chart displays data points connected by a line, showing trends over time. It is useful for visualizing continuous data and identifying patterns or trends. Users can track changes in values over time and compare multiple data series. This visual is ideal for time series analysis and monitoring performance over a period.

- **Area Chart:**

An area chart is similar to a line chart but with the area below the line filled in. This visual emphasizes the magnitude of change over time, making it easier to see the overall trend and the volume of data. It is useful for showing cumulative data and highlighting the area covered by each data series.

- **Stacked Area Chart:**

A stacked area chart displays multiple data series with the area below each line filled in and stacked on top of each other. This visual is useful for showing the cumulative effect of different categories over time. It allows users to see the contribution of each category to the total and how they change over time.

- **100% Stacked Area Chart:**

A 100% stacked area chart displays multiple data series with the area below each line filled in and stacked on top of each other, summing to 100%. This visual is useful for showing the proportional contribution of each category to the total over time, highlighting how each category's share evolves.

- **Line and Stacked Column Chart:**

A line and stacked column chart combine a line chart and a stacked column chart in one visual. It allows users to compare trends over time (line) while also showing the composition of categories (stacked columns) for each period. This visual is ideal for understanding both overall trends and the breakdown of individual components.

- **Line and Clustered Column Chart:**

A line and clustered column chart integrate a line chart with a clustered column chart. This visual enable users to compare trends over time (line) while simultaneously comparing individual data points across categories (clustered columns). It is useful for analyzing overall trends and detailed category comparisons in a single view.

- **Ribbon Chart:**

A ribbon chart shows data changes over time with ribbons connecting data points. It highlights the ranking of different categories, making it easy to see how categories move up or down over time. This visual is useful for comparing the relative performance of categories and identifying trends.

- **Waterfall Chart:**

A waterfall chart displays the cumulative effect of sequential positive and negative values. It is useful for understanding the impact of individual components on the total. Each bar represents a value that contributes to the overall total, showing how each component affects the final result.

- **Funnel Chart:**

A funnel chart shows a process with different stages, where each stage narrows down the data, like a funnel. Each section of the funnel represents a stage, and the width shows how much data is at that stage. A funnel chart can be used to compare different scenarios. Each segment of the funnel would show the proportion of leads converting at each stage.

- **Scatter Chart:**

A scatter chart plots data points on a Cartesian plane, showing the relationship between variables. It is useful for identifying trends, patterns, and outliers. Users can analyze the correlation between variables and see how data points are distributed. This visual is ideal for exploratory data analysis and regression analysis.

- **Pie Chart:**

A pie chart displays data as slices of a circle, showing proportions as a whole. Each slice represents a category, and the size of the slice corresponds to the value of the category. This visual is useful for comparing parts of a whole and visualizing the distribution of values.

- **Donut Chart:**

A donut chart is similar to a pie chart but with a hole in the center. This visual provides a different aesthetic for visualizing proportions and can include an additional layer of information in the center. It is useful for showing the composition of categories and comparing their relative sizes.

- **Tree Map:**

A tree map displays hierarchical data as nested rectangles, with each rectangle representing a category. The size of the rectangle corresponds to the value of the category. This visual is useful for showing the relative size of categories within a hierarchy and identifying patterns within the data.

- **Map:**

A map displays data points on a geographical map, allowing for spatial analysis. Users can visualize the distribution of data across different regions and analyze geographical patterns. This visual is useful for understanding the geographical context of data and identifying regional trends.

- **Filled Map:**

A filled map is similar to a map but with regions filled in based on data values. This visual emphasizes areas with higher or lower values, making it easy to see the distribution of data across regions. It is useful for comparing the performance of different regions and identifying areas of interest.

- **Azure Maps:**

Azure Maps in Power BI provides advanced spatial data visualizations, allowing users to analyze geographical data with rich map visuals. This integration enables users to visualize data points on a map, perform spatial analysis, and gain insights into location-based data. Azure Maps supports various map types, including filled maps and custom shapes, enhancing the ability to understand and interpret geographical patterns and relationships.

- **Gauge:**

A gauge displays a single value within a range, similar to a speedometer. It is useful for showing progress towards a target and comparing the current value to predefined thresholds. This visual is ideal for tracking key performance indicators (KPIs) and monitoring progress.

- **Card:**

A card displays a single data point, highlighting key metrics or KPIs. It is useful for presenting important numbers and providing a quick summary of performance. This visual is ideal for displaying key figures in a concise and visually appealing manner.

- **Multi-row Card:**

A multi-row card displays multiple data points in a card format. It is useful for showing multiple key metrics or KPIs at once, providing a quick overview of performance. This visual is ideal for summarizing key figures in a compact and organized layout.

- **KPI:**

A KPI visual displays key performance indicators, tracking progress towards specific goals or targets. It is useful for monitoring performance and identifying areas that need improvement. This visual is ideal for displaying key metrics in a visually engaging manner.

- **Slicer:**

A slicer in Power BI is an interactive filtering visual that allows users to dynamically segment and filter data within reports and dashboards. By adding slicers, users can easily refine their data views based on specific criteria, such as date ranges, categories, or measures. This enhances the interactivity and usability of reports, enabling users to explore data from different perspectives and make more informed decisions.

- **Table:**

A table displays data in a tabular format, useful for detailed data analysis and comparison. Users can sort and filter the data, making it easy to explore and analyze. This visual is ideal for presenting detailed information and allowing for in-depth analysis.

- **Matrix:**

A matrix is similar to a table but with additional features for displaying data hierarchically. It is useful for exploring data at different levels of detail and showing the relationships between categories. This visual is ideal for summarizing data and providing a clear overview of complex information.

- **R Script Visual:**

An R Script Visual allows for custom visuals created using R scripts. It enables advanced statistical analysis and visualization, leveraging the power of R's extensive library of packages. This visual is ideal for users who need advanced analytical capabilities and custom visualizations.

- **Python Visual:**

A Python Visual allows for custom visuals created using Python scripts. It provides flexibility for advanced data manipulation and visualization, utilizing Python's powerful libraries like pandas, matplotlib, and seaborn. This visual is ideal for users who need custom visualizations and advanced data analysis.

- **Key Influencers:**

Key influencers visual identifies factors that influence a particular metric, helping users understand what drives key outcomes. It uses machine learning algorithms to analyze the data and highlight the most significant influencers. This visual is ideal for analyzing customer behavior, sales performance, and other scenarios where understanding key drivers is crucial for making informed decisions.

- **Decomposition Tree:**

A decomposition tree breaks down a measure into its contributing factors, providing a visual representation of how different variables affect the measure. It is useful for root cause analysis and understanding the influence of different factors on performance. This visual is ideal for users who need to analyze the contributing factors to a specific outcome.

- **Q&A Visual:** A Q&A visual allows users to ask questions in natural language and get answers in the form of visuals. It makes data exploration more intuitive and accessible, enabling users to interact with their data using simple language. This visual is ideal for users who want to quickly find answers and explore their data without needing advanced technical skills.

- **Narrative:**

Automatically generates text summaries of data insights, providing contextual explanations alongside visualizations. This visual helps users quickly understand key takeaways and adds descriptive context to reports, enhancing data storytelling and comprehension.

- **Goals (Preview):**

The Goals (Preview) visual in Power BI helps users set, track, and manage key objectives directly within their reports. This visual provides an interactive way to monitor progress towards specific targets and Aligns team efforts by visualizing goal achievements.

- **Paginated Report:**

Paginated reports are highly formatted, pixel-perfect reports designed for printing or PDF generation. They display data in a structured and detailed layout, ideal for scenarios where exact placement and layout of elements are crucial, such as invoices or financial statements.

- **Card (New):**

The card visual displays a single data point, highlighting key metrics or KPIs. It is ideal for quickly conveying important information in a concise and visually appealing manner, making it easy to track essential performance indicators at a glance.

- **Tile Slicer:**

A tile slicer is an interactive filtering visual that uses a tiled interface. Each tile represents a unique value or category, allowing users to filter data dynamically. It provides an intuitive way to segment data and explore different subsets of information within reports.

- **ArcGIS for Power BI:**

ArcGIS for Power BI integrates advanced mapping and spatial analytics from Esri's ArcGIS platform into Power BI. This visual allows users to create detailed maps, perform spatial analysis, and visualize geographical data, enhancing location-based insights and decision-making.

- **Power Apps for Power BI:**

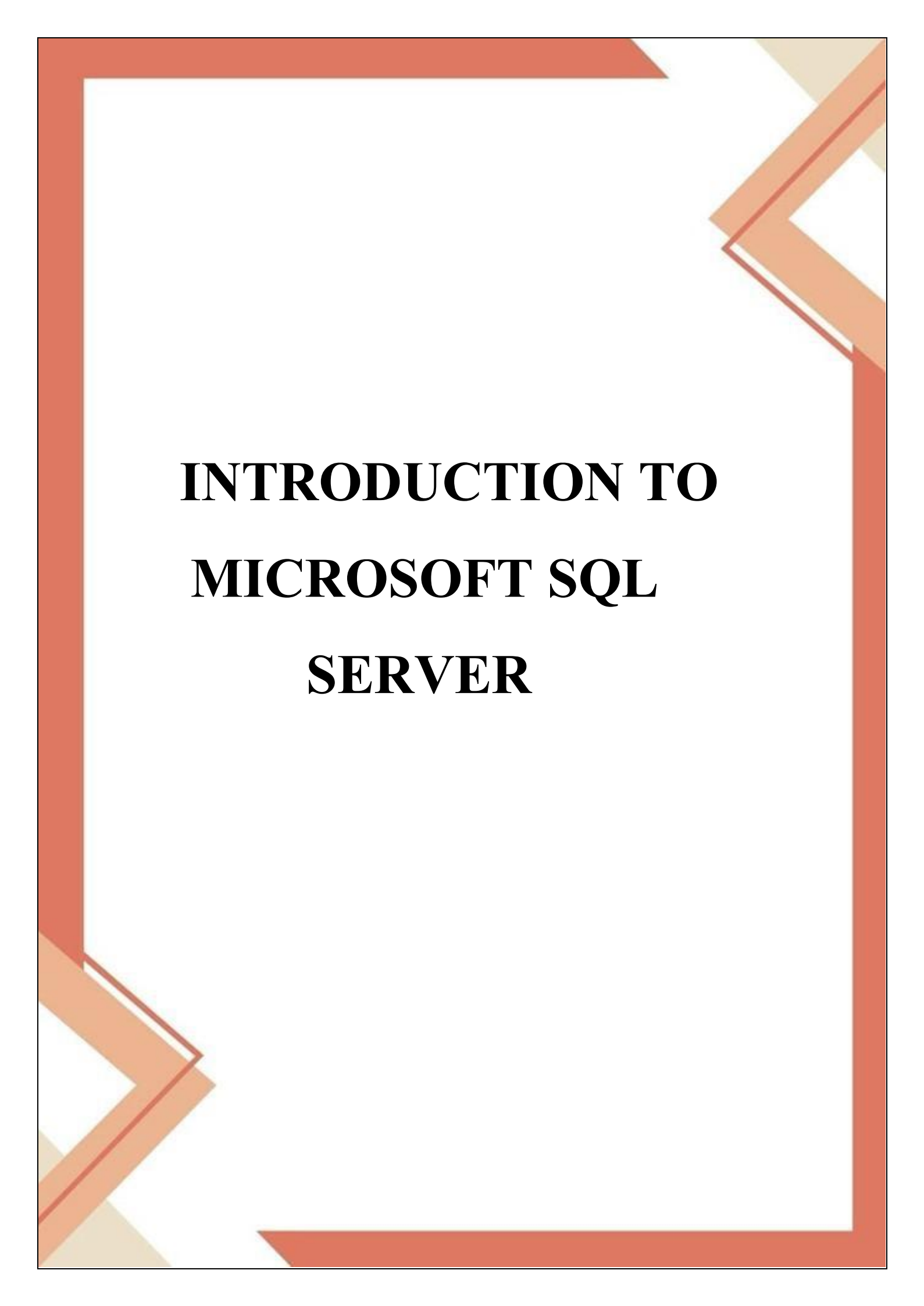
The Power Apps visual allows users to embed Power Apps directly into Power BI reports. This integration enables users to interact with their data, perform actions, and update records in real-time within the report, extending the functionality of Power BI with custom business applications.

- **Power Automate for Power BI:**

The Power Automate visual integrates Power Automate flows with Power BI, enabling automation of tasks and workflows directly from the report. Users can trigger actions based on data insights, streamlining processes and improving productivity by automating repetitive tasks.

- **Custom Visuals:**

Allows for the use of custom visuals created by the community or third parties, providing additional flexibility and customization options for data visualization.



INTRODUCTION TO MICROSOFT SQL SERVER

INTRODUCTION TO MICROSOFT SQL SERVER

Microsoft SQL Server (MS SQL Server) is a comprehensive relational database management system (RDBMS) developed by Microsoft. It is designed to store, retrieve, and manage data efficiently, providing a robust platform for a wide range of data-driven applications. MS SQL Server supports the Structured Query Language (SQL) for querying and managing databases, making it a powerful tool for data professionals and developers. One of the standout features of MS SQL Server is its ability to handle large volumes of data while maintaining high performance. It offers advanced data storage and retrieval capabilities, ensuring that queries are executed quickly and efficiently. With built-in support for indexing, query optimization, and in-memory processing, MS SQL Server delivers exceptional performance even under heavy workloads. This makes it an ideal choice for mission-critical applications that require fast and reliable data access.

Security is a top priority for MS SQL Server, and it includes a range of features to protect sensitive data. These features include encryption, authentication, and auditing, which help ensure that data is secure and that access is controlled. MS SQL Server also supports role-based security, allowing administrators to define and enforce access policies based on user roles and responsibilities. This helps organizations comply with industry standards and regulations, safeguarding their data from unauthorized access and breaches.

MS SQL Server is known for its high availability and disaster recovery capabilities. The features provide redundancy and fault tolerance, minimizing downtime and ensuring that data is always available when needed. This reliability is crucial for businesses that depend on continuous access to their data for daily operations and decision-making.

MS SQL Server can scale to accommodate growing data volumes and increasing user demands, making it suitable for organizations of all sizes. Whether deployed on-premises or in the cloud, MS SQL Server can handle large-scale data environments, ensuring that performance remains consistent as data and user numbers grow.

Integration with other Microsoft products and services is seamless with MS SQL Server. It works well with applications like Power BI, Azure, and Excel, enabling users to create comprehensive data solutions. This integration allows users to leverage their existing Microsoft ecosystem, enhancing productivity and providing a unified environment for data management, analysis, and reporting. For example, data can be imported from SQL Server

into Power BI for advanced analytics and visualization, or it can be integrated with Azure services for cloud-based data storage and processing.

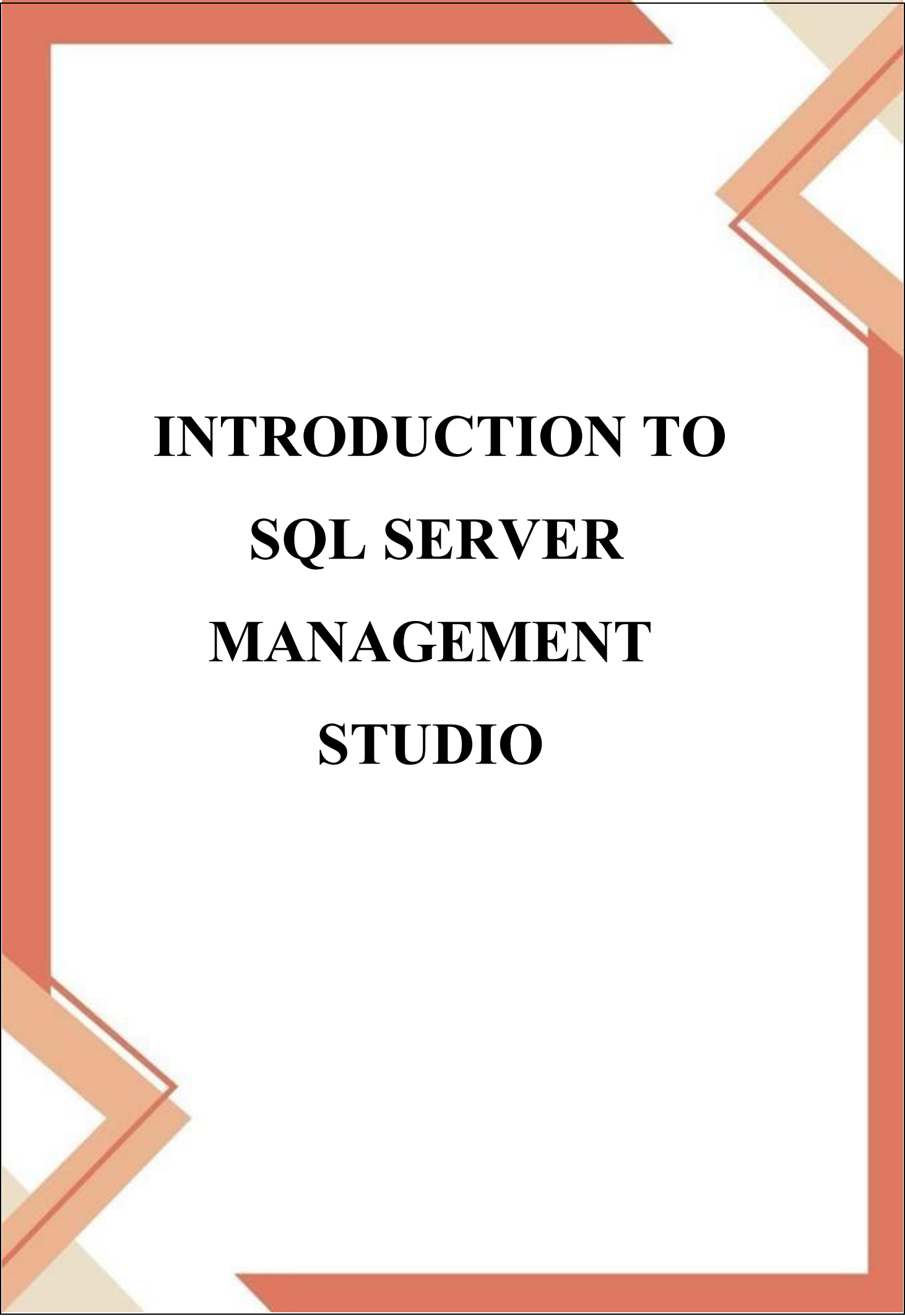
MS SQL Server also supports a variety of data types and complex queries, making it versatile for different use cases. It can handle transactional processing, business intelligence, and analytics, providing a single platform for diverse data needs. Advanced analytics and machine learning capabilities are built into the platform, enabling users to gain deeper insights from their data and drive innovation. These features allow organizations to perform predictive analytics, uncover hidden patterns, and make data-driven decisions.

Microsoft SQL Server is a powerful and reliable RDBMS that offers high performance, advanced security, high availability, scalability, and seamless integration with other Microsoft products. Its versatility and comprehensive feature set make it an ideal choice for organizations looking to manage and analyze their data effectively, ensuring both operational efficiency and strategic insights.

Benefits of MS SQL Server:

- **High Availability and Disaster Recovery:** MS SQL Server offers features like 'AlwaysOn' Availability Groups, ensuring databases remain accessible even during hardware failures or disruptions. This reliability is crucial for maintaining continuous business operations and safeguarding critical data.
- **Advanced Security:** It provides robust security measures, including encryption, authentication, and auditing, protecting sensitive information and ensuring compliance with industry standards and regulations. This helps organizations maintain data integrity and security.
- **Performance Optimization:** With in-memory processing, query optimization, and indexing, MS SQL Server delivers fast and efficient data retrieval. This allows for real-time analytics and decision-making, improving overall system performance.
- **Scalability:** MS SQL Server can handle growing volumes of data and increasing user demands, making it suitable for organizations of all sizes. Its scalability ensures that the system can grow with the business without compromising performance.

- **Seamless Integration:** It integrates smoothly with other Microsoft tools and services like Power BI, Azure, and Excel, enhancing productivity and providing a unified environment for data management, analysis, and reporting.
- **Versatility in Data Types and Queries:** MS SQL Server supports a variety of data types and complex queries, making it versatile for different use cases, from transactional processing to business intelligence and analytics.
- **Advanced Analytics and AI Integration:** The support for advanced analytics, machine learning, and artificial intelligence enables users to gain deeper insights from their data, driving innovation and informed decision-making.



INTRODUCTION TO SQL SERVER MANAGEMENT STUDIO

INTRODUCTION TO SQL SERVER MANAGEMENT STUDIO

SQL Server Management Studio (SSMS) is a powerful integrated environment designed for managing Microsoft SQL Server instances. SSMS provides tools for configuring, monitoring, and administering SQL Server databases, making it an essential application for database administrators and developers. With SSMS, users can execute queries, manage database objects, configure security settings, and perform various administrative tasks with ease.

One of the key features of SSMS is its intuitive user interface, which simplifies the management of SQL Server instances. The Object Explorer provides a hierarchical view of database objects, making it easy to navigate and manage databases, tables, views, stored procedures, and other elements. SSMS also includes Query Editor, a versatile tool for writing, debugging, and executing SQL queries. The editor supports syntax highlighting, IntelliSense, and query formatting, enhancing productivity and reducing errors.

Security is a critical aspect of database management, and SSMS offers robust security features to ensure data protection. Users can configure authentication modes, manage user roles and permissions, and set up encryption for sensitive data. SSMS also provides tools for auditing and monitoring database activities, helping administrators track changes and ensure compliance with security policies.

Performance tuning and optimization are essential for maintaining the efficiency of SQL Server instances, and SSMS includes a range of tools to support these activities. The Query Performance Insight feature helps identify slow-running queries and provides recommendations for improving performance. Users can also utilize the Database Tuning Advisor to analyze workloads and optimize indexing and query strategies.

SSMS integrates seamlessly with other Microsoft products and services, enhancing its functionality and providing a unified environment for database management. For example, users can connect SSMS to Azure SQL Database and manage cloud-based databases using the same tools and interface. Integration with Visual Studio allows developers to work on database projects and collaborate with application development teams.

Backup and recovery are crucial for data protection, and SSMS offers comprehensive tools for managing backup and restore operations. Users can configure automated backups, perform point-in-time restores, and manage disaster recovery plans. SSMS also supports

high availability features like 'AlwaysOn' Availability Groups, ensuring that databases remain accessible even in the event of hardware failures or other disruptions.

In addition to its administrative capabilities, SSMS supports advanced analytics and data visualization. Users can integrate SSMS with Power BI to create interactive reports and dashboards, leveraging the power of data visualization to gain insights and drive business decisions. SSMS also supports the execution of R and Python scripts, enabling advanced data analysis and machine learning within the SQL Server environment.

In summary, SQL Server Management Studio (SSMS) is a comprehensive and versatile tool for managing

SQL Server instances. Its intuitive interface, robust security features, performance optimization tools, seamless integration, and support for advanced analytics make it an indispensable application for database administrators and developers. SSMS empowers users to efficiently manage, secure, and optimize their SQL Server databases, ensuring both operational efficiency and strategic insights.

Benefits of SQL Server Management Studio (SSMS)

- **Intuitive User Interface:**

SSMS provides an easy-to-use interface with features like Object Explorer and Query Editor, simplifying navigation and query execution.

- **Robust Security:**

SSMS offers robust security features, including authentication modes, user roles, permissions, and encryption, ensuring data protection and compliance.

- **Performance Optimization:**

SSMS includes tools like Query Performance Insight and Database Tuning Advisor to help identify slow queries and optimize database performance.

- **Seamless Integration:**

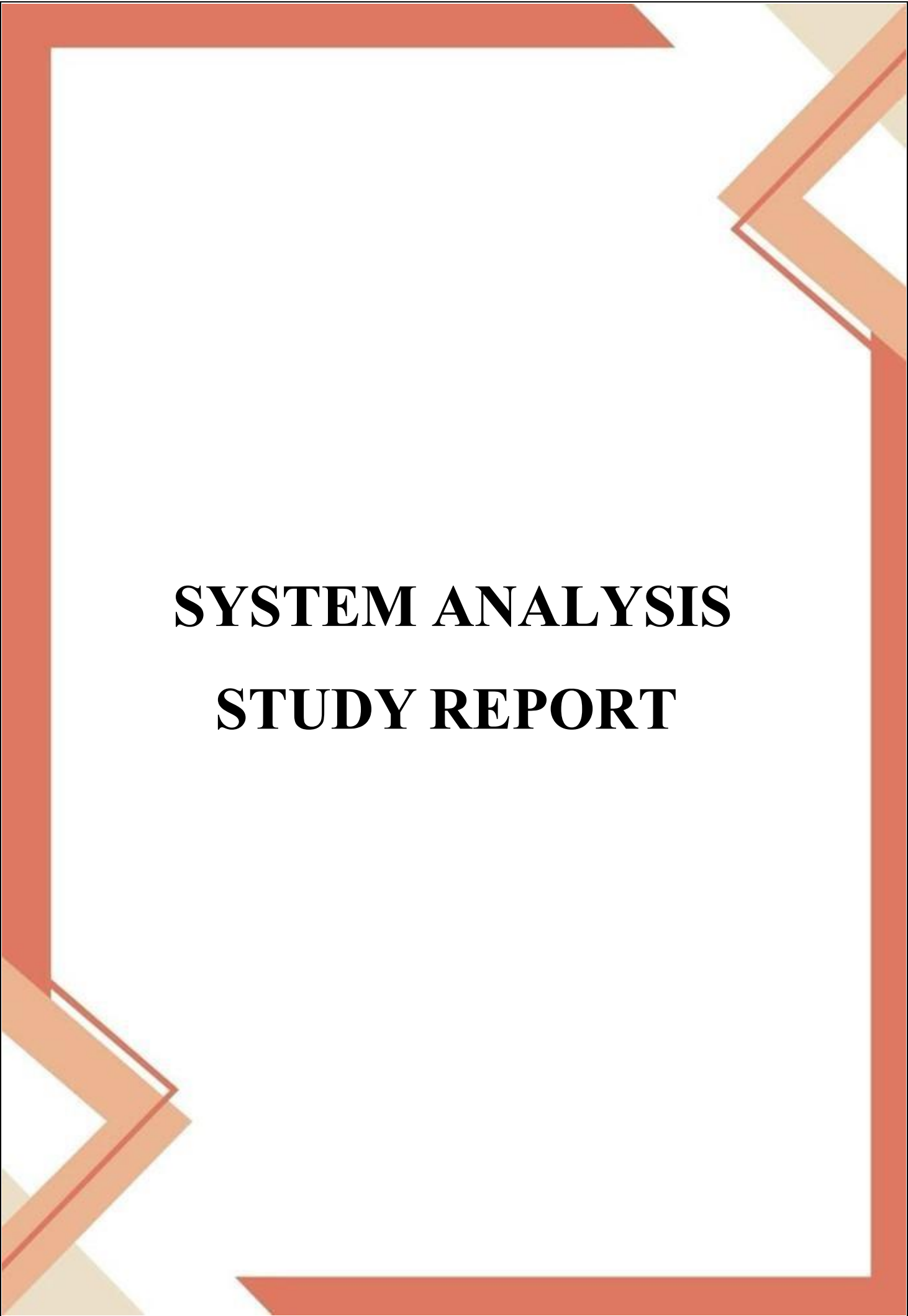
SSMS integrates with other Microsoft products like Azure SQL Database and Visual Studio, providing a unified environment for database management and development.

- **Comprehensive Backup and Recovery:**

SSMS offers tools for managing backup and restore operations, ensuring data protection and high availability.

- **Advanced Analytics and Visualization:**

SSMS supports integration with Power BI and execution of R and Python scripts, enabling advanced data analysis and visualization. These benefits make SQL Server Management Studio (SSMS) a powerful and indispensable tool for managing SQL Server instances effectively.



SYSTEM ANALYSIS STUDY REPORT

SYSTEM ANALYSIS STUDY REPORT

1. What is the objective of the project?

Ans: The objective of this project is to analyze and compare the performance of multiple social media platforms — Facebook, Instagram, YouTube, LinkedIn, Twitter, and Snapchat — by tracking key metrics such as Total Posts, Total Reach, Total Engagement, Total Views, and Engagement Rate %. The goal is to provide actionable insights that help optimize content strategy and improve digital marketing performance.

2. What data source is used for this analysis?

Ans: SQL Server is used as the primary data source. Social media data containing post-level information such as Platform, Category, Post Type, Reach, Engagement, Views, and Post Date is stored and managed in SQL Server.

3. What interactive features are included in the dashboard?

Ans: The dashboard includes interactive slicers such as Platform, Category, Post Type, and Date (Month-Year). Users can dynamically filter data to compare specific platforms, analyze particular content categories, and view performance trends over time.

4. How is customer behaviour analysed?

Ans: Audience behavior is analyzed using engagement metrics such as likes, comments, and shares (combined as Total Engagement). Performance is evaluated by platform, content category, post type, and monthly trends to understand which type of content attracts higher user interaction.

5. What role does DAX play in this project?

Ans: DAX (Data Analysis Expressions) is used to create calculated measures such as Total Posts (DISTINCTCOUNT), Total Reach (SUM), Total Engagement (SUM), Total Views (SUM), and Engagement Rate % (Engagement ÷ Reach using DIVIDE function). These measures allow dynamic and filter-responsive KPI calculations.

6. How is the Power BI dashboard published and shared?

Ans: The report is published to Power BI Service, where authorized users can access it through a web browser. It can also be shared with team members for collaborative analysis.

7. How does gender-based analysis help?

Ans: Platform-based analysis helps identify which social media platform generates the highest engagement, reach, and content effectiveness. This supports better allocation of marketing efforts and content planning strategies.

8. Can reports be accessed on mobile devices?

Ans: Yes, the dashboard can be accessed through the Power BI Mobile App, enabling users to monitor social media performance anytime and anywhere.

9. Why is Power BI used for this project?

Ans: Power BI is used because it provides interactive data visualization, real-time filtering, advanced DAX calculations, and effective dashboard storytelling. It helps convert raw social media data into meaningful insights.

10. What demographic insights are included?

Ans: The project includes analysis by Platform, Category, Post Type, and Monthly Trends. These dimensions allow multi-level performance comparison and deeper content analysis.

11. How is monthly performance analysis used?

Ans: Monthly analysis helps track engagement and reach trends over time, identify growth or decline patterns, and detect seasonal performance changes. This supports strategic content scheduling.

12. How does the project handle data updates?

Ans: The SQL Server connection allows scheduled or manual data refresh in Power BI Service. This ensures that the dashboard always reflects the latest performance data.

13. How does this project help in decision-making?

Ans: The dashboard provides data-driven insights that help marketing teams decide which platform to prioritize, which content category performs best, and which post type generates higher engagement rates. This improves strategic planning and ROI.

14. How does Power BI Service help in report updates?

Ans: Power BI Service allows scheduled refresh and manual refresh options, ensuring that updated SQL data is automatically reflected in the dashboard without manual intervention.

15. How is security handled in Power BI Service?

Ans: Row-Level Security (RLS) can be implemented to restrict data access based on user roles. This ensures that users only see relevant platform or performance data as per their authorization level.

16. What are the advantages of using Power BI Service for this project?

Ans: Advantages include cloud accessibility, automatic refresh, secure data sharing, real-time collaboration, and mobile compatibility.

17. How does Power BI Service support collaboration?

Ans: It allows multiple stakeholders to access, comment, and analyze the dashboard simultaneously, improving communication and decision-making efficiency.

18. What type of user roles can be assigned in Power BI Service?

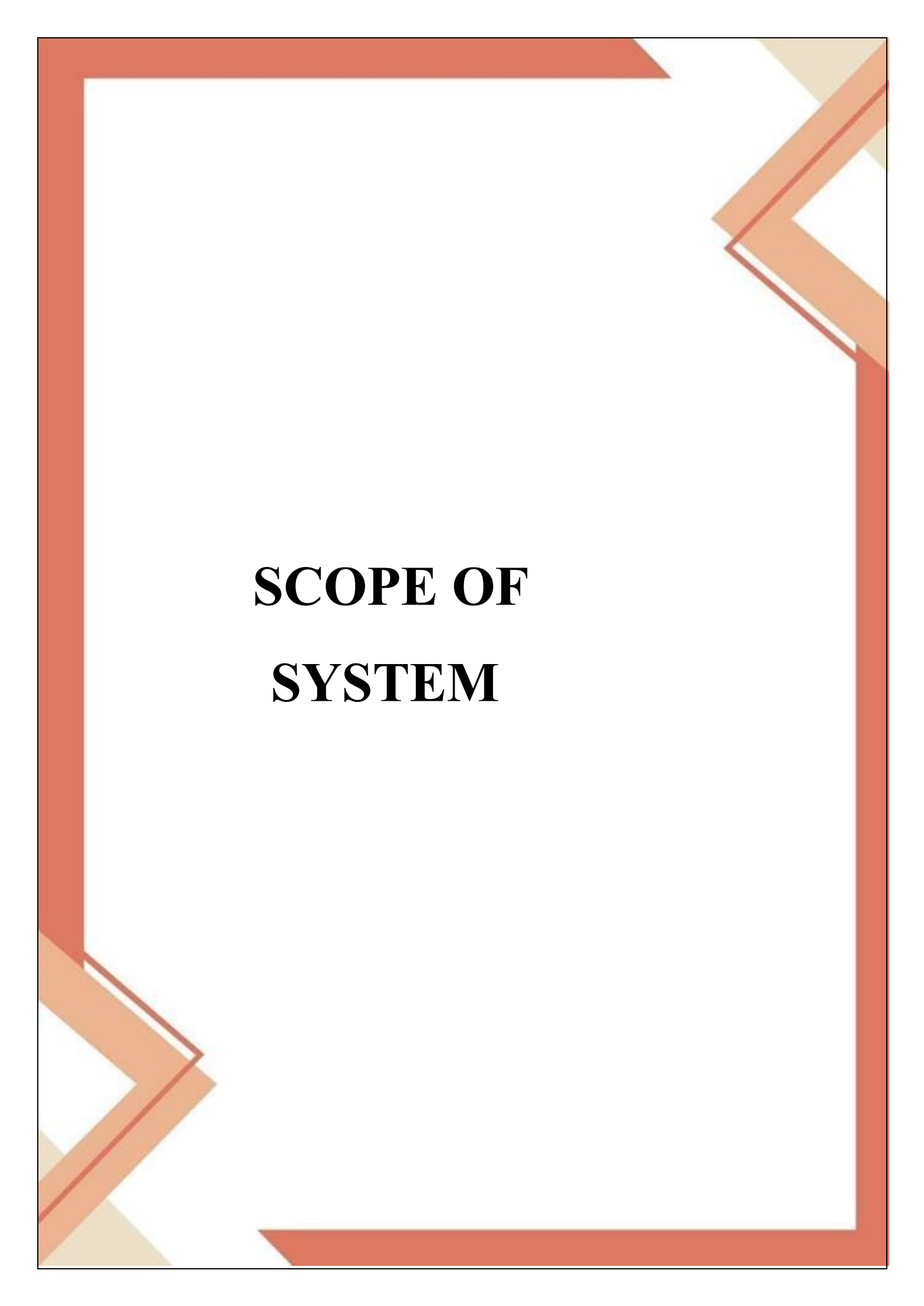
Ans: Roles such as Admin, Member, Contributor, and Viewer can be assigned to control access permissions and ensure secure data usage.

19. How does Power BI Service help in performance monitoring?

Ans: It enables real-time monitoring of key metrics such as engagement rate, reach trends, and platform-wise performance, allowing continuous tracking of social media effectiveness.

20. How can businesses benefit from this project?

Ans: Businesses can optimize content strategy, improve audience engagement, allocate marketing budgets effectively, and enhance overall digital presence using data-driven insights provided by the dashboard.



SCOPE OF SYSTEM

SCOPE OF SYSTEM

The Social Media ContentXplorer system provides a comprehensive, data-driven approach to understanding content trends. The scope of this system includes:

1. Multi-Platform Performance Analysis

The scope of this project includes analyzing and comparing performance across six major social media platforms — Facebook, Instagram, YouTube, LinkedIn, Twitter, and Snapchat. The dashboard integrates all platform data into a unified structure, enabling side-by-side comparison of reach, engagement, views, and posting activity.

2. KPI-Based Performance Measurement

The project focuses on calculating and analyzing key performance indicators (KPIs) such as:

- Total Posts
- Total Reach
- Total Engagement
- Total Views
- Engagement Rate %

These KPIs provide a measurable framework to evaluate digital marketing effectiveness and audience interaction levels.

3. Category-Wise Content Performance Analysis

The scope includes analysing how different content categories (e.g., Educational, Promotional, Informative, Entertainment) perform across platforms. This helps identify which type of content drives higher engagement and which platform is most suitable for specific categories.

4. Post Type Effectiveness Evaluation

The project analyses performance based on different post formats such as Live, Short, Image, and Video. It compares engagement, reach, and engagement rate by post type to determine the most effective content format for each platform.

5. Time-Based Trend Analysis

The scope includes monthly and date-wise performance tracking. This enables identification of growth patterns, seasonal trends, and fluctuations in engagement and reach over time.

6. Interactive Dashboard Development

The project includes the development of an interactive Power BI dashboard with slicers for:

- Platform
- Category
- Post Type
- Date (Month-Year)

These features allow dynamic filtering and multi-dimensional analysis.

7. SQL-Based Data Validation and Aggregation

The scope covers SQL query development for aggregating engagement, reach, and views data. SQL is used to validate KPI calculations and ensure data accuracy before visualization in Power BI.

8. Data Modeling and DAX Measure Creation

The project includes creating a structured data model in Power BI and developing DAX measures to calculate dynamic metrics that respond to filters and slicers.

9. Business Insight Generation

The dashboard is designed to generate actionable insights that support:

- Platform prioritization
- Content strategy improvement
- Engagement optimization
- Marketing performance evaluation



HARDWARE & SOFTWARE REQUIREMENT

HARDWARE & SOFTWARE REQUIREMENT

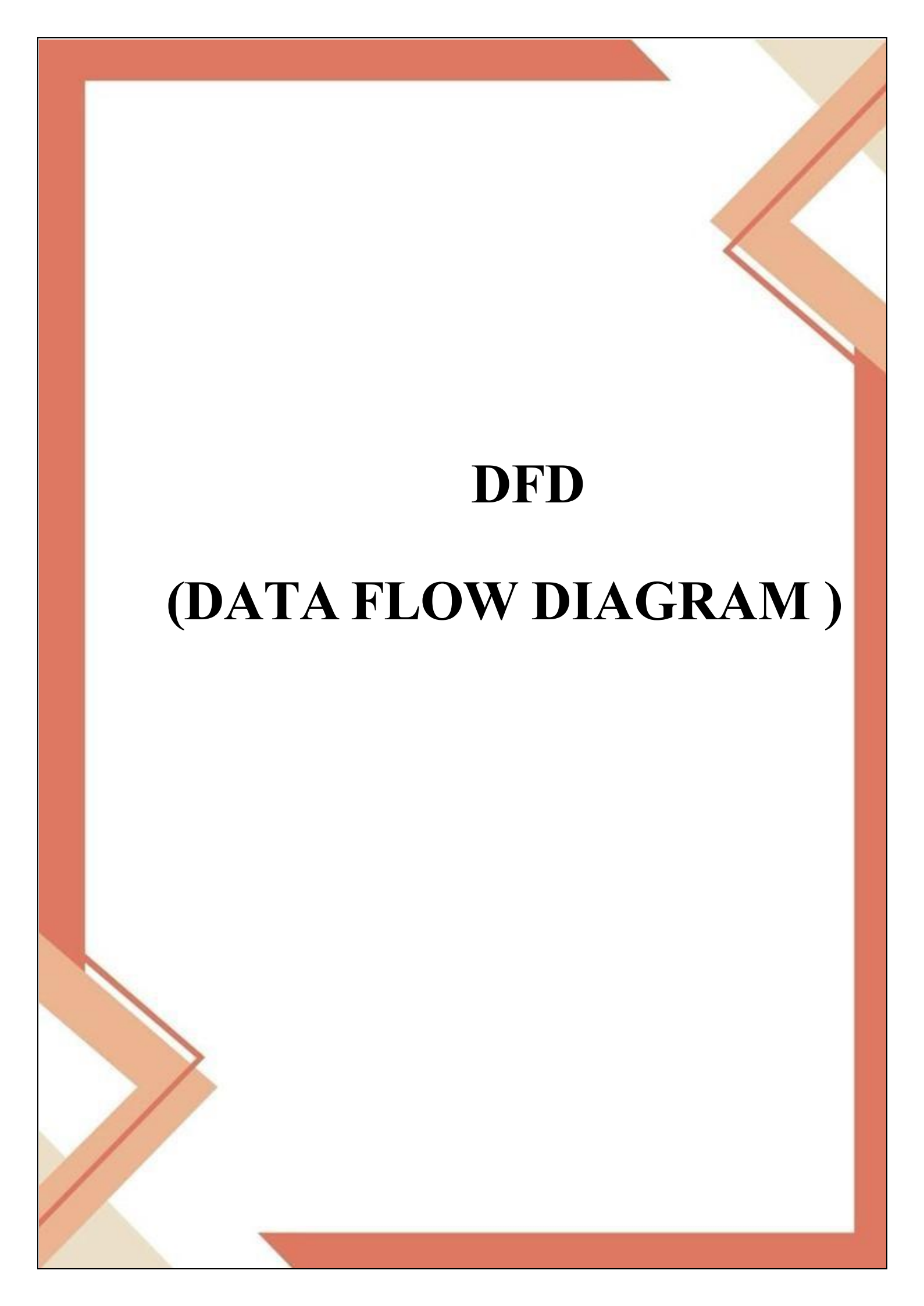
The Hardware and Software Requirements Specifications is produced at the culmination of the analysis task. The function and performance allocated to software as a part of system engineering are refined by establishment a complete information description, a detailed functional and behavioural description, an indication of performance requirements and design constraints, appropriate validation criteria and other data pertinent to requirements.

Hardware requirements:

Name of component	Specification
Processor	Intel/ Ryzen 3 Dual Core
RAM	128 MB
Hard Disk	20 GB
Monitor	15'Color Monitor
Keyboard	122 keys

Software requirements:

Name of the component	Specification
Language	DAX Queries, SQL Queries
Database	SQL Server
Browser	Chrome
Applications	Power BI Desktop, Power BI Service, MS Excel, MS SQL Server, MS SQL Server Management Studio
Operating system	Windows 10 and above



DFD

(DATA FLOW DIAGRAM)

DFD (DATA FLOW DIAGRAM)

It is easy to understand the flow of data through system with the right data flow diagram software. A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles, and arrows, plus short text labels, to show data inputs, outputs, storage point and the routes between each destination. Data flowchart can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyses an existing system or model a new one.

Symbols and Notations Used in DFDs

Data Flow Diagrams (DFDs) are used to visually represent the flow of data within a system, depicting how data moves from inputs to outputs through various processes. To create a clear and effective DFD, specific symbols and notations are used. Here are the key symbols and their meanings:

- **Process:**

Symbol: Rounded rectangle or circle.

Notation: Represents a process or function that transforms input data into output data. Processes are labeled with verbs to indicate the actions performed.

- **Data Flow:**

Symbol: Arrow.

Notation: Indicates the flow of data between processes, data stores, and external entities. Arrows are labeled with the name of the data being transferred.

- **Data Store:**

Symbol: Open-ended rectangle or two parallel lines.

Notation: Represents a place where data is stored within the system, such as a database or a file. Data stores are labeled with a noun indicating the type of data stored.

- **External Entity:**

Symbol: Rectangle.

Notation: Represents external sources or destinations of data, such as users, other systems, or organizations. External entities are labeled with a noun indicating the entity's name.

- **Data Store Line:**

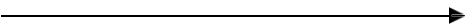
Symbol: Arrow pointing towards or away from an open-ended rectangle.

Notation: Indicates the direction of data flow between processes and data stores.

- **External Entity Connection:**

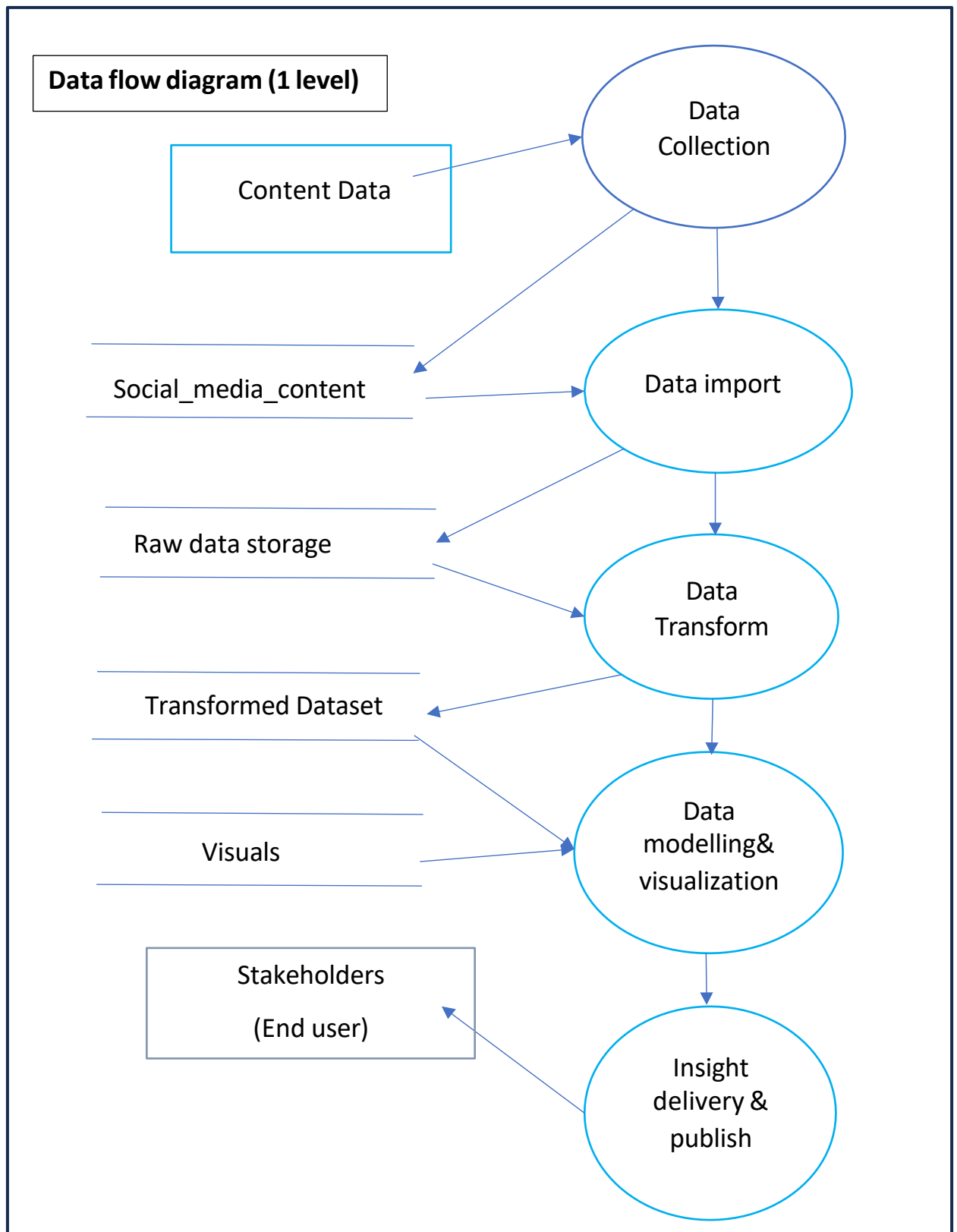
Symbol: Arrow connecting a rectangle to a process.

Notation: Shows the interaction between external entities and system processes.

Notation	Symbol
External Entity	
Process	
Data Flow	
Data Store	

DATA FLOW DIAGRAM (0 level)







ENTITY REALATIONSHIP (ER) DIAGRAM

ENTITY REALATIONSHIP (ER) DIAGRAM

The ER or (Entity Relational Model) is a high-level conceptual data model diagram. Entity-Relation model is based on the notion of real-world entities and the relationship between them. ER modelling helps you to analysis data requirements systematically to produce a well- designed database. So, it is considered a best practice to complete ER modelling before implementing your database.

Components of the ER Diagram

This model is based on three basic concepts:

- Entities
- Attributes
- Relationships

Entities:

A real-world thing either living or non-living that is easily recognizable and no recognizable. It is anything in the enterprise that is to be represented in our database. It may be a physical thing or simply a fact about the enterprise or an event that happens in the real world.

Attributes:

An attribute describes the property of an entity. An attribute is represented as Oval in an ER diagram.

There are four types of attributes:

- Key attribute.
- Composite attribute.
- Multivalued attribute.
- Derived attribute.

Relationships:

Defines the numerical attributes of the relationship between two entities or entity sets.

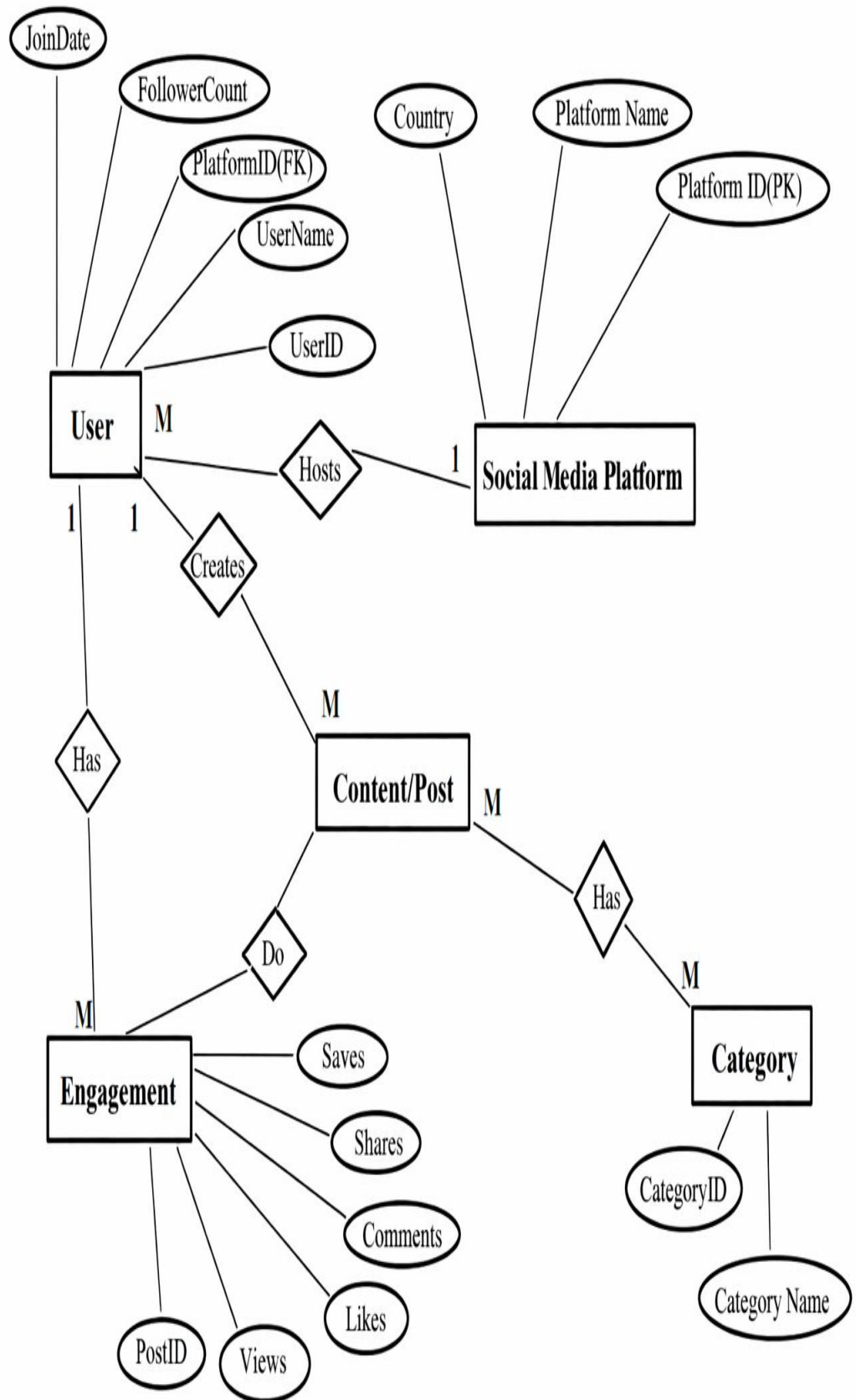
Different types of cardinal relationships are:

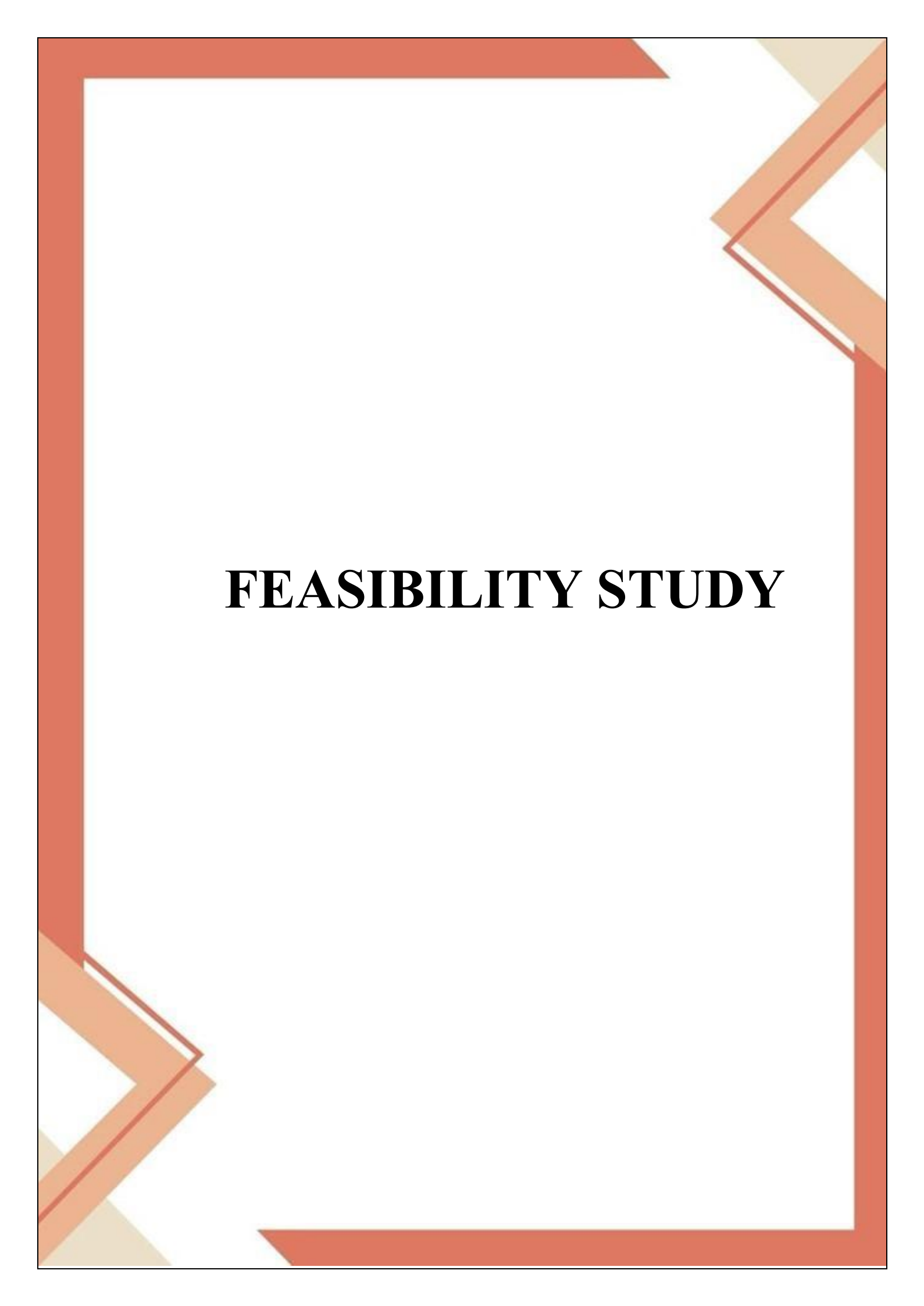
- One-to-One Relationships
- One-to-Many Relationships
- Many to One Relationships
- Many-to-Many Relationships

ER- Diagram Notations

ER- Diagram is a visual representation of data that describe how data is related to each other.

- **Rectangles:** This symbol represent entity types
- **Ellipses:** Symbol represent attributes
- **Diamonds:** This symbol represents relationship type
- **Lines:** It links attributes to entity types and entity types with other relationship types.
- **Primary key:** attributes are underlined
- **Double Ellipses:** Represent Multi-valued attribute
- **Dashed Ellipses:** Derived Attributes
- **Double Rectangles:** Weak Entity Sets
- **Double Lines:** Total participation of an entity in a relationship set





FEASIBILITY STUDY

FEASIBILITY STUDY

The Amazon Online Shopping Sales Analysis project in Power BI is evaluated based on environmental, technical, and economic feasibility to ensure its effectiveness and sustainability.

They are as follows:

➤ **Economic Feasibility:**

The project is economically feasible as it offers cost-effective solutions for data analytics and reporting. By using Power BI's cloud-based approach, businesses can significantly reduce expenses on IT infrastructure, hardware maintenance, and manual reporting efforts. Power BI offers affordable licensing options (Pro and Premium), making it accessible for businesses of all sizes. Additionally, the project leads to better business decision-making by analyzing sales performance, customer trends, and marketing effectiveness, which helps optimize pricing strategies, improve inventory management, and increase overall revenue. Automating reports also reduces manual labor costs and minimizes errors, further improving operational efficiency.

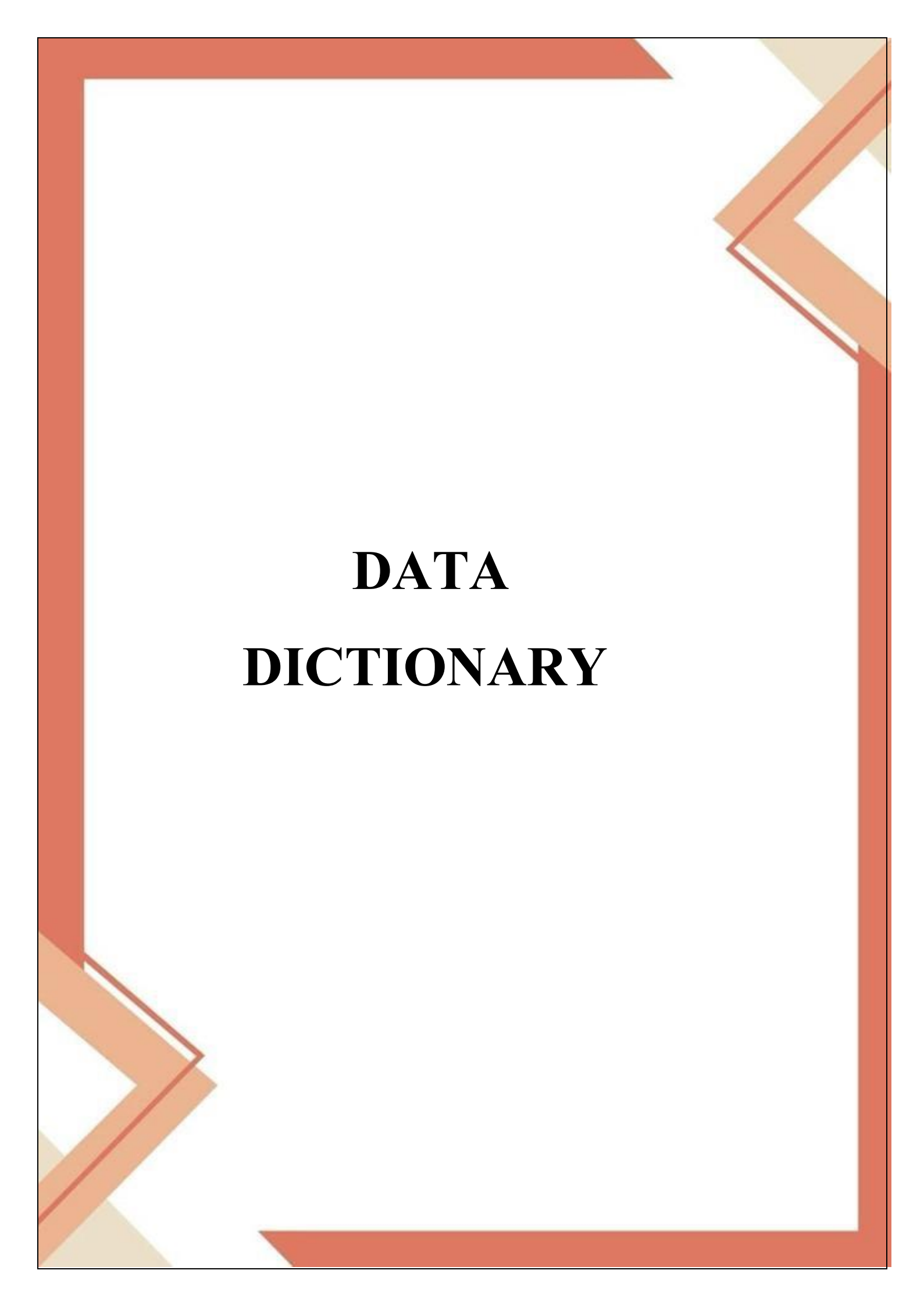
➤ **Environmental Feasibility:**

The project is environmentally sustainable as it utilizes cloud-based technology, reducing the need for physical servers and on-premise infrastructure, which lowers energy consumption. By implementing Power BI Service, businesses can access reports digitally, eliminating the need for paper-based reporting and contributing to a paperless, eco-friendly approach. Additionally, Power BI operates on Microsoft Azure, which follows energy-efficient practices and supports scalable computing resources, ensuring minimal environmental impact by only using resources when required. This approach reduces carbon footprints and supports sustainable business operations.

➤ **Technical Feasibility:**

The project is technically viable as Power BI seamlessly integrates with SQL Server, allowing real-time data extraction, transformation, and visualization. Power Query Editor enables data cleaning and transformation, while DAX (Data Analysis Expressions) supports complex calculations for in-depth analysis. Reports can be accessed on desktop,

web, and mobile devices, ensuring flexibility and remote access. Power BI Service offers cloud-based data storage, automated refresh, and sharing capabilities, making collaboration easy. Security is ensured with Row-Level Security (RLS), encryption, and compliance with data protection standards. The system is scalable, reliable, and well-suited for analysing large datasets.



DATA DICTIONARY

DATA DICTIONARY

- CONTENT DATA TABLE(Main Fact Table)**

COLUMN	DATA TYPE	SIZE	CONSTRAINTS
POST ID	VARCHAR	25	PRIMARY KEY
PLATFORM	VARCHAR	50	NOT NULL
CATEGORY	VARCHAR	50	NOT NULL
POST TYPE	VARCHAR	50	NOT NULL
POST DATE	DATE	-	NOT NULL
REACH	INT	-	NOT NULL
ENGAGEMENT	INT	-	NOT NULL
VIEWS	INT	-	NOT NULL
IMPRESSIONS	INT	-	NOT NULL

- DATA DIMENSION TABLE**

COLUMN	DATA TYPE	SIZE	CONSTRAINTS
DATE_ID	INT	-	PRIMARY KEY
FULL_DATE	DATE	-	NOT NULL
DAY	INT	-	NOT NULL
MONTH	VARCHAR	20	NOT NULL
MONTH_NUMBER	INT	-	NOT NULL
YEAR	INT	-	NOT NULL
QUARTER	VARCHAR	10	NOT NULL

- **PLATFROM DIMENSION TABLE**

COLUMN	DATA TYPE	SIZE	CONSTRAINTS
PLATFROM_ID	INT	-	PRIMARY KEY
PLATFORM_NAME	VARCHAR	50	NOT NULL

- **CAREGORY DIMENSION TABLE**

COLUMN	DATA TYPE	SIZE	CONSTRAINTS
CATEGORY ID	INT	-	PRIMARY KEY
CATEGORY NAME	VARCHAR	50	NOT NULL

- **POST TYPE DIMENSION TABLE**

COLUMN	DATA TYPE	SIZE	CONSTRAINTS
POSTTYPE ID	INT	-	PRIMARY KEY
POSTYPE NAME	VARCHAR	30	NOT NULL

**PROBLEM
STATEMENTS
WITH
SQL QUERIES AND
OUTPUT**

PROBLEM STATEMENTS WITH SQL QUERIES AND OUTPUT

INSTAGRAM DASHBOARD

Visual 1: Sum of Engagement by Month

Problem Statement: Analyze monthly engagement performance on Instagram to identify which months generate the highest user interaction.

What This Query Does: It calculates total engagement month-wise for Instagram posts.

Explanation

- MONTH(PostDate) → Extracts month number (1–12).
- DATENAME(MONTH, PostDate) → Extracts month name (January, February...).
- SUM(Engagement) → Adds all engagement values.
- WHERE PlatformID = 2 → Filters only Instagram data.
- GROUP BY → Groups data month-wise.
- ORDER BY MonthNumber → Ensures correct chronological order.

Query :

```
-- sum of engagement by month (instagram)
SELECT
    MONTH(PostDate) AS MonthNumber,
    DATENAME(MONTH, PostDate) AS MonthName,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE PlatformID = 2
GROUP BY
    MONTH(PostDate),
    DATENAME(MONTH, PostDate)
ORDER BY MonthNumber;
```

OUTPUT :

100 %

Results		Messages	
	MonthNumber	MonthName	Total_Engagement
1	1	January	168512
2	2	February	196804
3	3	March	375216
4	4	April	235045
5	5	May	263617
6	6	June	206564
7	7	July	222468
8	8	August	236424
9	9	September	326108
10	10	October	304645
11	11	November	226942
12	12	December	353581

Visual 2: Sum of Engagement and Sum of Reach

Problem Statement

Measure overall performance of Instagram by comparing total engagement and total reach.

Explanation

- SUM(Engagement) → Total interactions.
- SUM(Reach) → Total audience reached.
- WHERE Platform = 'Instagram' → Filters Instagram only.

Query :

```
-- sum of engagement and sum of reach
select
sum(Engagement) as Sum_of_Engagement ,
sum(Reach) as Sum_of_Reach
from contentdata
where Platform='Instagram';
```

OUTPUT :

	Sum_of_Engagement	Sum_of_Reach
1	3115926	73801711

Visual 3: Total Engagement by PostType

Problem Statement

Identify which type of Instagram content (Reels, Posts, Stories) generates the highest engagement.

Explanation

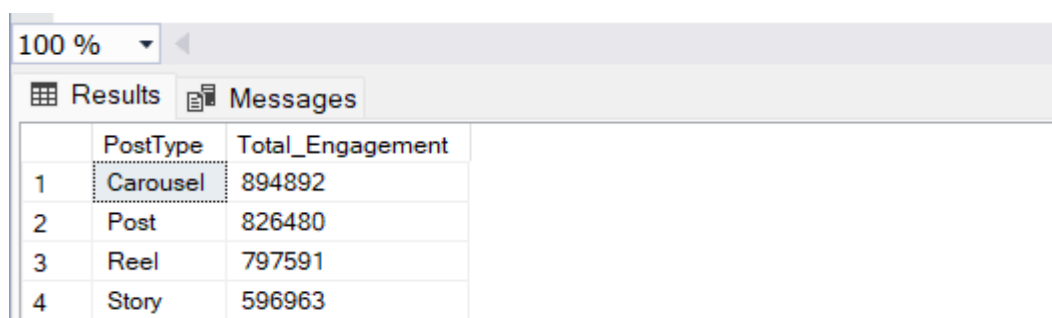
- Groups by PostType
- Sums engagement per type

Query :

-- Total Engagement by PostType

```
select PostType,SUM(Engagement) as Total_Engagement
from contentdata
where Platform = 'Instagram'
group by PostType;
```

OUTPUT :



The screenshot shows a database interface with a 'Results' tab selected. It displays a table with two columns: 'PostType' and 'Total_Engagement'. The table contains four rows of data, numbered 1 to 4. The 'PostType' values are 'Carousel', 'Post', 'Reel', and 'Story'. The corresponding 'Total_Engagement' values are 894892, 826480, 797591, and 596963.

	PostType	Total_Engagement
1	Carousel	894892
2	Post	826480
3	Reel	797591
4	Story	596963

Visual 4: Total Engagement by Category

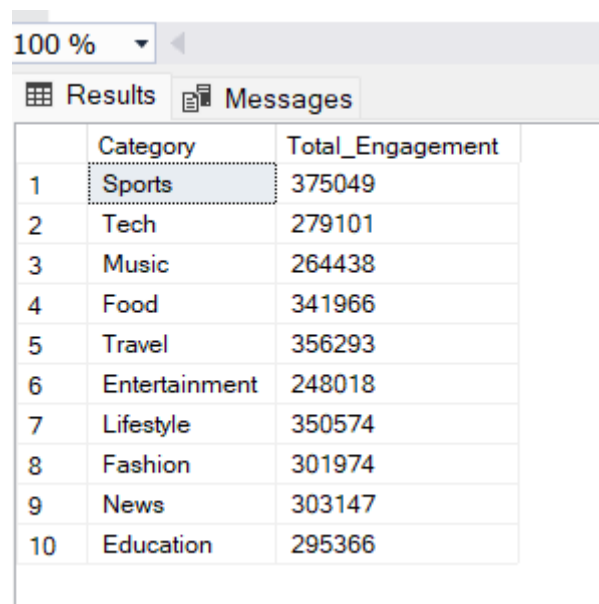
Problem Statement

Analyze which content category (Education, Promotion, Announcement) drives maximum engagement.

Query :

```
-- Total Engagement by Category
select Category,SUM(Engagement) as Total_Engagement
from contentdata
where Platform = 'Instagram'
group by Category;
```

OUTPUT :



The screenshot shows a database interface with a 'Results' tab selected. The table displays the following data:

	Category	Total_Engagement
1	Sports	375049
2	Tech	279101
3	Music	264438
4	Food	341966
5	Travel	356293
6	Entertainment	248018
7	Lifestyle	350574
8	Fashion	301974
9	News	303147
10	Education	295366

SNAPCHAT DASHBOARD

Visual 1: Total Reach vs Total Engagement by Month

Problem Statement

Compare monthly reach and engagement trends on Snapchat.

Explanation

- Calculates monthly totals for:
 - Reach
 - Engagement
- Uses MONTH() and DATENAME() for proper ordering.
- MONTH(PostDate) extracts the month number for correct chronological order.
- DATENAME(MONTH, PostDate) extracts the month name for readable labels.
- SUM(Reach) calculates the total audience reached each month.
- SUM(Engagement) calculates the total user interactions each month. WHERE Platform = 'Snapchat' filters data for Snapchat only.
- GROUP BY aggregates all posts month-wise.
- ORDER BY MonthNumber ensures January to December sorting.

Query :

-- Total Reach VS Total Engagement in MonthName

SELECT

MONTH(PostDate) AS MonthNumber,

DATENAME(MONTH, PostDate) AS MonthName,

SUM(Reach) AS Total_Reach,

SUM(Engagement) AS Total_Engagement

FROM contentdata

WHERE Platform = 'Snapchat'

GROUP BY

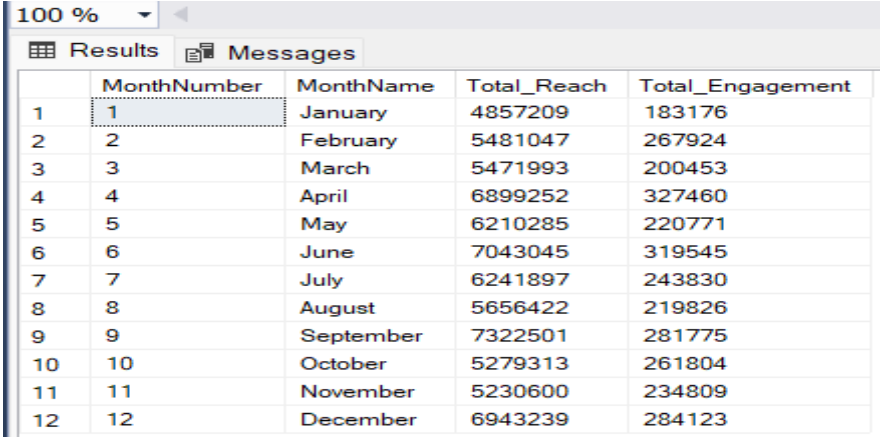
MONTH(PostDate),

DATENAME(MONTH, PostDate)

ORDER BY

MonthNumber;

OUTPUT :



	MonthNumber	MonthName	Total_Reach	Total_Engagement
1	1	January	4857209	183176
2	2	February	5481047	267924
3	3	March	5471993	200453
4	4	April	6899252	327460
5	5	May	6210285	220771
6	6	June	7043045	319545
7	7	July	6241897	243830
8	8	August	5656422	219826
9	9	September	7322501	281775
10	10	October	5279313	261804
11	11	November	5230600	234809
12	12	December	6943239	284123

Visual 2: Total Engagement by PostType

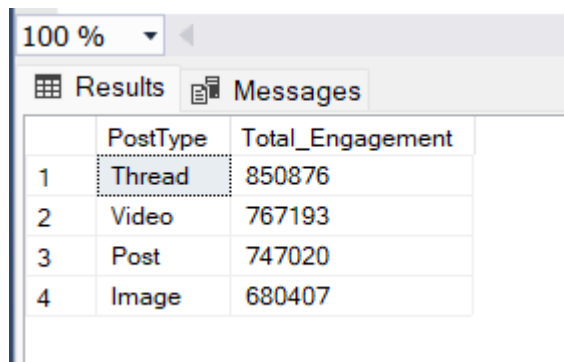
Problem

Find best-performing Snapchat content type.

Query:

```
--Total Engagement by PostType
SELECT
    PostType,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'Snapchat'
GROUP BY PostType
ORDER BY Total_Engagement DESC;
```

OUTPUT :



The screenshot shows a software interface with a zoom level of 100%. Below the zoom bar are two tabs: 'Results' (active) and 'Messages'. The 'Results' tab displays a table with two columns: 'PostType' and 'Total_Engagement'. The table contains four rows of data, numbered 1 to 4.

	PostType	Total_Engagement
1	Thread	850876
2	Video	767193
3	Post	747020
4	Image	680407

Visual 3: Engagement Rate by PostType

Problem Statement

Measure effectiveness of Snapchat content types.

Formula

$$EngagementRate = (TotalEngagement \div TotalReach) \times 100$$

- NULLIF(SUM(Reach),0) prevents divide-by-zero error.
- ROUND(...,2) limits to 2 decimal places.

Query :

-- Engagement Rate by PostType(snapchat)

SELECT

PostType,

ROUND(

(SUM(Engagement) * 100.0) / NULLIF(SUM(Reach), 0),

2

) AS Engagement_Rate_Percent

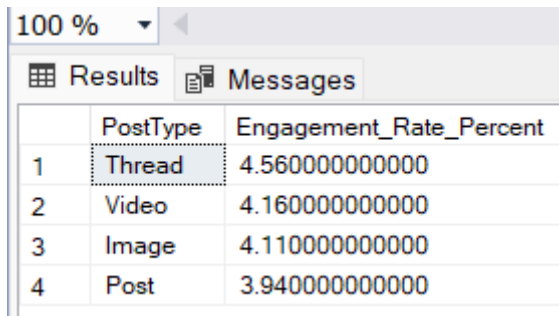
FROM contentdata

WHERE Platform = 'Snapchat'

GROUP BY PostType

ORDER BY Engagement_Rate_Percent DESC;

OUTPUT :



A screenshot of a data table interface. At the top, there is a zoom dropdown set to '100 %' and a left arrow. Below this are two tabs: 'Results' (active) and 'Messages'. The table has two columns: 'PostType' and 'Engagement_Rate_Percent'. It contains four rows of data.

	PostType	Engagement_Rate_Percent
1	Thread	4.560000000000
2	Video	4.160000000000
3	Image	4.110000000000
4	Post	3.940000000000

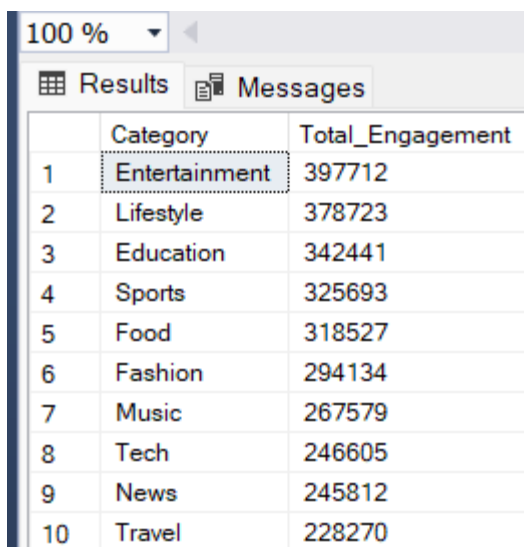
Visual 4 : Total Engagement by Category

Query :

-- Total Engagement by Category

```
SELECT
    Category,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'Snapchat'
GROUP BY Category
ORDER BY Total_Engagement DESC;
```

OUTPUT :



A screenshot of a data table interface. At the top, there is a zoom dropdown set to '100 %' and a left arrow. Below this are two tabs: 'Results' (active) and 'Messages'. The table has two columns: 'Category' and 'Total_Engagement'. It contains ten rows of data, ordered by Total_Engagement in descending order.

	Category	Total_Engagement
1	Entertainment	397712
2	Lifestyle	378723
3	Education	342441
4	Sports	325693
5	Food	318527
6	Fashion	294134
7	Music	267579
8	Tech	246605
9	News	245812
10	Travel	228270

TWITTER DASHBOARD

Visual 1: Total Reach by PostType

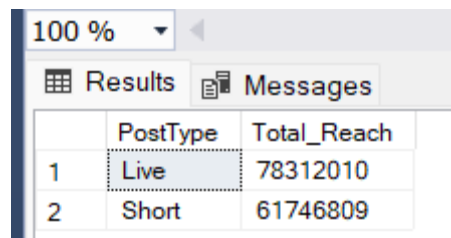
Problem

Identify which Twitter post format gets maximum visibility.

Query :

```
SELECT
    PostType,
    SUM(Reach) AS Total_Reach
FROM contentdata
WHERE Platform = 'Twitter'
GROUP BY PostType
ORDER BY Total_Reach DESC;
```

OUTPUT :



	PostType	Total_Reach
1	Live	78312010
2	Short	61746809

Visual 2: Monthly Reach & Engagement

Query :

```
SELECT
    MONTH(PostDate) AS MonthNumber,
    DATENAME(MONTH, PostDate) AS MonthName,
    SUM(Reach) AS Total_Reach,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'Twitter'
```

GROUP BY

MONTH(PostDate),

DATENAME(MONTH, PostDate)

ORDER BY MonthNumber;

OUTPUT :

	MonthNumber	MonthName	Total_Reach	Total_Engagement
1	1	January	13643711	498834
2	2	February	9839611	418164
3	3	March	8273007	381985
4	4	April	15183825	699765
5	5	May	8194372	361790
6	6	June	7764701	332286
7	7	July	9617020	367255
8	8	August	10063459	380984
9	9	September	12748307	467063
10	10	October	22604862	848616
11	11	November	13539124	561111
12	12	December	8586820	410009

Visual 3: Category vs PostType

Problem Statement

Compare Live and Short post reach across categories and also calculate grand totals

Query :

SELECT

Category,

SUM(CASE WHEN PostType = 'Live' THEN Reach ELSE 0 END) AS Live,

SUM(CASE WHEN PostType = 'Short' THEN Reach ELSE 0 END) AS Short,

SUM(Reach) AS Total

FROM contentdata

WHERE Platform = 'Twitter'

GROUP BY Category

UNION ALL

SELECT

```

'Total',
SUM(CASE WHEN PostType = 'Live' THEN Reach ELSE 0 END),
SUM(CASE WHEN PostType = 'Short' THEN Reach ELSE 0 END),
SUM(Reach)
FROM contentdata
WHERE Platform = 'Twitter';

```

OUTPUT :

	Category	Live	Short	Total
1	Sports	14414121	8302181	22716302
2	Tech	3738819	5942052	9680871
3	Music	7532275	7596824	15129099
4	Food	6570634	3211847	9782481
5	Travel	11788013	6042390	17830403
6	Entertainment	9148933	7284076	16433009
7	Lifestyle	7339902	7014971	14354873
8	Fashion	6442738	5304207	11746945
9	News	5956039	4797963	10754002
10	Education	5380536	6250298	11630834
11	Total	78312010	61746809	140058819

Visual 4: Total Reach by Category

Problem Statement :

Shows which category drives maximum impressions.

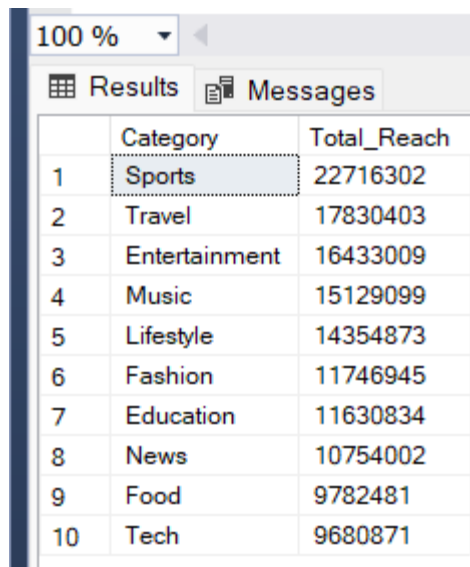
Query :

```

SELECT
    Category,
    SUM(Reach) AS Total_Reach
FROM contentdata
WHERE Platform = 'Twitter'
GROUP BY Category
ORDER BY Total_Reach DESC;

```

OUTPUT :



	Category	Total_Reach
1	Sports	22716302
2	Travel	17830403
3	Entertainment	16433009
4	Music	15129099
5	Lifestyle	14354873
6	Fashion	11746945
7	Education	11630834
8	News	10754002
9	Food	9782481
10	Tech	9680871

YOUTUBE DASHBOARD

Visual 1: Total Views & Total Engagement by MonthName

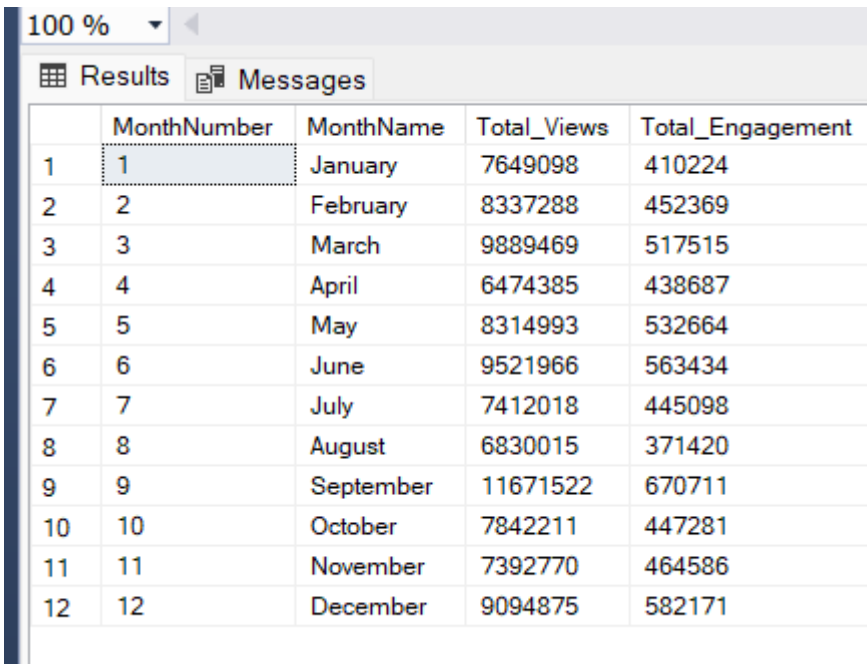
Problem Statement:

Analyze video performance trend month-wise.

Query:

```
SELECT
    MONTH(PostDate) AS MonthNumber,
    DATENAME (MONTH, PostDate) AS MonthName,
    SUM(Views) AS Total_Views,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'YouTube'
GROUP BY
    MONTH(PostDate),
    DATENAME(MONTH, PostDate)
ORDER BY MonthNumber;
```

OUTPUT :



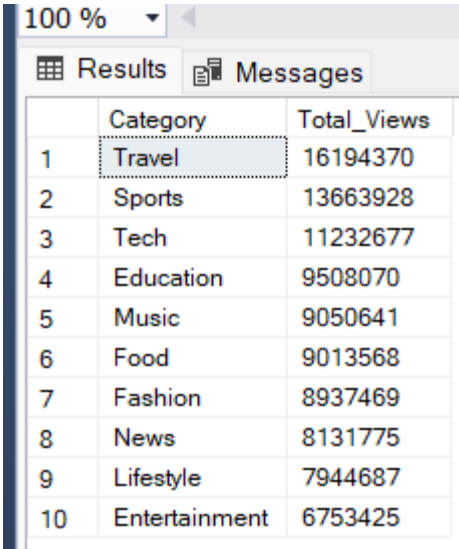
	MonthNumber	MonthName	Total_Views	Total_Engagement
1	1	January	7649098	410224
2	2	February	8337288	452369
3	3	March	9889469	517515
4	4	April	6474385	438687
5	5	May	8314993	532664
6	6	June	9521966	563434
7	7	July	7412018	445098
8	8	August	6830015	371420
9	9	September	11671522	670711
10	10	October	7842211	447281
11	11	November	7392770	464586
12	12	December	9094875	582171

Visual 2 : Total Views by Category

Query :

```
SELECT
    Category,
    SUM(Views) AS Total_Views
FROM contentdata
WHERE Platform = 'YouTube'
GROUP BY Category
ORDER BY Total_Views DESC;
```

OUTPUT :



	Category	Total_Views
1	Travel	16194370
2	Sports	13663928
3	Tech	11232677
4	Education	9508070
5	Music	9050641
6	Food	9013568
7	Fashion	8937469
8	News	8131775
9	Lifestyle	7944687
10	Entertainment	6753425

Visual 3: Total Engagement by PostType

Problem Statement :

Shows which format performs best (Live, Shorts).

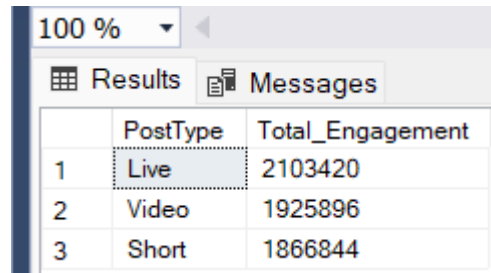
Query :

```
SELECT
    PostType,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'YouTube'
```

GROUP BY PostType

ORDER BY Total_Engagement DESC;

OUTPUT :



A screenshot of a SQL query result table. The table has two columns: 'PostType' and 'Total_Engagement'. It contains three rows of data. The first row is 'Live' with a value of 2103420. The second row is 'Video' with a value of 1925896. The third row is 'Short' with a value of 1866844. The table is displayed in a window with a '100 %' zoom level and tabs for 'Results' and 'Messages'.

	PostType	Total_Engagement
1	Live	2103420
2	Video	1925896
3	Short	1866844

Visual 4: Engagement Rate by PostType

Formula : $(Engagement \div Views) \times 100$

Query :

PostType,

ROUND(

(SUM(Engagement) * 100.0) / NULLIF(SUM(Views), 0),

2

) AS Engagement_Rate_Percent

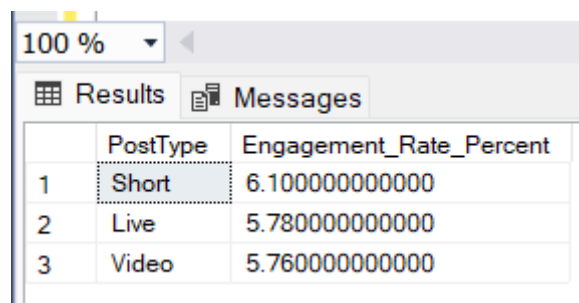
FROM contentdata

WHERE Platform = 'YouTube'

GROUP BY PostType

ORDER BY Engagement_Rate_Percent DESC;

OUTPUT :



A screenshot of a SQL query result table. The table has two columns: 'PostType' and 'Engagement_Rate_Percent'. It contains three rows of data. The first row is 'Short' with a value of 6.100000000000. The second row is 'Live' with a value of 5.780000000000. The third row is 'Video' with a value of 5.760000000000. The table is displayed in a window with a '100 %' zoom level and tabs for 'Results' and 'Messages'.

	PostType	Engagement_Rate_Percent
1	Short	6.100000000000
2	Live	5.780000000000
3	Video	5.760000000000

FACEBOOK DASHBOARD

Visual 1 : Total Reach & Total Engagement by MonthName

Query :

```
SELECT
    MONTH(PostDate) AS MonthNumber,
    DATENAME(MONTH, PostDate) AS MonthName,
    SUM(Reach) AS Total_Reach,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'Facebook'
GROUP BY
    MONTH(PostDate),
    DATENAME(MONTH, PostDate)
ORDER BY MonthNumber;
```

OUTPUT :

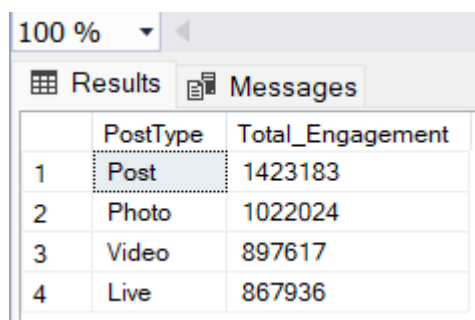
	MonthNumber	MonthName	Total_Reach	Total_Engagement
1	1	January	16985994	823791
2	2	February	5156096	226731
3	3	March	9415967	352432
4	4	April	4591645	244632
5	5	May	9655088	353405
6	6	June	4807839	170949
7	7	July	8395893	385072
8	8	August	8834520	368223
9	9	September	9093429	324557
10	10	October	8140148	348258
11	11	November	9384063	365142
12	12	December	5814830	247568

Visual 2: Total Engagement by PostType

Query:

```
SELECT
    PostType,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'Facebook'
GROUP BY PostType
ORDER BY Total_Engagement DESC;
```

OUTPUT:



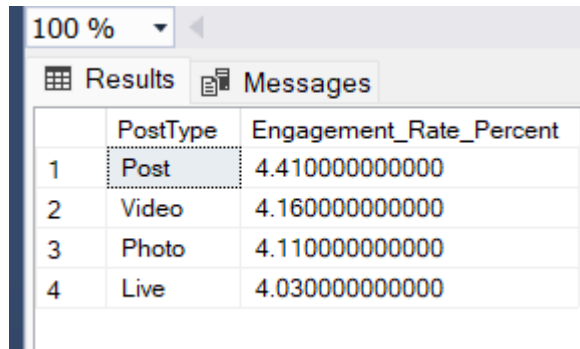
	PostType	Total_Engagement
1	Post	1423183
2	Photo	1022024
3	Video	897617
4	Live	867936

Visual 3 : Engagement Rate % by PostType

Query :

```
SELECT
    PostType,
    ROUND(
        (SUM(Engagement) * 100.0) / NULLIF(SUM(Reach), 0),
        2
    ) AS Engagement_Rate_Percent
FROM contentdata
WHERE Platform = 'Facebook'
GROUP BY PostType
ORDER BY Engagement_Rate_Percent DESC;
```

OUTPUT :



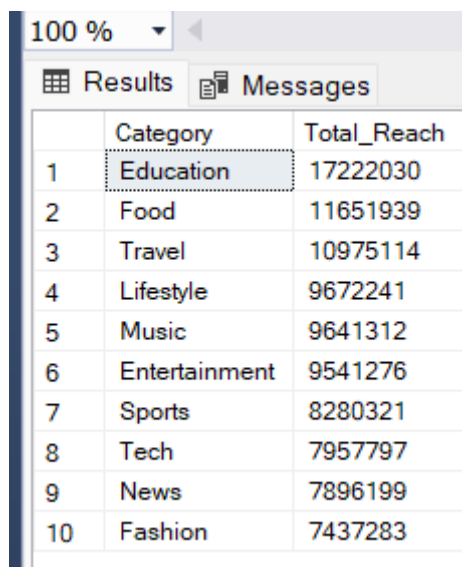
	PostType	Engagement_Rate_Percent
1	Post	4.410000000000
2	Video	4.160000000000
3	Photo	4.110000000000
4	Live	4.030000000000

Visual 4 : Total Reach by Category

Query :

```
SELECT
    Category,
    SUM(Reach) AS Total_Reach
FROM contentdata
WHERE Platform = 'Facebook'
GROUP BY Category
ORDER BY Total_Reach DESC;
```

OUTPUT :



	Category	Total_Reach
1	Education	17222030
2	Food	11651939
3	Travel	10975114
4	Lifestyle	9672241
5	Music	9641312
6	Entertainment	9541276
7	Sports	8280321
8	Tech	7957797
9	News	7896199
10	Fashion	7437283

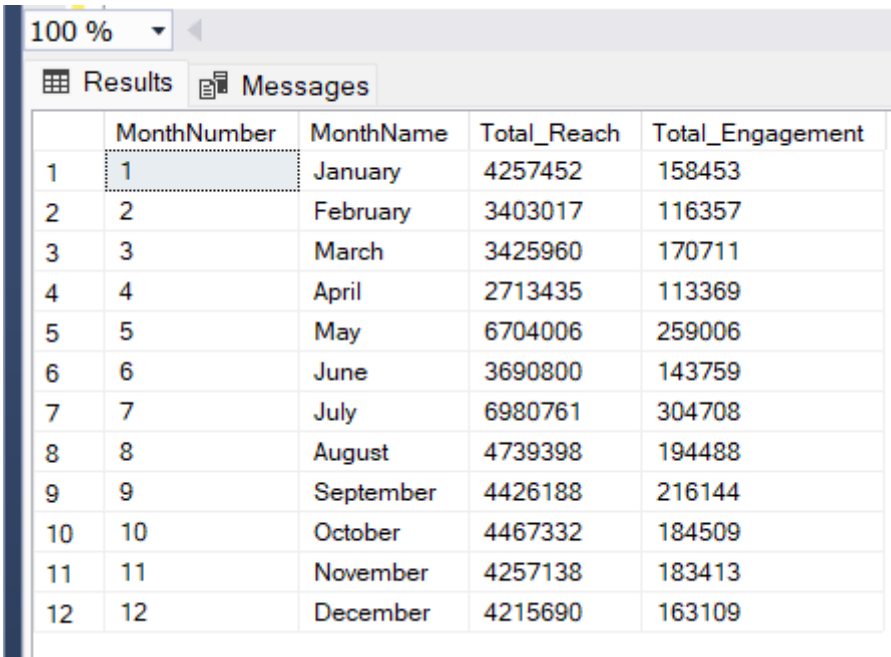
LINKEDIN DASHBOARD

Visual 1 : Total Reach & Total Engagement by MonthName

Query :

```
SELECT
    MONTH(PostDate) AS MonthNumber,
    DATENAME(MONTH, PostDate) AS MonthName,
    SUM(Reach) AS Total_Reach,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'LinkedIn'
GROUP BY
    MONTH(PostDate),
    DATENAME(MONTH, PostDate)
ORDER BY MonthNumber;
```

OUTPUT :



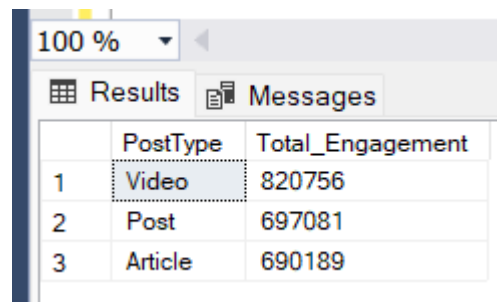
	MonthNumber	MonthName	Total_Reach	Total_Engagement
1	1	January	4257452	158453
2	2	February	3403017	116357
3	3	March	3425960	170711
4	4	April	2713435	113369
5	5	May	6704006	259006
6	6	June	3690800	143759
7	7	July	6980761	304708
8	8	August	4739398	194488
9	9	September	4426188	216144
10	10	October	4467332	184509
11	11	November	4257138	183413
12	12	December	4215690	163109

Visual 2 : Total Engagement by PostType

Query :

```
SELECT
    PostType,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
WHERE Platform = 'LinkedIn'
GROUP BY PostType
ORDER BY Total_Engagement DESC;
```

OUTPUT :



	PostType	Total_Engagement
1	Video	820756
2	Post	697081
3	Article	690189

Visual 3 : Total Reach by Category

Query :

```
SELECT
    Category,
    SUM(Reach) AS Total_Reach
FROM contentdata
WHERE Platform = 'LinkedIn'
GROUP BY Category
ORDER BY Total_Reach DESC;
```

OUTPUT :

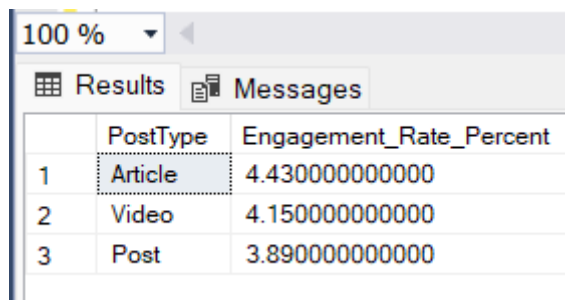
	Category	Total_Reach
1	Tech	7033543
2	Entertainment	6115240
3	Food	6062375
4	Travel	5588945
5	Lifestyle	5300783
6	Fashion	4945253
7	Sports	4788041
8	Education	4757375
9	Music	4531680
10	News	4157942

Visual 4 : Engagement Rate % by PostType

Query :

```
SELECT
    PostType,
    ROUND(
        (SUM(Engagement) * 100.0) / NULLIF(SUM(Reach), 0),
        2
    ) AS Engagement_Rate_Percent
FROM contentdata
WHERE Platform = 'LinkedIn'
GROUP BY PostType
ORDER BY Engagement_Rate_Percent DESC;
```

OUTPUT :



The screenshot shows a SQL Server query results window. At the top, there is a zoom level of 100% and two tabs: 'Results' (active) and 'Messages'. Below the tabs is a table with three columns: 'PostType', 'Engagement_Rate_Percent', and an unlabeled column. The table contains three rows of data.

	PostType	Engagement_Rate_Percent
1	Article	4.430000000000
2	Video	4.150000000000
3	Post	3.890000000000

COMPARISON DASHBOARD

Visual 1: Engagement by Category & Platform

Problem Statement :

Compare performance across all platforms and categories.

Query :

```
SELECT
    Platform,
    Category,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
GROUP BY Platform, Category
ORDER BY Platform, Total_Engagement DESC;
```

Results		Messages	
	Platform	Category	Total_Engagement
1	Facebook	Education	804814
2	Facebook	Travel	477636
3	Facebook	Food	446363
4	Facebook	Music	418759
5	Facebook	Entertainment	396710
6	Facebook	Sports	354847
7	Facebook	Tech	347592
8	Facebook	Lifestyle	332370
9	Facebook	News	329004
10	Facebook	Fashion	302665
11	Instagram	Sports	375049
12	Instagram	Travel	356293
13	Instagram	Lifestyle	350574
14	Instagram	Food	341966
15	Instagram	News	303147
16	Instagram	Fashion	301974
17	Instagram	Education	295366
18	Instagram	Tech	279101
19	Instagram	Music	264438
20	Instagram	Entertainment	248018
21	LinkedIn	Tech	300128
22	LinkedIn	Entertainment	265808
23	LinkedIn	Lifestyle	244161
24	LinkedIn	Fashion	230074
25	LinkedIn	Travel	216281
26	LinkedIn	Food	212005
27	LinkedIn	Music	197438
28	LinkedIn	Sports	189540
29	LinkedIn	Education	187654
30	LinkedIn	News	164937
31	Snapchat	Entertainment	397712
32	Snapchat	Lifestyle	378723
33	Snapchat	Education	342441
34	Snapchat	Sports	325693
35	Snapchat	Food	318527

35	Snapchat	Food	318527
36	Snapchat	Fashion	294134
37	Snapchat	Music	267579
38	Snapchat	Tech	246605
39	Snapchat	News	245812
40	Snapchat	Travel	228270
41	Twitter	Sports	866944
42	Twitter	Travel	821091
43	Twitter	Entertainment	650298
44	Twitter	Lifestyle	597884
45	Twitter	Music	532877
46	Twitter	Fashion	500220
47	Twitter	Education	476679
48	Twitter	Food	457888
49	Twitter	News	428638
50	Twitter	Tech	395343
51	YouTube	Travel	854921
52	YouTube	Sports	813358
53	YouTube	Music	587517
54	YouTube	Fashion	585117
55	YouTube	Tech	579442
56	YouTube	Food	573068
57	YouTube	Education	551578
58	YouTube	News	508343
59	YouTube	Lifestyle	439882
60	YouTube	Entertainment	402934

Visual 2 : Total Reach by Platform

Query :

```

SELECT
    Platform,
    SUM(Reach) AS Total_Reach
FROM contentdata
GROUP BY Platform
ORDER BY Total_Reach DESC;
```


OUTPUT :

Results Messages		
	Platform	Total_Reach
1	YouTube	144145858
2	Twitter	140058819
3	Facebook	100275512
4	Instagram	73801711
5	Snapchat	72636803
6	LinkedIn	53281177

Visual 3: Engagement Rate by Platform

Problem Statement :

Shows most efficient platform.

Query :

```
SELECT
    Platform,
    ROUND(
        (SUM(Engagement) * 100.0) / NULLIF(SUM(Reach), 0),
        2
    ) AS Engagement_Rate_Percent
FROM contentdata
GROUP BY Platform
ORDER BY Engagement_Rate_Percent DESC;
```

OUTPUT :

Results Messages		
	Platform	Engagement_Rate_Percent
1	Instagram	4.220000000000
2	Facebook	4.200000000000
3	Snapchat	4.190000000000
4	LinkedIn	4.140000000000
5	YouTube	4.090000000000
6	Twitter	4.090000000000

Visual 4: Total Engagement by Platform

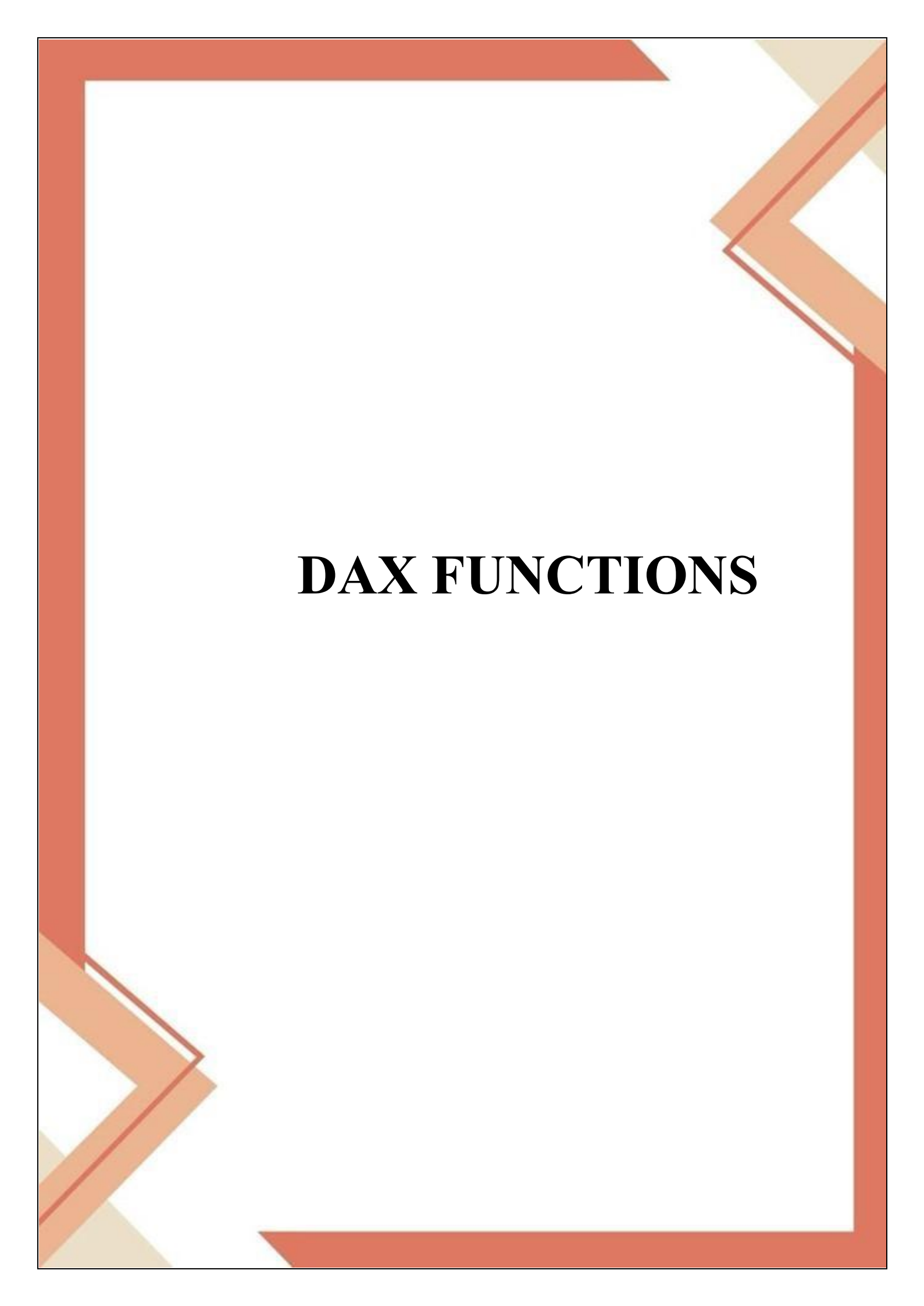
Shows overall interaction comparison.

Query :

```
SELECT
    Platform,
    SUM(Engagement) AS Total_Engagement
FROM contentdata
GROUP BY Platform
ORDER BY Total_Engagement DESC;
```

OUTPUT :

Results		Messages
	Platform	Total_Engagement
1	YouTube	5896160
2	Twitter	5727862
3	Facebook	4210760
4	Instagram	3115926
5	Snapchat	3045496
6	LinkedIn	2208026



DAX FUNCTIONS

DAX FUNCTIONS

Home page

KPI 1: Total Engagement

DAX FUNCTION

SUM (Fact_PostPerformance[Engagement])

Measures overall content performance

Used to compare platforms, categories, posts

KPI 2 : Engagement Rate (%)

How engaging the content is compared to Reach

DAX FUNCTION

Engagement Rate (%) =

DIVIDE(
SUM (Fact_PostPerformance[Engagement]),
SUM (Fact_PostPerformance[Reach]),
0
)

KPI 3: Total Reach

What it shows

Total number of unique users reached

DAX Formula

Total Reach =

SUM (Fact_PostPerformance[Reach])

Instagram

1. Total Posts

Total Posts = COUNT (Instagram_Data[PostID])

2. Total Engagement

Total Engagement = SUM (Instagram_Data[Engagement])

3. Engagement Rate (%)

Engagement Rate (%) =

DIVIDE(
SUM (Instagram_Data[Engagement]),
SUM (Instagram_Data[Reach]),
0
)

Format as **Percentage**

4. Total Reach

Total Reach = SUM (Instagram_Data[Reach])

5. Avg Watch Time per View

Avg Watch Time =

DIVIDE (
SUM (Instagram_Data[WatchTimeSeconds]),
SUM (Instagram_Data[Views]),
0
)

Snapchat

1. TOTAL SNAPS (POSTS)

Total Snaps =

COUNT (Snapchat_Data[PostID])

2. TOTAL ENGAGEMENT

Total Engagement =

SUM (Snapchat_Data[Engagement])

3. ENGAGEMENT RATE (%)

Engagement Rate (%) =

DIVIDE(
SUM (Snapchat_Data[Engagement]),
SUM (Snapchat_Data[Reach]),
0
)

Formula meaning

$$\text{Engagement Rate} = \frac{\text{Total Engagement}}{\text{Total Reach}}$$

4. TOTAL REACH

Total Reach =

SUM (Snapchat_Data[Reach])

Twitter

1.TOTAL REACH (PRIMARY KPI)

Total Reach =

SUM (Twitter_Data[Reach])

Function Explanation

2.TOTAL IMPRESSIONS

Total Impressions =

SUM (Twitter_Data[Impressions])

3)ENGAGEMENT RATE (%)

Engagement Rate % =

DIVIDE(
SUM (Twitter_Data[Engagement]),
SUM (Twitter_Data[Impressions]),
0
)

Formula Meaning

$$\text{Engagement Rate} = \frac{\text{Total Engagement}}{\text{Total Impressions}}$$

4)REACH PER TWEET (ONLY AVERAGE KPI)

Reach per Tweet =

DIVIDE(
SUM (Twitter_Data[Reach]),
DISTINCTCOUNT (Twitter_Data[PostID]),
0
)

Mathematical Formula

$$\text{Reach per Tweet} = \frac{\text{Total Reach}}{\text{Number of Tweets}}$$

5) Total Engagement (Twitter) =

Total Engagement (Twitter) =

SUM (Twitter_Data[Engagement])

6) Reach Efficiency (%)

Reach Efficiency =

DIVIDE(

SUM (Twitter_Data[Reach]),

SUM (Twitter_Data[Impressions]),

0

) * 100

YOUTUBE

1. Avg Watch Time (YouTube)

Avg Watch Time(Youtube) =
DIVIDE(
 SUM (YouTube_Data[WatchTimeSeconds]),
 SUM (YouTube_Data[Views]),
 0
)

2. Total Videos

Total Videos = DISTINCTCOUNT (YouTube_Data[PostID])

3. Total Views

Total Views = SUM (YouTube_Data[Views])

4. Engagement Rate %

Engagement Rate % =
DIVIDE(
 SUM (YouTube_Data[Engagement]),
 SUM (YouTube_Data[Views]),
 0
) * 100

5. Total Engagement (YouTube) =

Total Engagement(Youtube) = SUM (YouTube_Data[Engagement])

Facebook

1) Total Posts (Facebook)

Total Posts (Facebook) =
DISTINCTCOUNT (Facebook_Data[PostID])

2) Total Engagement (Facebook)

Total Engagement (Facebook) =
SUM (Facebook_Data[Engagement])

3) Total Reach (Facebook)

Total Reach (Facebook) =
SUM (Facebook_Data[Reach])

4) Engagement Rate % (Facebook)

Engagement Rate % (Facebook) =
DIVIDE(
 SUM (Facebook_Data[Engagement]),
 SUM (Facebook_Data[Reach]),
 0
) * 100

5) Avg Engagement per Post (Facebook)

Avg Engagement per Post (Facebook) =
DIVIDE(
 SUM (Facebook_Data[Engagement]),
 DISTINCTCOUNT (Facebook_Data[PostID]),
 0
)

LinkedIn

1) Total Posts (LinkedIn)

Total Posts (LinkedIn) =
DISTINCTCOUNT (LinkedIn_Data[PostID])

2) Total Reach (LinkedIn)

Total Reach (LinkedIn) =
SUM (LinkedIn_Data[Reach])

3) Total Engagement (LinkedIn)

Total Engagement (LinkedIn) =
SUM (LinkedIn_Data[Engagement])

4) Engagement Rate % (LinkedIn)

Engagement Rate % (LinkedIn) =
DIVIDE(
 SUM (LinkedIn_Data[Engagement]),
 SUM (LinkedIn_Data[Reach]),
 0
) * 100

Comparison Dashboard

1). Total Posts :

Total Posts = `DISTINCTCOUNT ( contentdata[PostID] )`

2). Total Reach :

Total Reach = `SUM ( contentdata[Reach] )`

3). Total Engagement :

Total Engagement = `SUM ( contentdata[Engagement] )`

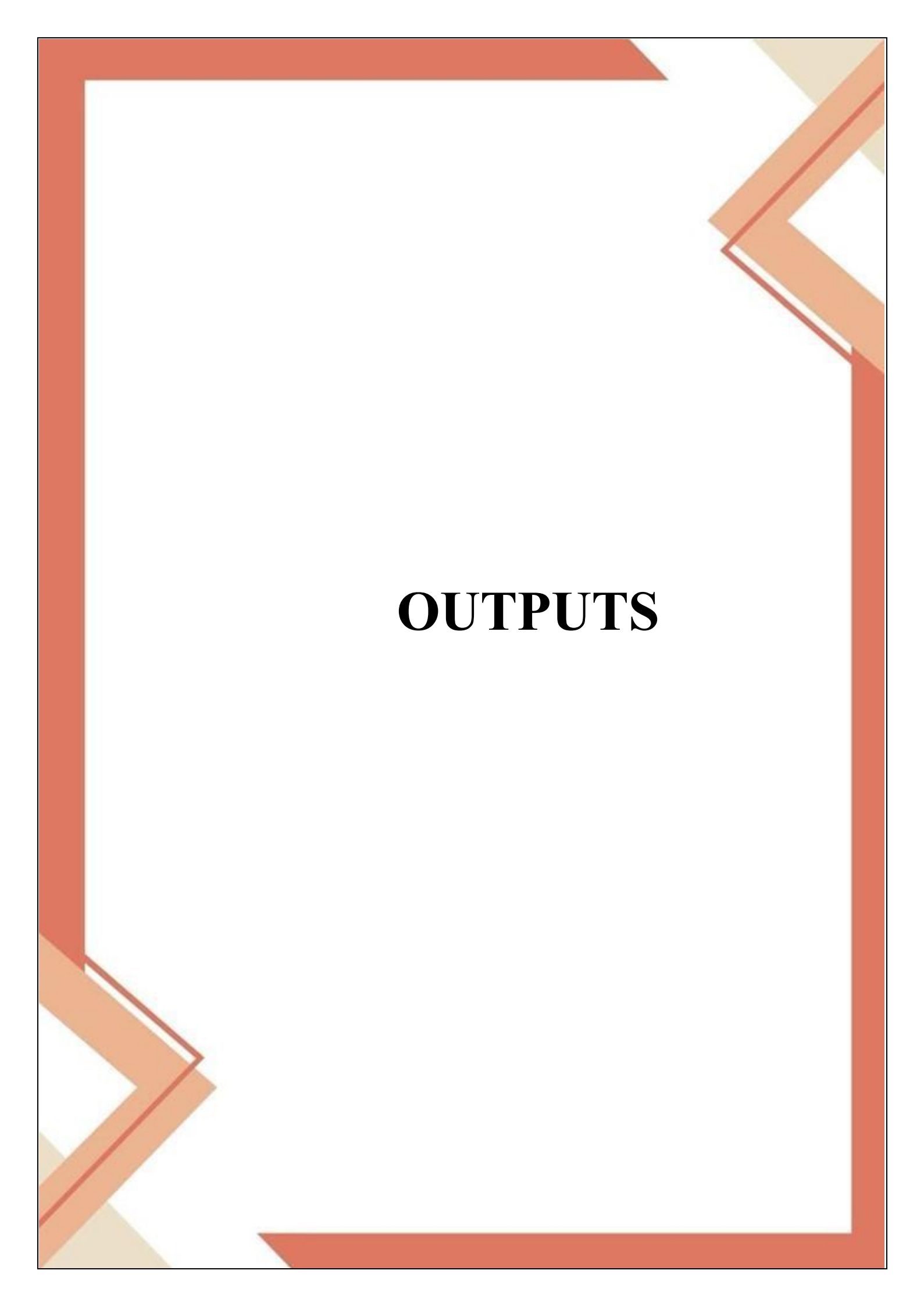
4). Engagement Rate % :

Engagement Rate % =

`DIVIDE(`
 `SUM ( contentdata[Engagement] ),`
 `SUM ( contentdata[Reach] ),`
 `0`
`) * 100`

5). Total Views

Total Views = `SUM ( contentdata[Views] )`



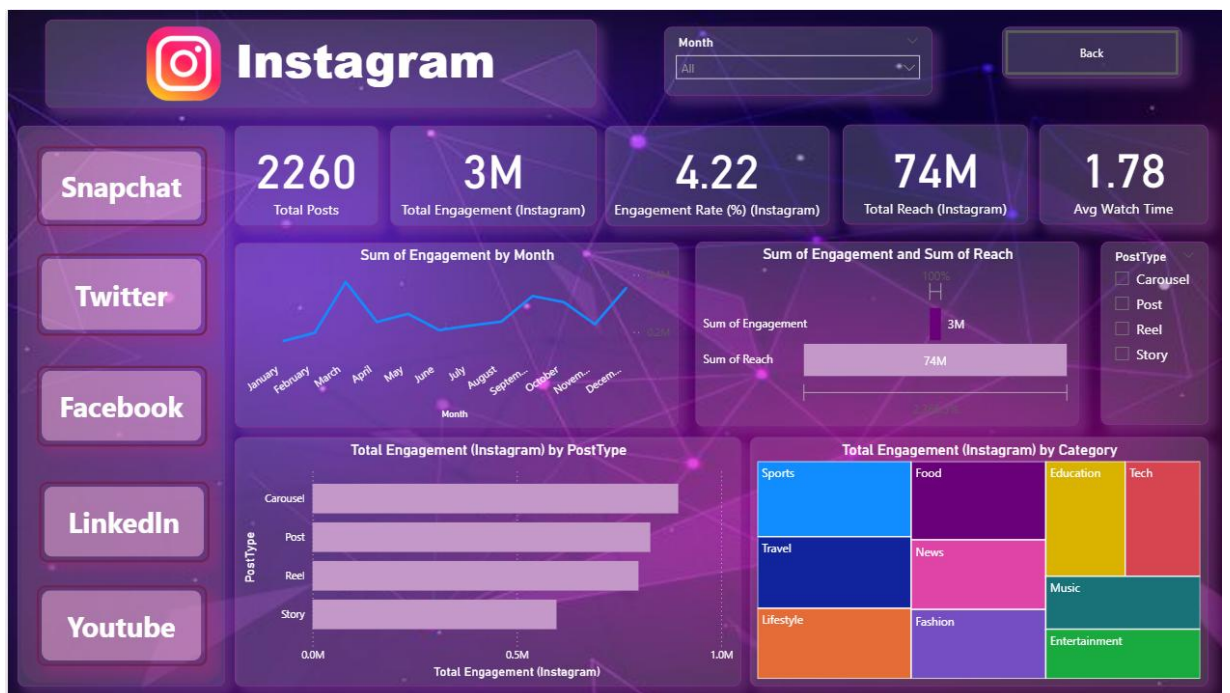
OUTPUTS

OUTPUTS

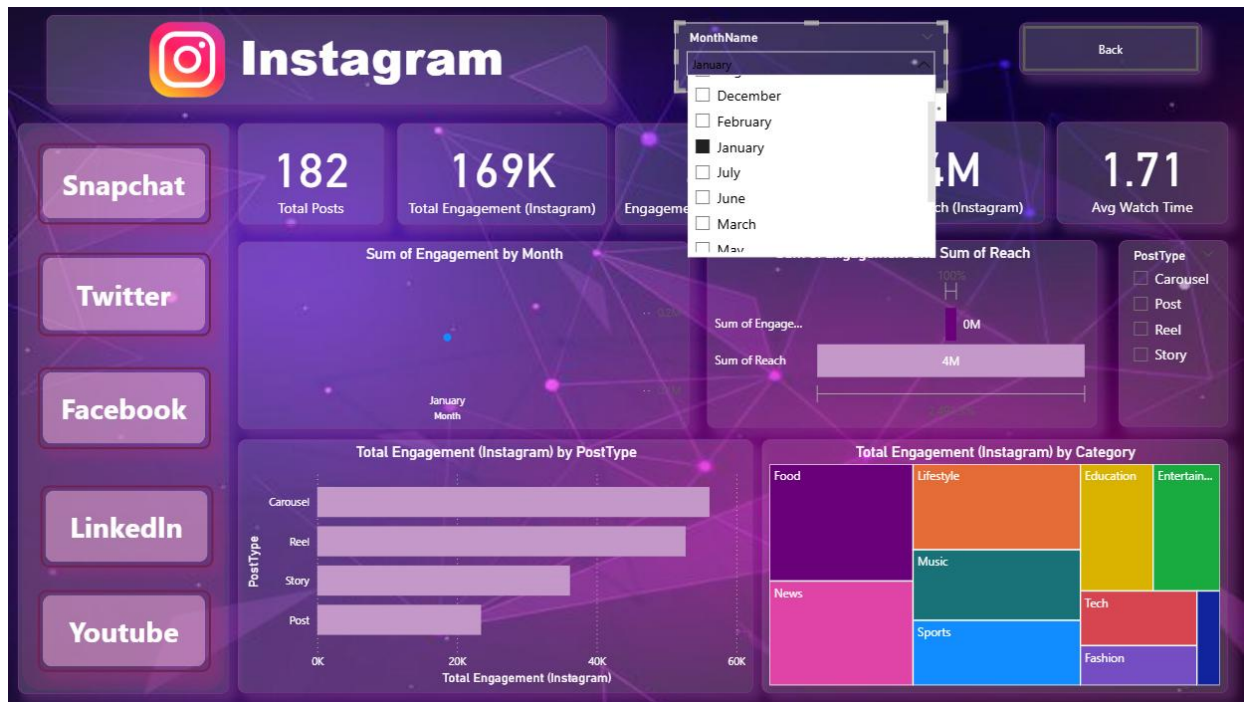
1) HOME PAGE



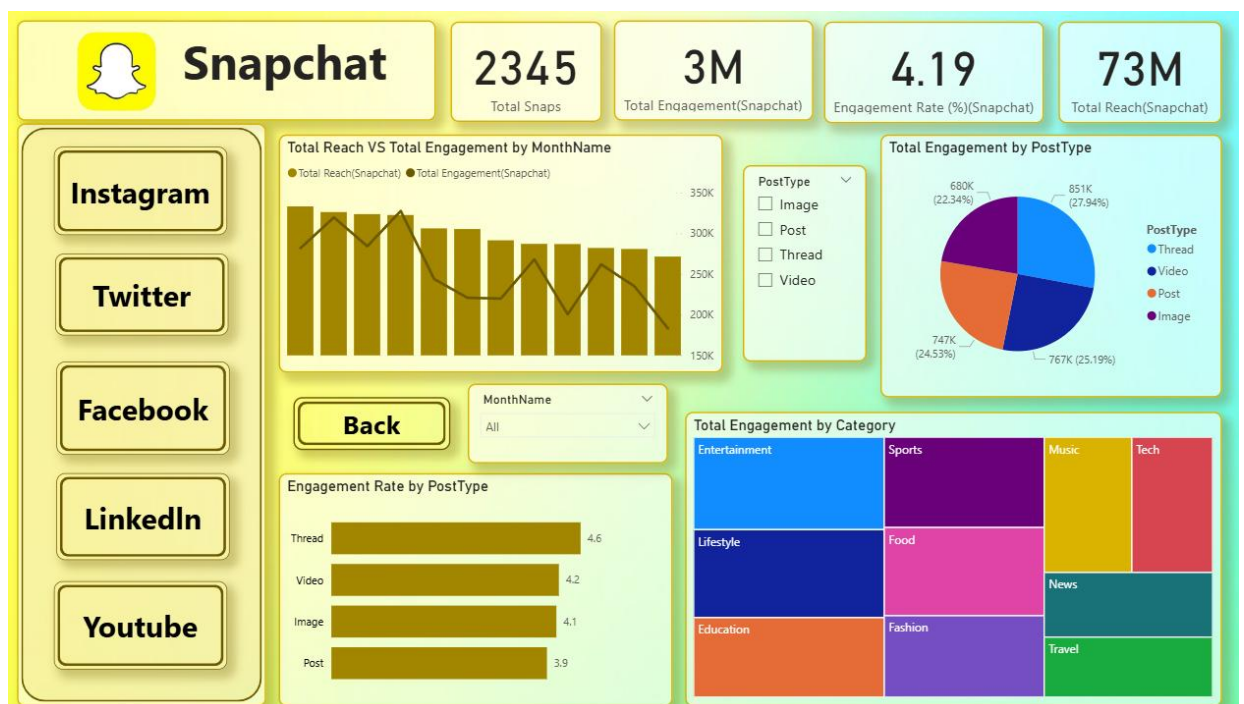
2) INSTAGRAM DASHBOARD



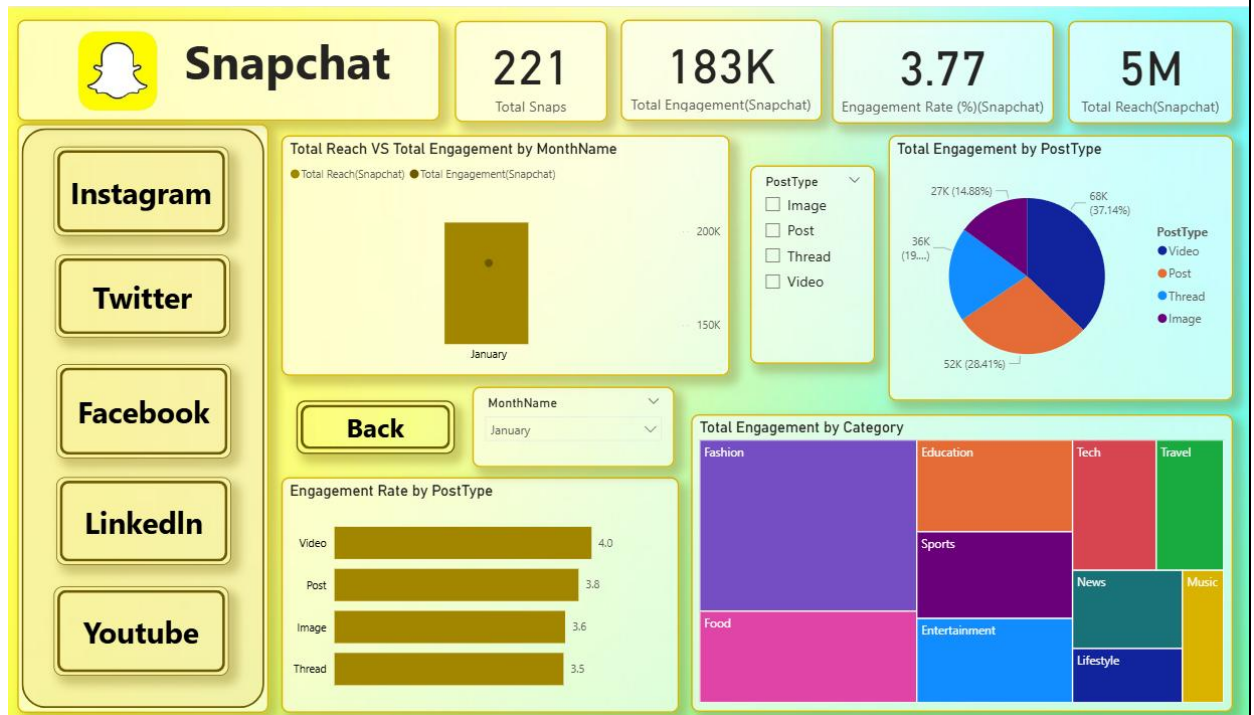
- Instagram page filter apply – month January



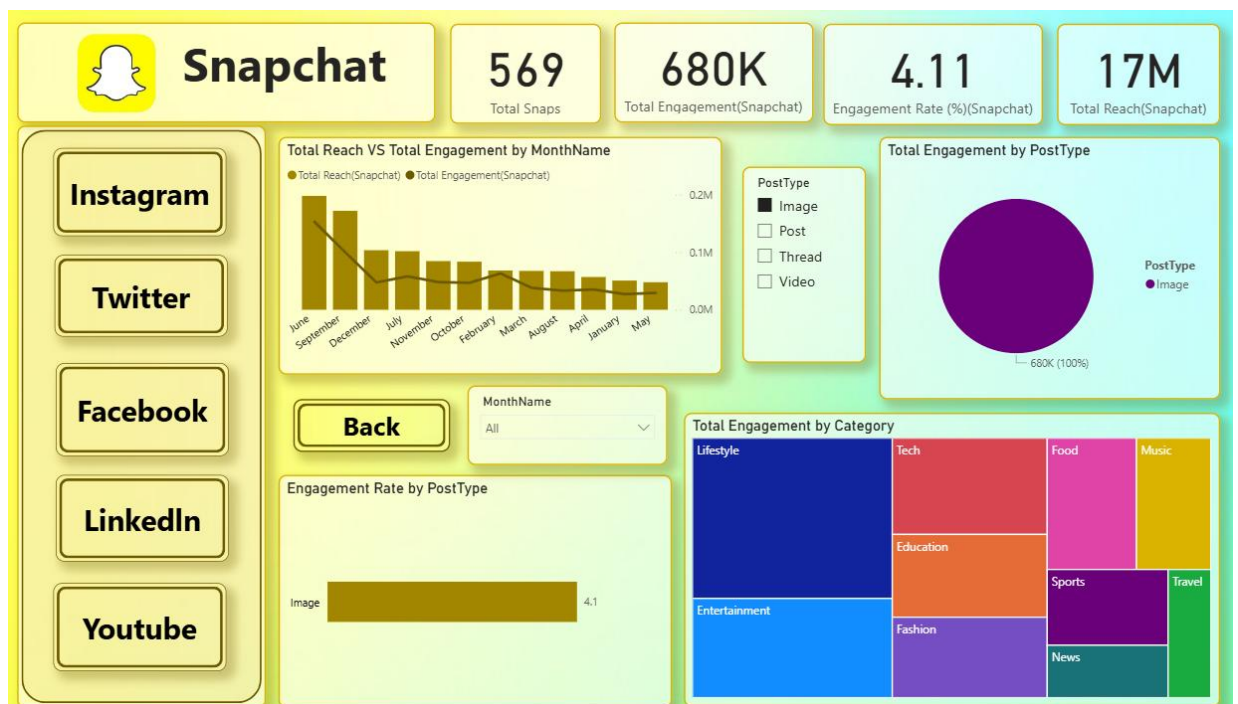
3)SNAPCHAT DASHBOARD



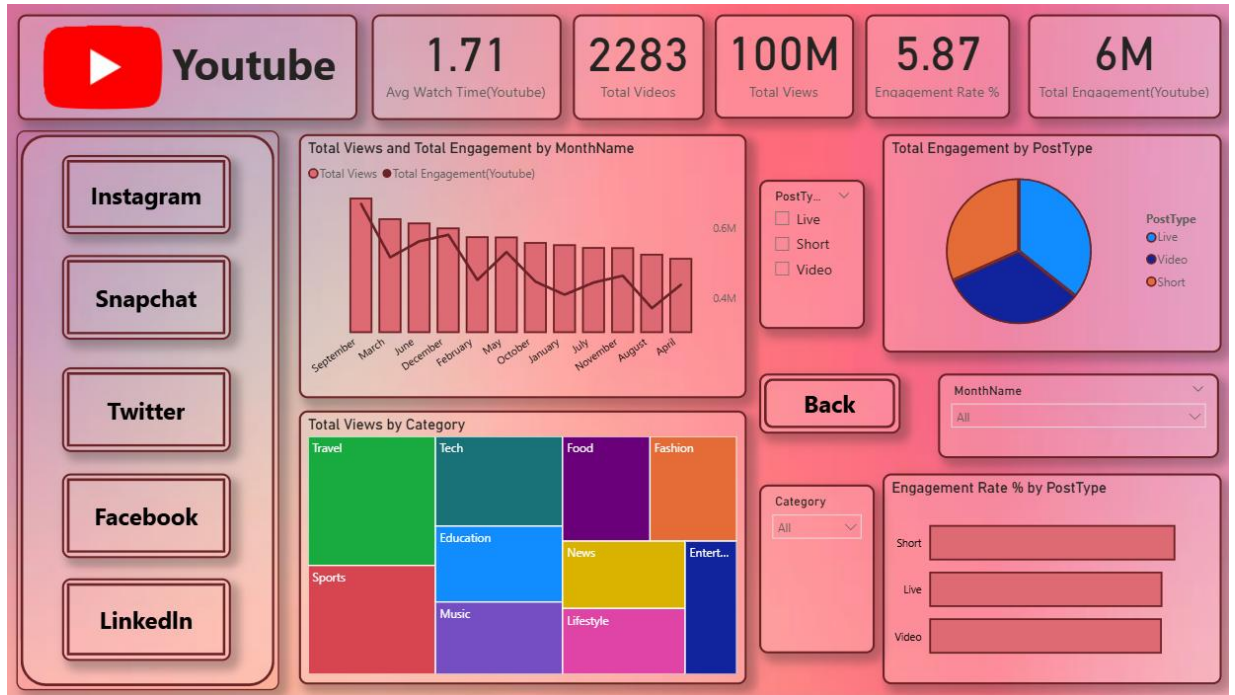
- Snapchat page filter apply - month January



- Snapchat page filter apply – PostType “Image”



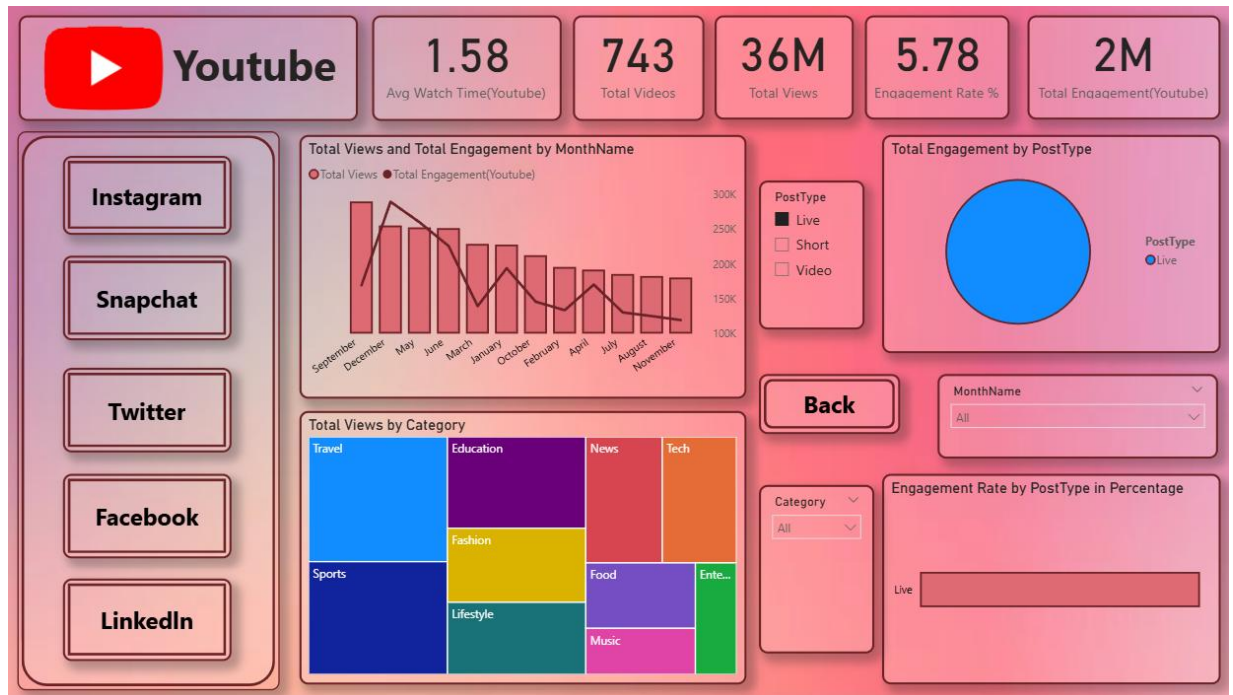
4) YOUTUBE DASHBOARD



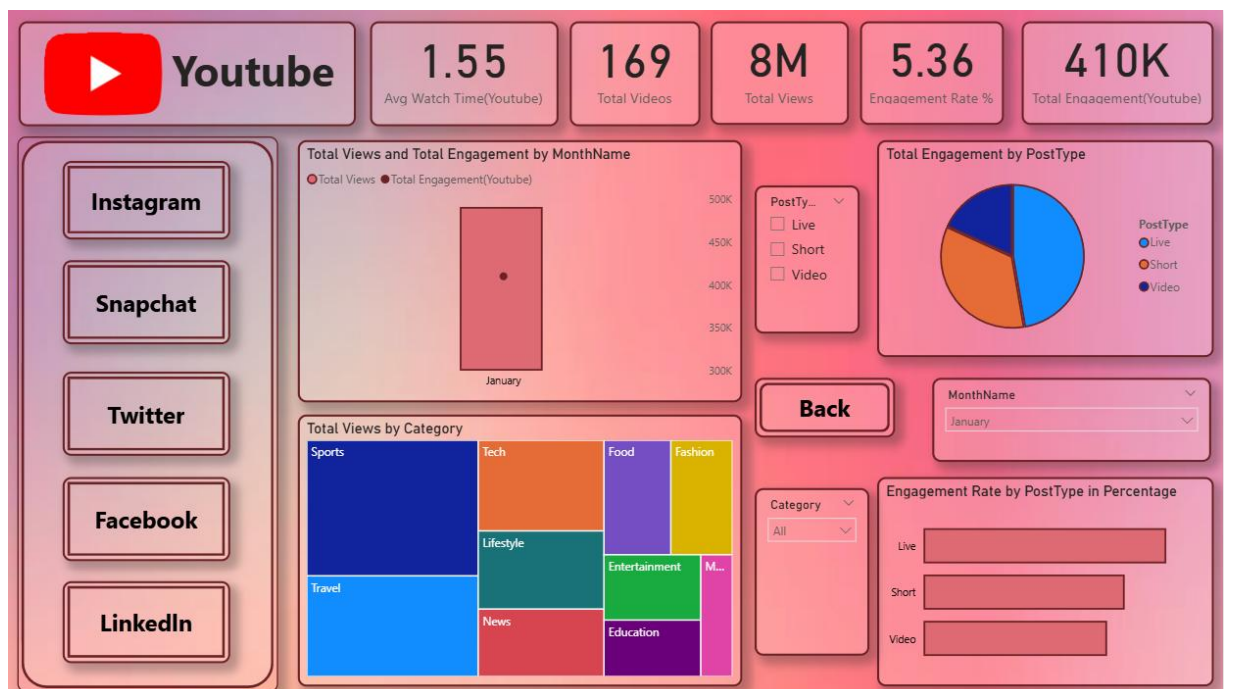
- Youtube Page filter apply – Category “News”



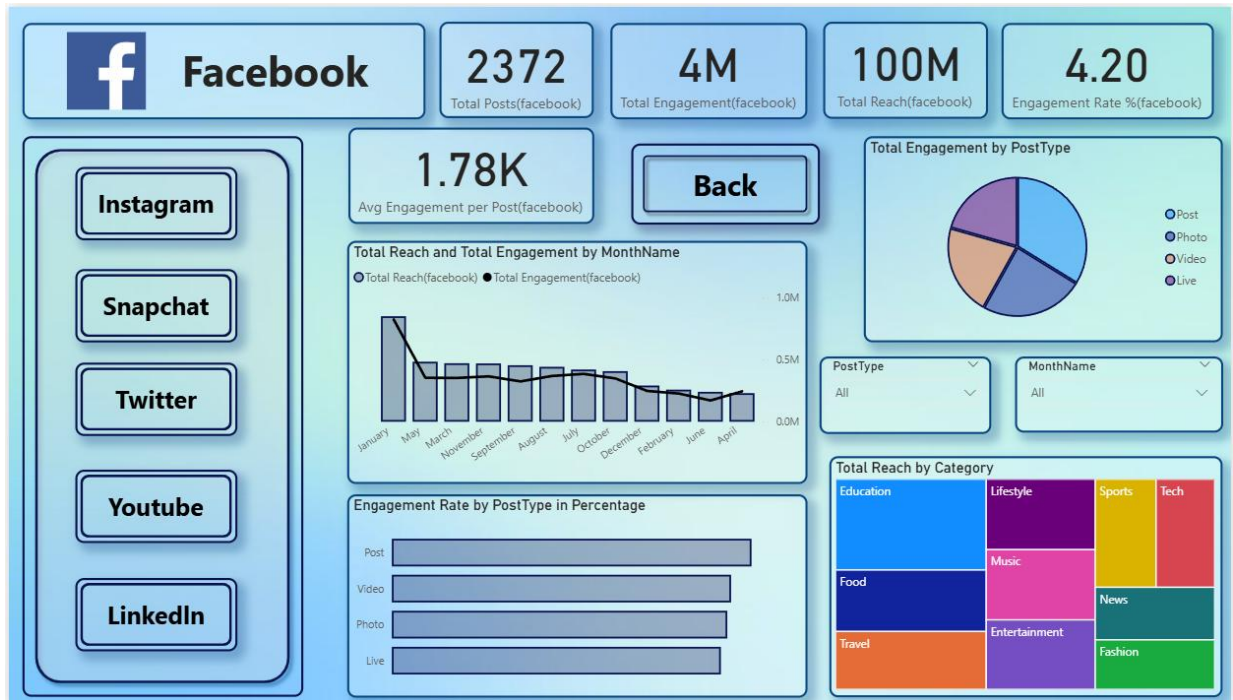
- Youtube Page filter apply – PostType “Live”



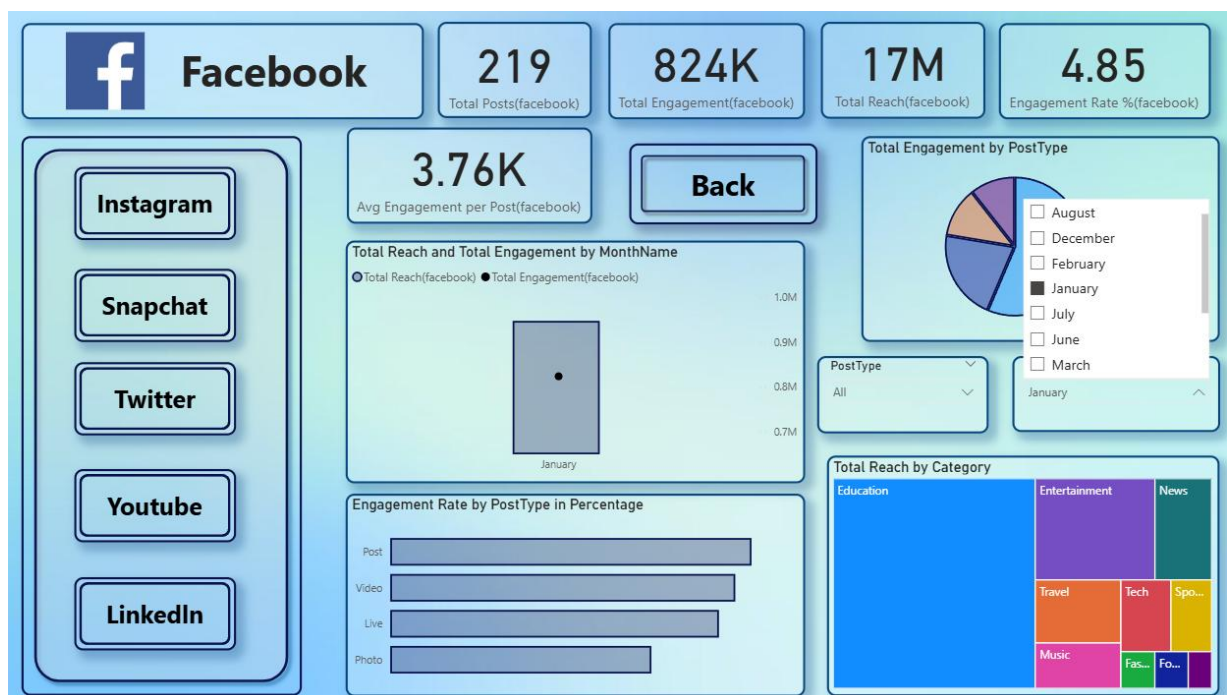
- Youtube Page filter apply – Month “January”



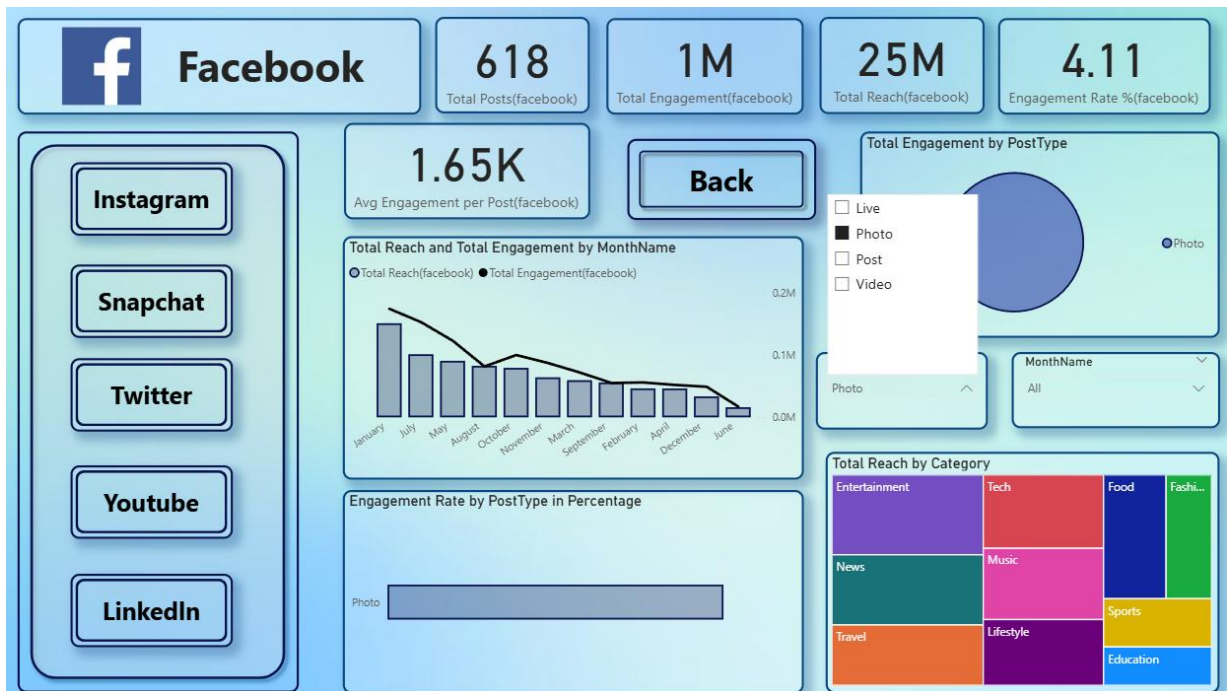
5) FACEBOOK DASHBOARD



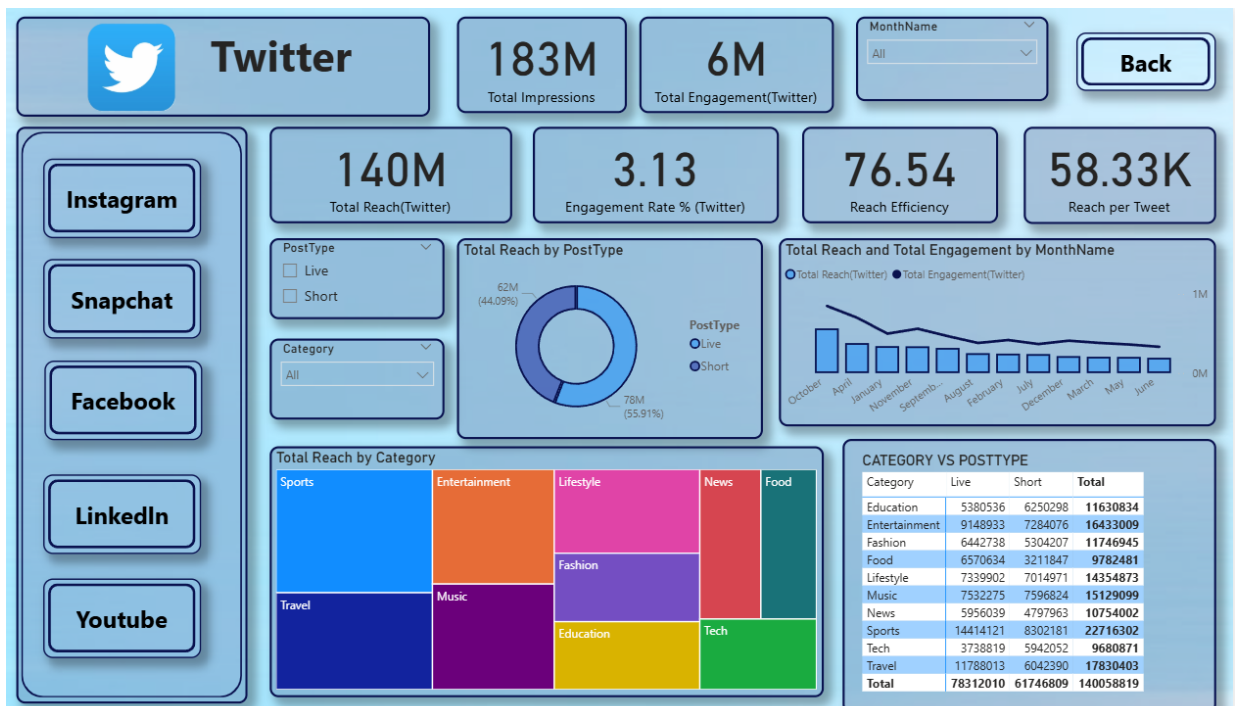
- Facebook page filter apply – month “January”



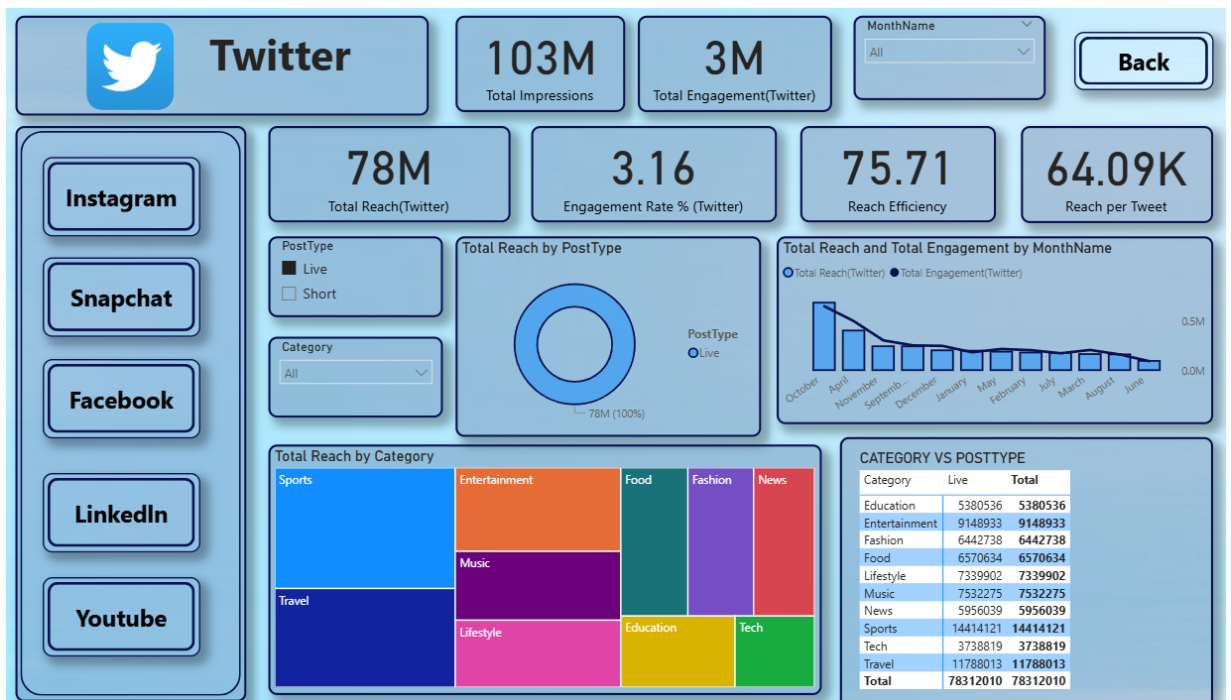
- Facebook page filter apply – PostType “Photo”



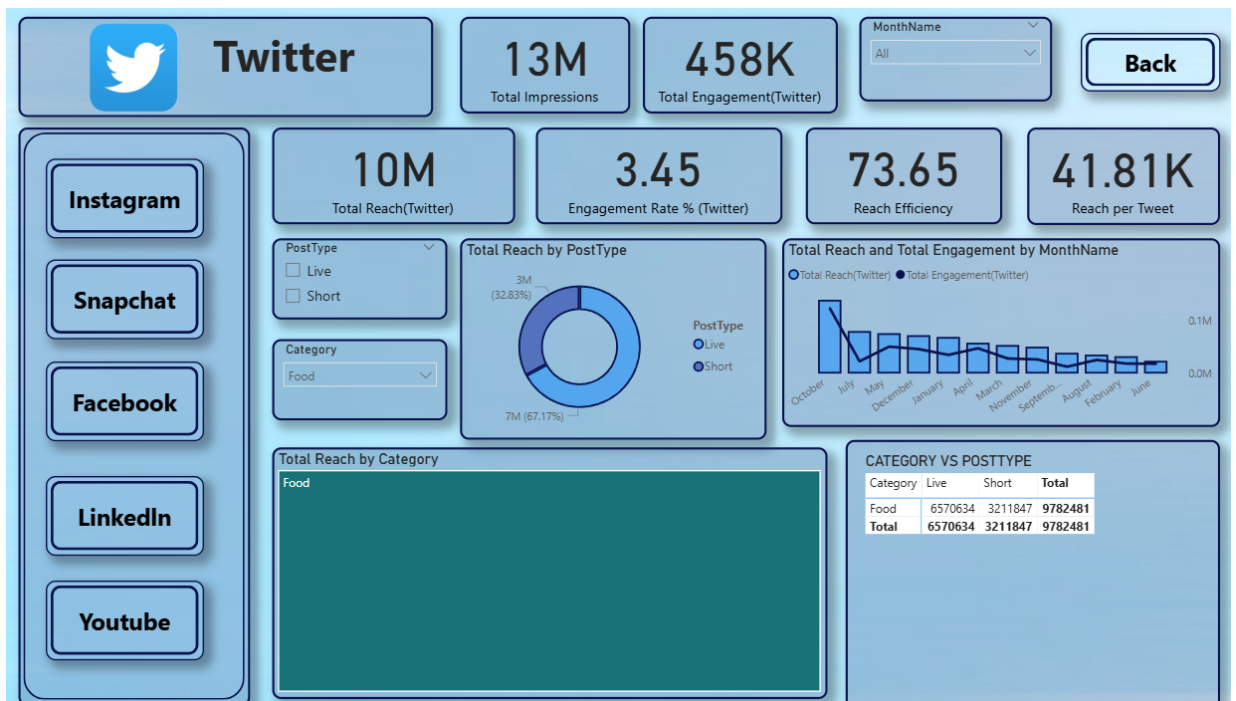
6) TWITTER DASHBOARD



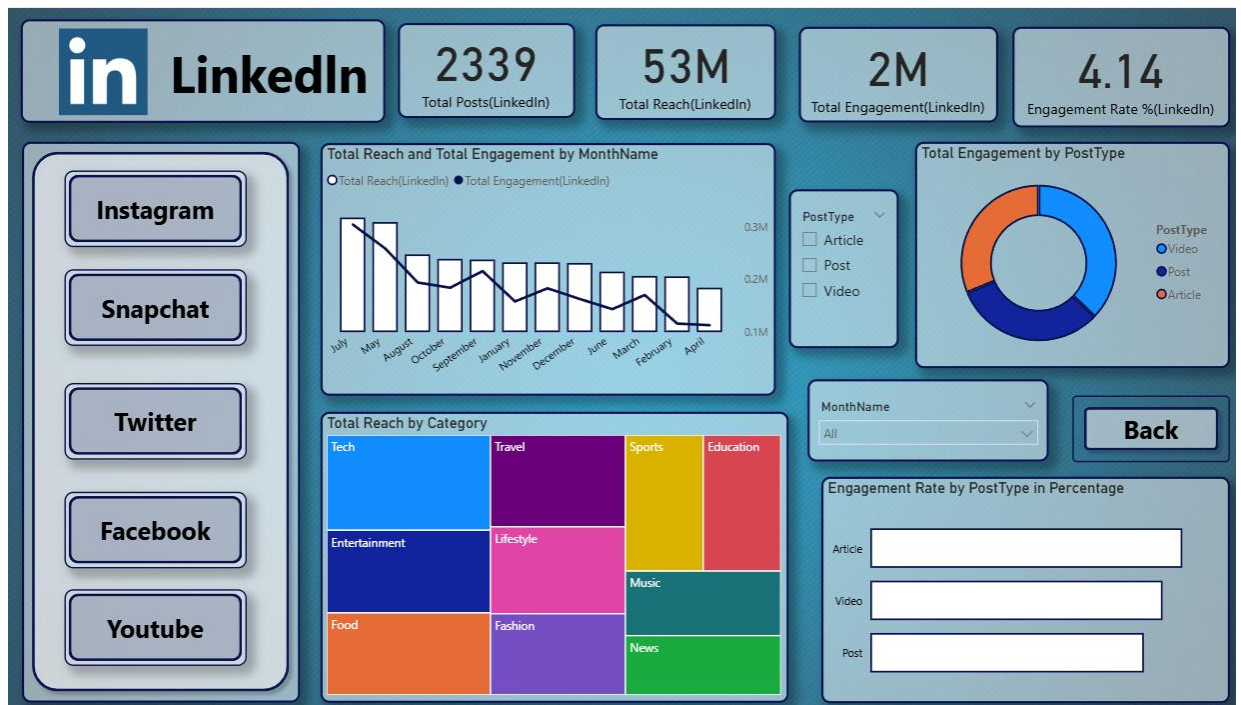
- Twitter page filter apply – PostType “Live”



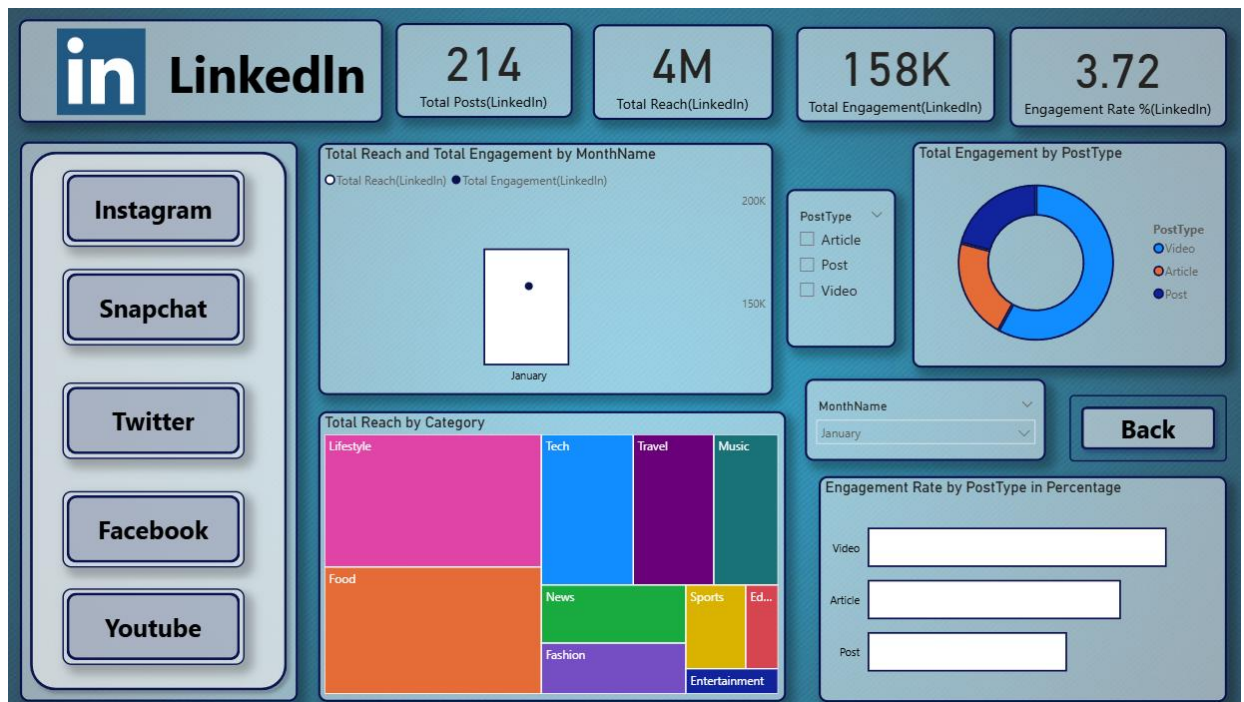
- Twitter page filter apply – category “Food”



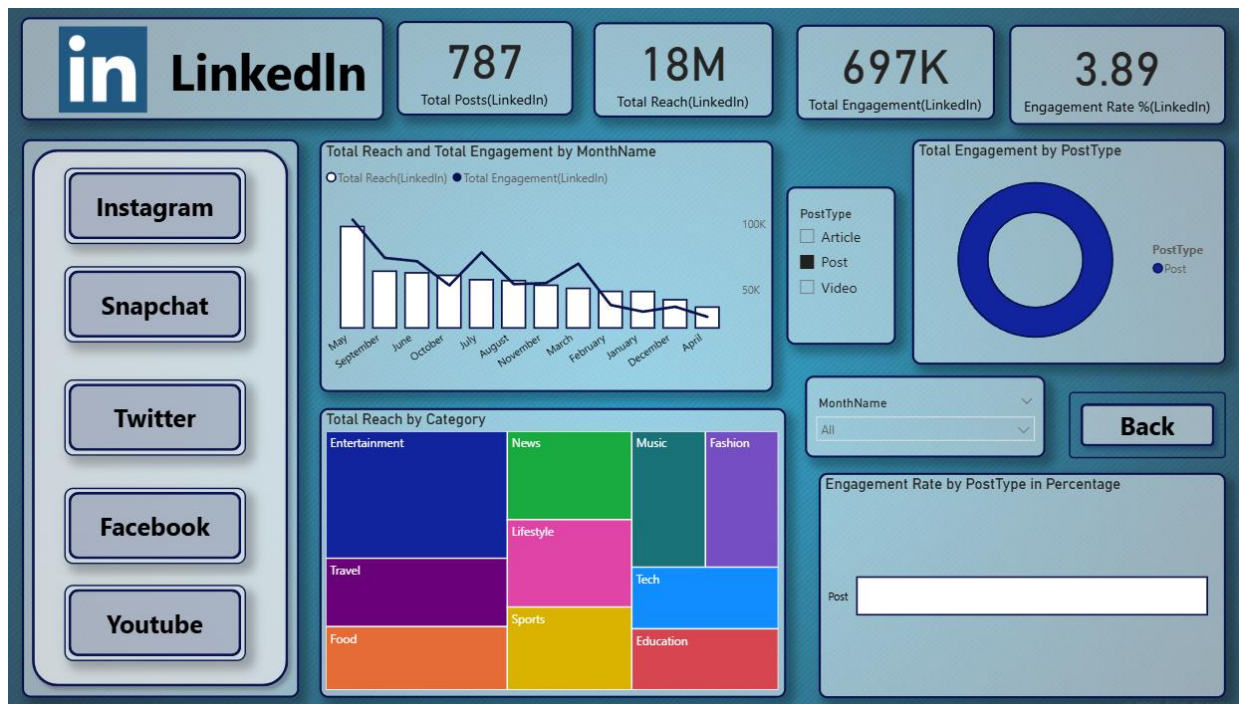
7) LINKEDIN DASHBOARD



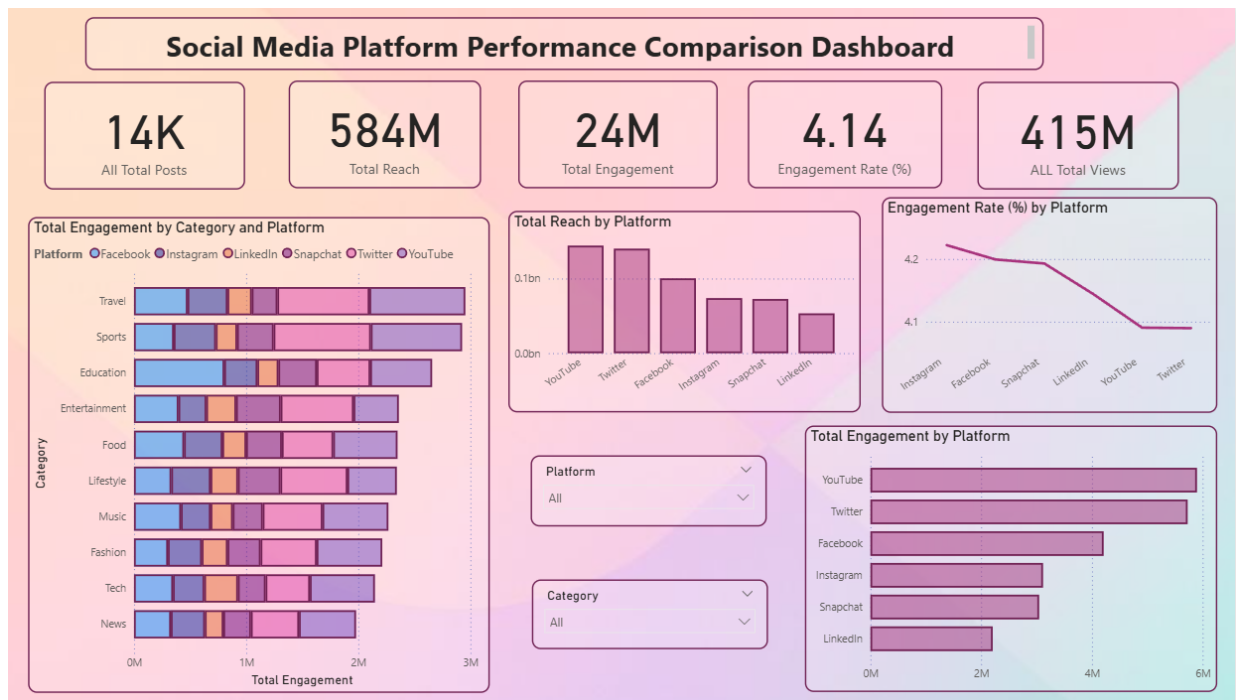
- LinkedIn Page filter apply – month “January”



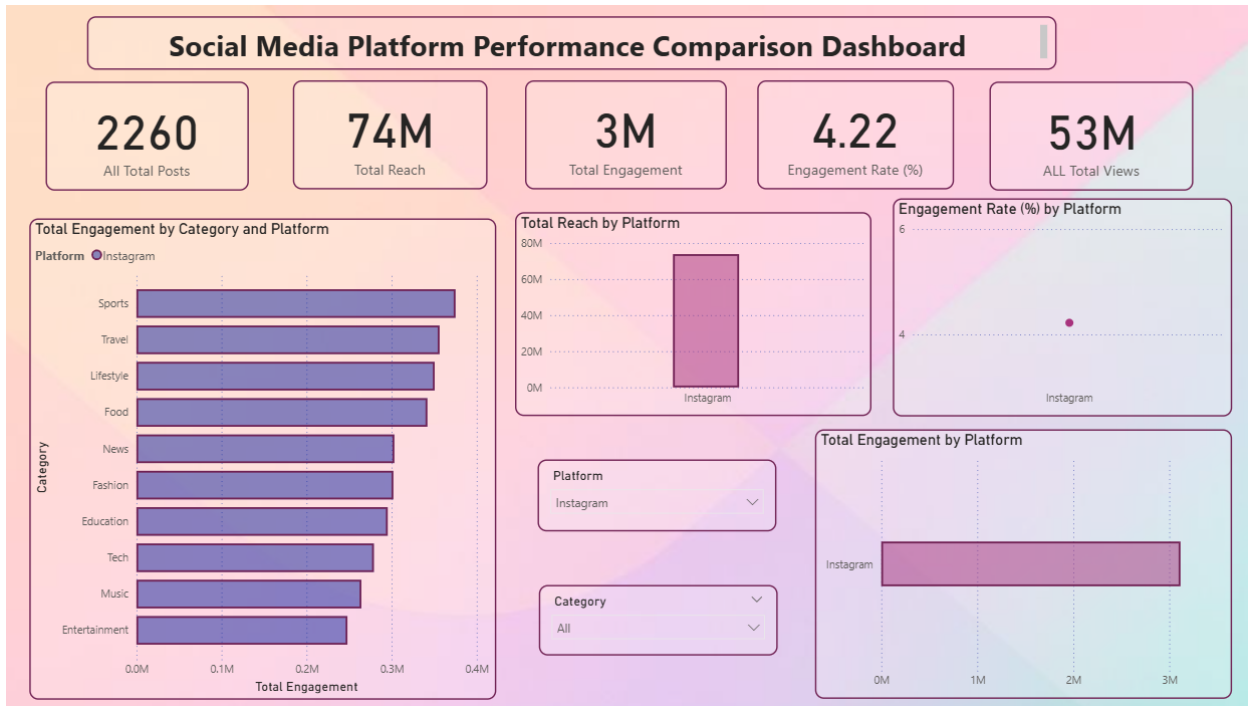
- LinkedIn Page filter apply – PostType “Post”



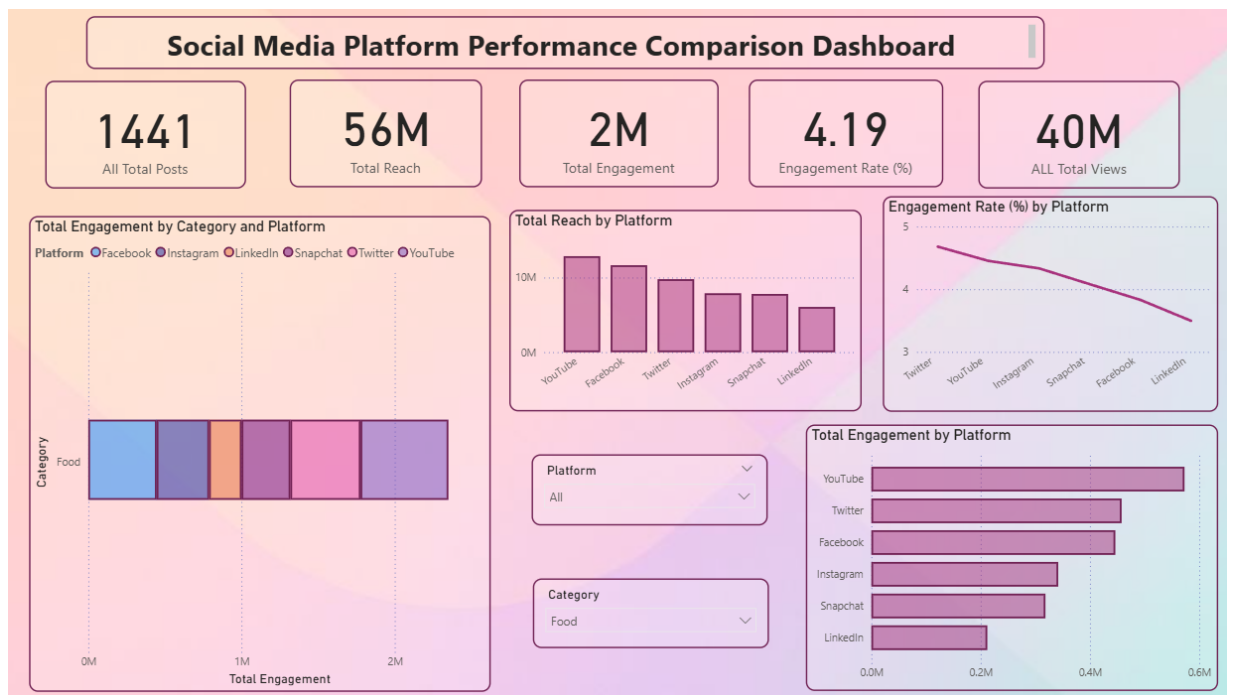
8) COMPARISON DASHBOARD

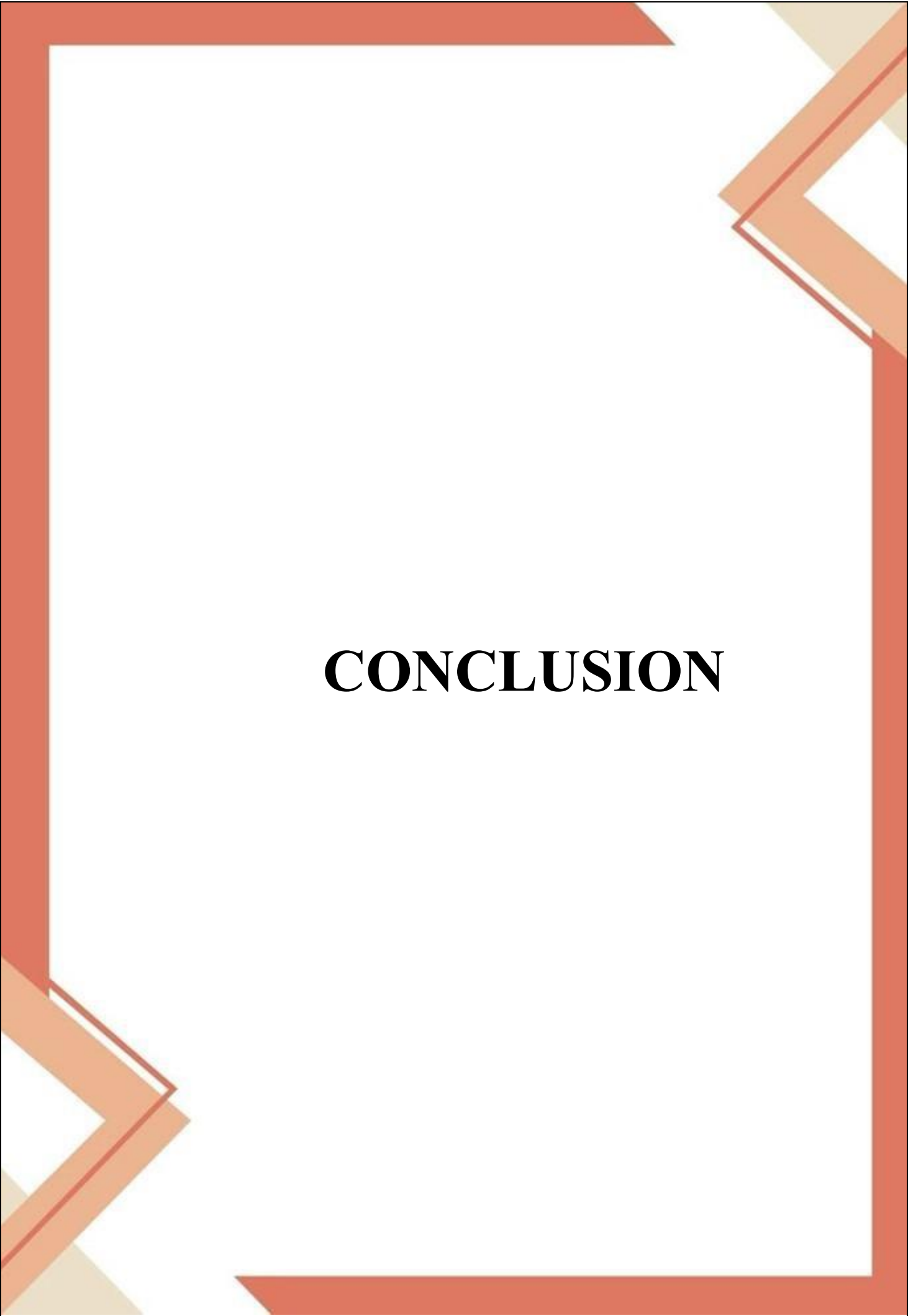


- Comparison dashboard filter apply – Platform “Instagram”



- Comparison dashboard filter apply – Category “Food”





CONCLUSION

CONCLUSION

The ContentXplorer Dashboard project successfully demonstrates how social media data can be transformed into meaningful insights using SQL and data visualization tools. This project focused on analyzing performance metrics across multiple platforms such as Instagram, YouTube, Facebook, LinkedIn, Twitter, and Snapchat. By structuring the database into Platform, Content, and Performance tables, the project ensured proper data organization and relational integrity. SQL queries were used to calculate key performance indicators including Total Reach, Total Engagement, Views, Engagement Rate, and category-wise performance. These metrics helped in identifying trends, comparing platform performance, and evaluating content effectiveness.

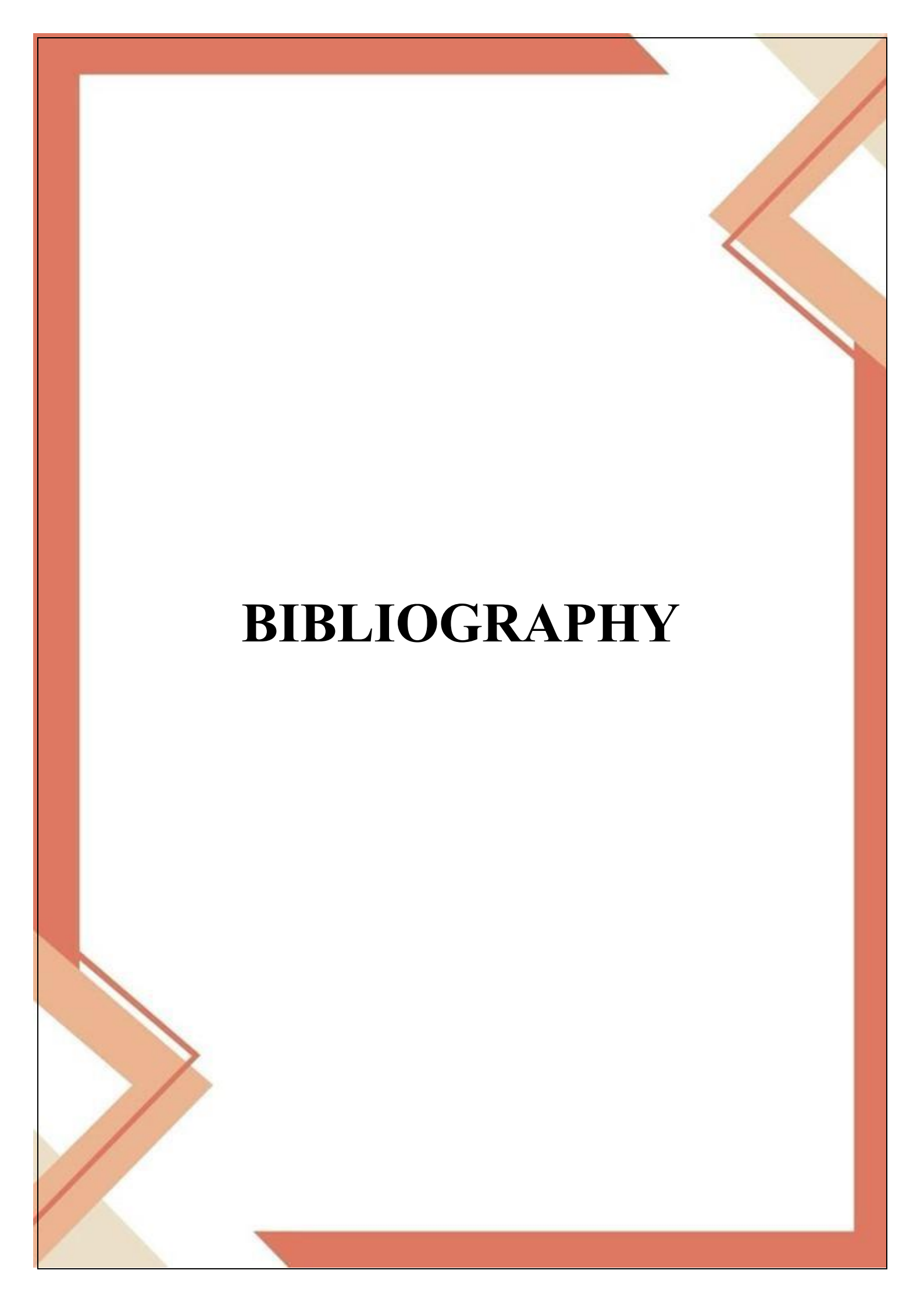
The dashboard provides a clear comparative analysis of six social media platforms on a single page, enabling stakeholders to quickly understand which platform generates the highest engagement, which content type performs best, and how monthly performance trends fluctuate. The use of aggregation functions, grouping techniques, calculated metrics, and conditional logic enhanced analytical accuracy and reliability. Engagement Rate calculations further helped in measuring audience interaction efficiency rather than just raw numbers.

From a business perspective, the dashboard enables cross-platform comparison across Instagram, YouTube, Facebook, LinkedIn, Twitter, and Snapchat within a unified analytical view. This consolidated comparison allows decision-makers to identify high-performing platforms, optimize content categories, evaluate post-type effectiveness, and monitor month-over-month growth trends. Instead of relying solely on vanity metrics like views or reach, the inclusion of engagement rate and interaction-based metrics ensures performance is measured in terms of audience quality and content effectiveness.

The project supports strategic digital marketing decisions by highlighting which content formats maximize engagement, which platforms provide the highest return in audience interaction, and how content strategy can be adjusted based on measurable insights. By transforming unstructured social media data into actionable intelligence, the ContentXplorer Dashboard demonstrates the practical application of SQL-based analytics, performance benchmarking, and business intelligence principles. Overall, the system enhances operational efficiency, improves content optimization strategies, and supports data-backed decision-

making in competitive digital environments.

Overall, this project highlights the importance of data-driven decision-making in digital marketing and content strategy. By converting raw social media data into structured insights, the ContentXplorer Dashboard supports better strategic planning, optimized content creation, and improved audience targeting. It reflects strong skills in database design, SQL querying, data modeling, and analytical thinking, making it a practical and industry-relevant analytics solution.



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BIBLIOGRAPHY

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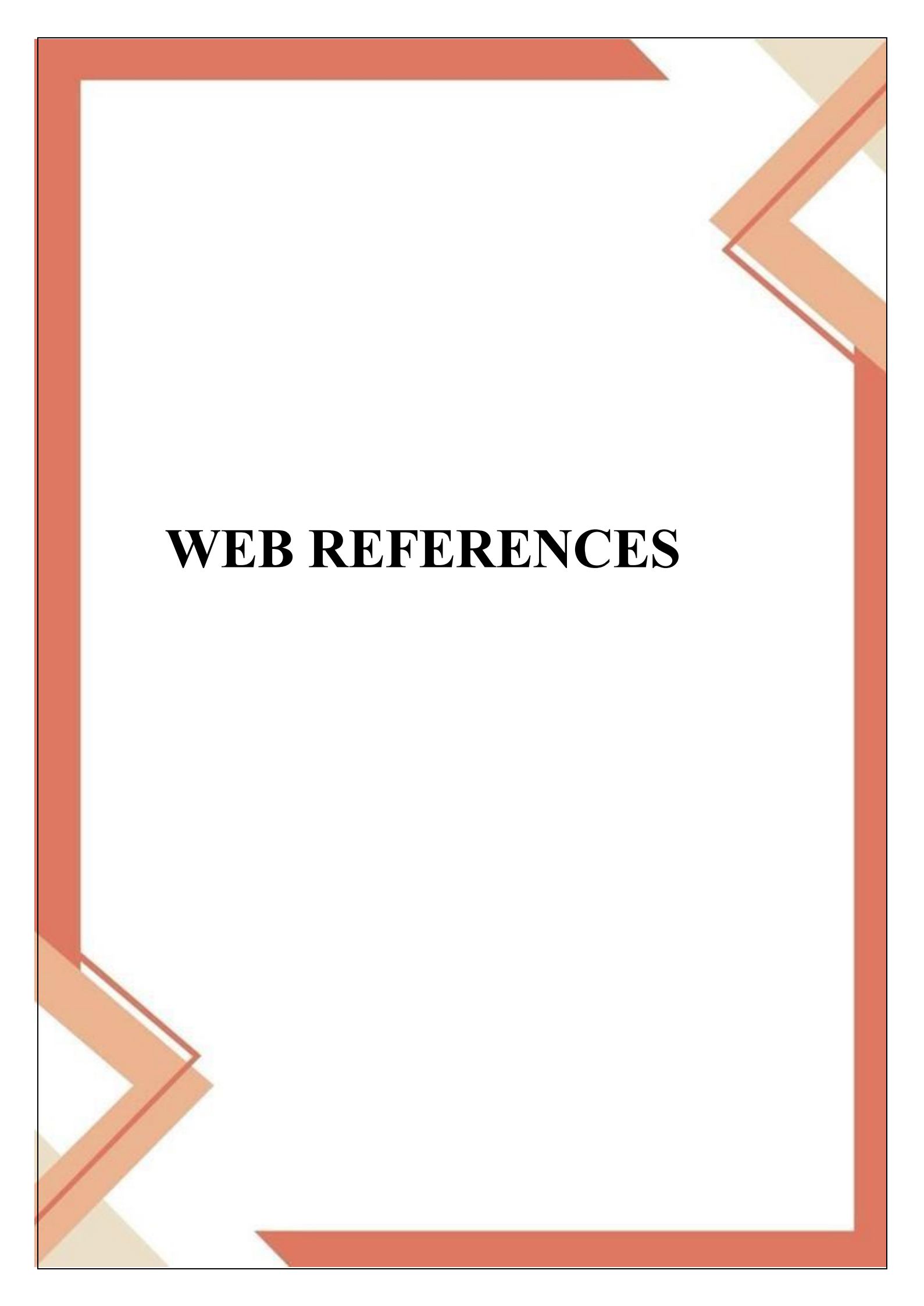
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